The Price of "Free" Advice: Measuring Agency Problems in the Insurance Broker Market*

Elaine Shen† Margaret Kallus

October 26, 2026

Click **here** for the most recent version

Abstract

Governments often rely on private intermediaries to help people navigate complex public programs, but the very information frictions that make advice valuable can also prevent consumers from evaluating advice quality. We study how commissions, information frictions, and competitive pressure shape agency problems in Medicare, a public health insurance program serving 70 million seniors and people with disabilities. In a large-scale audit study, we hire senior "secret shoppers" to make 637 calls to randomly assigned brokers. Brokers steer seniors toward higher-commission plans on 52% of calls. Dominated recommendations carry a 29% markup, costing seniors $377 per year and over $12,000 in present-value lifetime wealth. Advice is driven by incentives and information frictions: when a natural experiment eliminates broker pay, steering to the high-commission plan that does not match senior preferences falls by 47%, but does not fall to zero. A randomly assigned treatment designed to increase perceived competition causes brokers to recommend lower-priced plans, increasing consumer surplus when plans are homogeneous, but decreasing it when plans differ in coverage. Fitting a model of broker incentives and misperceptions to the data, we find that supply-side reforms, such as equalizing commissions and capping premium markups, generate larger consumer surplus gains than changes to consumer search. The findings show that competitive pressure can shift recommendations in complex markets without aligning them with consumer interests, highlighting the importance of well-designed intermediary incentives.

*IRB of record: UC Berkeley Protocol #2024-04-17373 (Audit), 2024-08-17690 (Senior Survey), 2024-03-17226 (Broker Survey); AEA Trial Registration #AEARCTR-0014329 (Audit), #AEARCTR-0018834 (Broker Survey).
We thank professors Ned Augenblick, Sydnee Caldwell, Stefano DellaVigna, Ben Handel, Supreet Kaur, Pat Kline, Jon Kolstad, Ulrike Malmendier, Terry Odean, David Sraer, Dmitry Taubinsky, and Chris Walters for their immensely valuable feedback. We are also very grateful to Jason Abaluck, Luisa Cefalà, Olivia Natan, Gautam Rao, Sharath Sonti, Karthik Tadepalli, Carly Urban, seminar participants at the NBER Summer Institute Workshop on Aging, the Stanford Institute for Theoretical Economics (SITE), the UC Berkeley research community, and the Initiative for Financial Decision-Making at Stanford University for helpful comments and conversations. We gratefully acknowledge funding from J-PAL North America's US Health Care Delivery Initiative (HCDI), the UC Berkeley Experimental Social Science Laboratory, the Commonwealth Fund, the USC Dornsife Center for Economic Research Understanding America Study (UAS), and the Stanford Initiative for Financial Decision-Making. This work was also supported by the Robert Wood Johnson Foundation, grant DOI https://doi.org/10.62700/qmhCqlQeHQ. All errors are our own.

†Shen: UC Berkeley (elaineshen@berkeley.edu); Kallus: UC Berkeley (margaret_kallus@berkeley.edu).

1 Introduction

Governments often rely on private intermediaries to help consumers navigate complex public programs in healthcare, housing, and welfare. Advice from expert intermediaries can facilitate consumer choice, but can also create agency problems. Governments aim to constrain agency problems through regulation and monitoring. Yet a fundamental tension in intermediated markets remains: the very information frictions that make expert advice valuable can also prevent consumers from evaluating the quality of the advice itself.

Understanding the costs and causes of agency problems is especially important in Medicare, a public health insurance program serving 70 million seniors and individuals with disabilities. Medicare is the largest public healthcare program in the US: the government spent over $1.2 trillion on Medicare in 2025 (CMS, 2026). Over half of Medicare beneficiaries rely on brokers to help them choose between Medicare plans that differ in premiums, provider networks, and long-term restrictions on care. Beyond shaping access to healthcare, the choice is also an important financial decision; despite near-universal Medicare coverage, roughly one in four seniors report carrying medical debt, one in three report forgoing medical care because of costs, and healthcare costs make up roughly 14% of household budgets.[1]

The consequential nature of the decision makes outsourcing to licensed experts appealing. Brokers are ostensibly "free" to consult since brokers are paid by private insurance companies for each enrollment. Mandatory sales disclosures, regulatory oversight, the threat of sanctions and fines, and commission caps should help protect the vulnerable population, but payments to brokers still exceeded $10 billion in 2022 alone (Meyers et al., 2026). The lack of data on broker use and difficulty measuring plan choice quality against opaque consumer preferences have made it challenging to evaluate the functioning of this market. Thus the scale of agency problems – including how heavily they distort advice, the cost to consumers, and whether current policies alleviate or exacerbate them – remains an open empirical question.

Using a large-scale audit study and a series of experiments, we study how financial incentives, information frictions, and competition shape agency problems in the Medicare broker market. We hire 109 senior secret shoppers to make 637 phone calls to 554 Medicare insurance brokers. This allows us to generate a dataset of broker advice for the two main decisions most beneficiaries face. In Decision 1, beneficiaries choose between lower-cost, more restrictive plans (Medicare Advantage), and more comprehensive coverage with no network restrictions and higher up-front costs (Traditional Medicare plus Medigap). To detect agency problems in Decision 1, we assign the shoppers preferences that match the plan that pays the broker less. Specifically, we assign shoppers preferences that match all the reasons *not* to enroll in Medicare Advantage listed on Medicare.gov (e.g., preferences for network freedom and comprehensive coverage).

[1] Jacobson and Blumenthal (2022); Gaffney et al. (2024); Gunja et al. (2024); Ochieng et al. (2024).

A key innovation is that Decision 2 allows us to quantify the cost of broker advice in monetary terms. In Decision 2, beneficiaries choose between homogeneous plans with standardized, identical coverage due to regulations (with a Medigap plan letter). Because Medigap plans are homogeneous within a plan letter, we are able to measure lost consumer surplus using the Medigap markup, or the amount the senior overpays for a plan with identical coverage due to a dominated recommendation. Since brokers directly enroll beneficiaries in the plans they recommend, our audit study outcomes are reasonable proxies for the enrollment outcomes seniors face.

We find that brokers recommend a higher-commission but less suitable plan on 52% of calls. In Decision 1, brokers recommend a Medicare Advantage plan, which pays them roughly $400 more but does not match consumer preferences, on 32% of calls. In Decision 2, where brokers are paid a percent of the Medigap premium but higher-premium plans have identical coverage and are strictly dominated, the average Medigap markup is $377 per year, a 29% markup. We estimate that dominated Medigap recommendations cost seniors at least $12,000 in present-value lifetime wealth given average premium growth rates. Using a back-of-the-envelope calculation, we estimate the recommendations we observe would cost seniors $1.25 billion a year in lost consumer surplus if scaled to the 1.41 million seniors who turn 65 and use a broker each year. Strictly dominated Medigap plan recommendations alone would cost these seniors $210 million a year.

Two experiments allow us to test the drivers of broker advice. We first take advantage of an unexpected commission change during the audit study to test how broker advice responds to commissions. In this natural experiment, Experiment A, every insurance company that call-center brokers could sell in one of our audit regions stopped paying commissions for Medicare Advantage (MA) enrollments. These plans originally paid higher commissions, but did not match consumer preferences. We collect audit study data before and after this commission change. For the exact same consumer profile, the rate of unsuitable MA recommendations falls by 47%, or 21 percentage points. However, the rate does not fall to zero. Call-center brokers continue to recommend MA on 24% of calls when the plan both does not match consumer preferences and does not pay a commission. The results provide causal evidence that recommendations are driven by both commissions and information frictions.

In Experiment B, we experimentally randomize a treatment designed to test whether perceived competition disciplines advice. Within the audit study, we randomly assign secret shoppers to tell brokers that they plan to talk to multiple brokers. The treatment is motivated by the fact that 66% of seniors only talk to one broker before enrolling in Medicare. We find that advice is elastic to the perceived competition treatment; brokers respond to the treatment by shifting their recommendations to the lower-price plan in both outcomes. In Outcome 1, brokers raise the rate of MA recommendations by 11 percentage points (SE 5). In Outcome 2 the Medigap markup falls by $84.15 a year (SE 20.01). When plans are homogeneous and vary in price alone (as with the

Medigap markup in Outcome 2), the treatment increases consumer surplus. When plans vary in both price and coverage (Outcome 1), however, the broker response to perceived competition also lowers plan coverage and leads to a net reduction in consumer surplus, according to broker beliefs from a survey of currently licensed Medicare brokers.

To evaluate which broker attributes consumers value, after each call we also ask the secret shoppers to rate how likely they are to recommend the broker to a friend or family member enrolling in Medicare. This allows us to test whether broker ratings are correlated with our two measures of broker recommendation quality. Broker ratings are uncorrelated with our main measures of broker quality, but are correlated with call length. This suggests that seniors' ratings do not distinguish good from bad advice by objective quality and instead reflect other heuristics. The results highlight how consumer information frictions can limit feedback mechanisms and allow low-quality advice to persist.

Finally, we estimate a structural model by combining the audit study data with a broker survey. We use the model to compare consumer surplus gains under supply-side policies that can be implemented by the government to changes in how seniors search for advice. The model estimates suggest that 38%–49% of MA recommendations are driven by commissions and that information frictions account for 41% of MA recommendations. Lost consumer surplus from broker advice is approximately $2,154 a year for a senior with our audit-study profile. Equalizing first-year Medicare Advantage and Medigap commissions recovers 18% of lost consumer surplus. Capping the Medigap markup at 25% recovers 6% of lost consumer surplus. In contrast, policies that change consumer search and rely on consumer feedback to discipline advice, such as competitive pressure (signaling plans to solicit multiple recommendations) and limiting search to highly rated brokers, recover far less. A "consumer education" counterfactual recovers 20% of lost consumer surplus.

We provide some of the first direct evidence of agency problems in a new, important financial market serving an older population. The paper contributes to the literature on expert intermediation as well as a growing literature on aging. Existing research has documented agency problems and broker steering in other predominantly private markets for financial advice (Malmendier and Shanthikumar, 2007; Bhattacharya et al., 2012; Gennaioli et al., 2015; Egan, 2019; Robles-Garcia, 2019; Linnainmaa et al., 2021; Bucher-Koenen et al., 2025). A related literature shows that older adults face difficulty making financial decisions (Agarwal et al., 2009; Kézdi and Willis, 2014; Lusardi et al., 2014; Kim et al., 2021; Ameriks et al., 2023; Mazzonna and Peracchi, 2024). We build on the literature by showing how misaligned incentives shape a consequential insurance decision faced by this vulnerable population.

Second, we build on a literature using audit studies as a measurement tool to translate broker advice into monetary loss. Furthermore, by experimentally randomizing a treatment increasing

competitive pressure within the audit study, we are able to directly detect strategic broker behavior. The paper builds on existing research using audit studies to identify discrimination and misconduct that is difficult to detect in observational data (Bertrand and Mullainathan, 2004; Bertrand and Morse, 2011; Gneezy et al., 2012; Mullainathan et al., 2012; Anagol et al., 2017; Alsan et al., 2019; Kessler et al., 2019; Kline et al., 2022). This allows us to measure the cost of agency problems in a setting where data are scarce and quality is difficult to measure.

Third, we contribute to the literature on complexity and mistakes in health insurance choice by showing that demand-side frictions can create an economic environment in which supply-side distortions can generate additional losses. Prior work has documented several demand-side frictions: consumers overweight premiums (Abaluck and Gruber, 2011), choose dominated plans (Afendulis et al., 2015; Bhargava et al., 2017), misunderstand coverage (Brot-Goldberg et al., 2017), and remain in unsuitable plans due to inattention (Ericson, 2014; Ketcham et al., 2015; Heiss et al., 2021; Brot-Goldberg et al., 2023). This paper builds on the literature studying the supply-side response to consumer information frictions by highlighting the importance of well-designed regulations in the face of these frictions (Aizawa and Kim, 2018; Cabral et al., 2018; Gruber et al., 2020).

Finally, this paper speaks to the theoretical and experimental literature on expert advice in credence-good markets and physician-induced demand by testing the model predictions in a field setting (Pitchik and Schotter, 1987; McGuire and Pauly, 1991; Wolinsky, 1993; Inderst and Ottaviani, 2009, 2012; Dulleck et al., 2011; Carroll, 2019; Balafoutas and Kerschbamer, 2020; Karakoç et al., 2024). In these models competition can either raise or lower the probability of conflicted recommendations; we test these predictions empirically within the audit study.

2 Background on Medicare insurance plan choice

This section provides institutional context on Medicare and the Medicare broker market. In a nationally representative survey, we find that 51% of current Medicare beneficiaries enrolled through a broker. This number may be an underestimate since many seniors think they are talking to "Medicare" when they enroll and do not realize they are enrolling using a broker.[2]

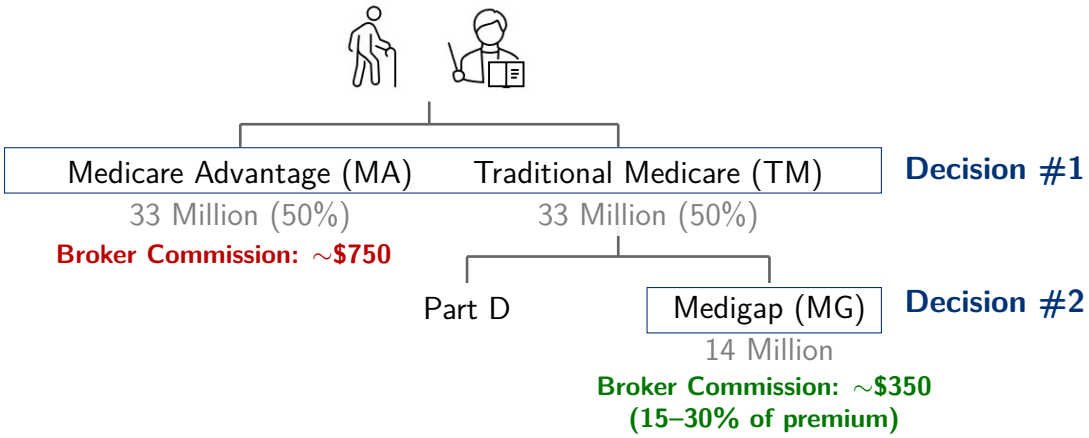
When seniors first join Medicare Part B (typically, when they turn 65 or retire), they qualify for a 7-month Initial Enrollment Period when they can choose any Medicare insurance plan.[3] This initial choice can be difficult to reverse; depending on the plan chosen, medical underwriting can

[2]Data on the use of brokers or agents is limited and mostly comes from surveys, including our own survey conducted through the Understanding America Study and described in Appendix B.4. Other surveys of broker use include Leonard et al. (2022) and Freed et al. (2023). Broker use is also expected to expand since Republican administrations have expressed support for expanding MA, which is primarily sold through brokers. See https://www.npr.org/sections/shots-health-news/2024/11/07/nx-s1-5183092/trump-election-2024-medicare-advantage.

[3]Seniors who are still working and delay Medicare coverage because they have insurance through an employer get a Special Enrollment Period when they retire and begin Part B coverage.

leave seniors stuck in their initial plan type, with long-term implications for access to healthcare, health outcomes, and retirement savings. Figure 1 provides an overview of the choice most seniors face when they enroll in Medicare.[4] The audit study focuses on two key parts of the decision tree: Decision 1, the choice between MA and Traditional Medicare plus Medigap; and Decision 2, the specific Medigap plan choice.[5]

Figure 1: Medicare Choice Landscape



Notes: Figure 1 shows the choice set a beneficiary faces upon enrolling in Medicare and the two decisions we study. Decision #1 is the choice between Medicare Advantage and Traditional Medicare; Decision #2 is the choice of Medigap plan among beneficiaries with Traditional Medicare. Dollar amounts show the approximate first-year commission a broker earns from each enrollment: a ~$750 lump sum for a Medicare Advantage enrollment and 15–30% of the annual premium (~$350) paid monthly for a Medigap enrollment. Section 2 details the full commission schedule.

Decision 1: Medicare Advantage vs Traditional Medicare + Medigap (TM+MG) The first choice all Medicare beneficiaries make is whether to enroll in an MA plan or Traditional Medicare (TM, also known as Original Medicare or Fee-For-Service Medicare). About half of Medicare beneficiaries have MA and half have Traditional Medicare. Beneficiaries enrolled in Traditional Medicare may also choose to enroll in a Medicare Supplement or Medigap (MG) plan, described further below, to cover out-of-pocket costs not covered by Traditional Medicare such as copays, coinsurance, or deductibles. Each option has its own costs and benefits summarized in Figure 2.

MA plans are offered by private insurance companies. Traditional Medicare and Medigap are administered by the federal government (specifically, the Centers for Medicare and Medicaid Services, or CMS). Figure 2 summarizes how the two options differ. Since MA plans are not

[4]We focus on the individual market faced by the majority of Americans. This is different from the group market, which includes a different set of non-standard (often subsidized) employer or union-sponsored retiree plans. These plans are usually managed by a former employer and not available to the general public. The choice set is also different for individuals eligible by reason of disability and under the age of 65, who have limited Medigap access, beneficiaries who are dual-eligible with Medicaid, and people with access to other supplemental care (e.g. veterans).

[5]Most beneficiaries with Traditional Medicare and Medigap also enroll in a separate Medicare Part D prescription drug plan. We do not focus on Part D in this paper since it is typically the smallest share of broker compensation: brokers are paid roughly $100 for an initial Part D enrollment.

Figure 2: Medicare Plan Tradeoffs by Category

Category	Medicare Advantage (MA)	Traditional Medicare+Medigap
Premium	**Low, sometimes $0/month**	Higher (∼$140/month)
Drug coverage	**Sometimes bundled in**	Separate Part D plan ($0-$50/month)
Extra benefits	**Often dental, vision, hearing, gym**	Not included
Restrictions on care	Network restrictions, referrals, prior authorization	**Any doctor that takes Medicare (nationwide coverage)**
Cost sharing (OOP)	Unpredictable; $14,000 cap	**Predictable; low (<$300)**
Switching	Underwriting required, hard to switch	**Switch to MA easily**

standardized, coverage, network size, and claims denials vary by plan, geography, and insurer (Biniek et al., 2024; Gaffney et al., 2024; Vabson et al., 2025).

MA and Medigap plans also differ in their flexibility to change coverage. Outside of the initial 7-month enrollment window, seniors with Traditional Medicare and Medigap can easily switch to MA without medical underwriting every year during special enrollment windows such as the Annual Enrollment Period (AEP), but seniors cannot switch from MA to a Medigap plan without medical underwriting. In practice, many seniors end up being stuck in lower-coverage MA plans or are forced to pay significantly higher premiums when they switch to Medigap (Boccuti et al., 2018; Curto, 2023; USC Schaeffer Center, 2024). For this reason, MA is sometimes described as a "one-way door." The asymmetric nature of switching between products in this market makes a poor initial plan choice difficult and sometimes impossible to reverse for seniors who would not pass medical underwriting.

Decision 2: Medicare Supplement (Medigap or MG) Plan Medicare Supplement (Medigap) plans help cover out-of-pocket costs such as copays, coinsurance, and deductibles that are not covered by Traditional Medicare. For example, Traditional Medicare covers 80% of the cost of most outpatient appointments and procedures. Medigap plans help cover the remaining cost sharing expenses. (Additional details on the market are included in Appendix A.)

A key feature of Medigap plans is that the plans are standardized, homogeneous products within a plan letter due to a regulation passed in 1990. Plans are sold by private insurance companies, but coverage terms come from a menu set by the government, as shown in Appendix A. As explicitly stated on Medicare.gov, the only difference between Medigap plans with the same plan letter is price. CMS is the primary payer and all Medigap claims are approved by CMS. Beneficiaries essentially never interact with the private insurance company, leaving little room for differences in

claims denials or customer service.[6]

About 14 million people today have a Medigap plan. Given their less restrictive coverage and higher quality of care, these plans are a popular choice for experts in the health insurance field (Ochieng et al., 2023). In interviews, several brokers stated "anyone who can afford to should have Traditional Medicare plus Medigap" and surveys of Medicare brokers found that brokers were more likely to choose Traditional Medicare plus Medigap for themselves (Leonard et al., 2023).

The Medicare market is so complex that even beneficiaries who are already enrolled in a plan struggle to correctly report their plan information. In a survey of 2,015 nationally-representative Medicare beneficiaries fielded through the Understanding America Study (UAS) (Appendix B.4), at least 40% of respondents describe their own current coverage in a way that does not match the Medicare plans offered. Specifically, these respondents report 1) a Medigap premium far below the minimum premium in the market; 2) an MA premium significantly higher than average premiums ($\geq$\$100/month); 3) a Medigap plan letter that is not part of the federal menu; or 4) not knowing which Medicare plan type they have. This is surprising for a longitudinal panel of study participants who may be positively selected. The survey results also highlight the difficulties evaluating broker quality using survey data: a beneficiary who cannot correctly identify their Medicare plan is unlikely to correctly evaluate plan attributes and plan quality.

Regulating broker incentives Medicare Advantage is regulated at the federal level by CMS; Medigap is primarily regulated by state Department of Insurance offices. The regulations that govern broker sales practices and broker commissions follow the same split. Table 1 summarizes the commission each product pays. CMS sets a cap for the lump-sum commission which adjusts annually and varies by state. MA commissions were roughly \$750 per enrollment in our study. Medigap commissions are 15–30% of plan premiums; we use 20% in the structural model. Commissions after year 6 are often much lower and have high variability.[7]

Insurance Company Appointments Medicare insurance brokers are paid by the private insurance companies they are licensed, or "appointed," to work with. Payments to Medicare insurance brokers exceeded \$10 billion in 2022 alone (Meyers et al., 2026). Broker commissions are set by insurance companies, but are subject to the caps described above. Brokers may or may not work with every insurance company or sell every Medicare product. Unlike in other intermediated markets,

[6]To test whether brand preferences might contribute to price dispersion among the identical products, we survey a representative group of senior Medicare beneficiaries through the Understanding America Study, described in Appendix B.4. 88% of seniors say there is no insurance company they are trying to avoid, and 61% have no positive or negative brand associations.

[7]Medigap commissions are guaranteed through year 6, but are allowed to continue after year 6. MA commissions were originally limited to years 1-6, but the 6-year guidance was dropped in a rule codified in 2021 (CMS, 2021). In practice, the amount the broker takes home from any of these enrollments may be less than the stated amounts since broker compensation structures depend on the hierarchy of their Field Marketing Organization (FMO).

Table 1: Broker commission schedule by Medicare product

Product	Year 1 commission	Renewal (years 2-6)	Payment
Medicare Advantage	$\sim$ \$750	50% of year 1	Lump sum
Medigap	$\sim$20% of premium ($\approx$ \$350)	50% of year 1	12 monthly
Part D	\$109	\$55	12 monthly

Notes: Table 1 reports broker commissions by Medicare product. CMS sets the maximum Medicare Advantage (MA) and Part D commissions at fair-market value (FMV) and caps renewals at 50% of the year-1 amount (42 C.F.R. §§422.2274(d), 423.2274(d)) (CMS, 2026). The MA FMV cap varies by state and year. The national rate, which applies in Texas, was \$611 in 2024, \$626 in 2025, and \$694 in 2026. The California cap was \$762, \$780, and \$864 in those years. Most major carriers pay the cap, so MA commissions do not differ across MA plans. Medigap commissions are set by the insurer rather than CMS. Industry schedules pay roughly 15–30% of the annual premium in year 1 and half that in renewal years, or about \$350 at the mean premium (Ali et al., 2021). The structural model uses 20% of the call-specific premium in year 1, with renewals at 10%. In general, renewal commissions are not guaranteed after year 6.

the appointment constraints are rarely binding since the majority of states allow brokers to obtain an appointment right before a sale is made through Just-In-Time (JIT) appointments.[8] Brokers may be appointed to sell for one or multiple insurance companies and are directly compensated by the insurance company for every beneficiary they enroll in a plan.

Government oversight of the broker market The Medicare broker market is highly regulated and closely monitored in part due to the government's financial stake in the program: the federal government spent over \$1.2 trillion on Medicare in 2025, over half of which was subsidies to private insurance companies (CMS, 2026). Brokers are required to pass certification exams and hold a state license to sell health insurance.[9] Limits on broker pay are set by federal and state agencies. While brokers are not subject to "fiduciary rules," state laws require insurance agents to "make reasonable efforts" to determine whether a recommendation is appropriate (NAIC, 2008). CMS can impose sanctions on insurers who contract with brokers that violate CMS rules, including suspending federal payments or imposing monetary fines.

Direct monitoring of Medicare broker activity is, at least on paper, more extreme than other intermediated markets. CMS requires Medicare brokers to record all sales calls and store the recordings for up to 10 years.[10] CMS also requires brokers to disclose that they may not work with all available insurance companies and refer the senior to Medicare.gov or free State Health Insurance Assistance Program (SHIP) resources within the first minute of a call (Appendix A). All broker and insurance company marketing material must be approved by regulators prior to distribution. Brokers that violate these rules can face regulatory action or lose their license.

Disciplining market forces In addition to regulatory concerns, traditional market forces such as reputation concerns, referrals, and competition might be thought to limit agency problems.

[8]Both states in our audit study, California and Texas, allow just-in-time appointments.

[9]Brokers who sell MA or Part D plans must pass Medicare training and testing every year with a score of at least 85 percent (42 C.F.R. § 422.2274(b)).

[10]This requirement was changed to 6 years in contract year 2027 (CMS, 2026).

With thousands of licensed brokers in each state, most of whom are licensed in multiple states, the advice market is highly competitive. Brokers are not paid by seniors per consultation and are free for seniors to consult, lowering search costs. Additionally, information on Medicare plans is freely available and must be disclosed to prospective clients on every call: a senior can check a broker's recommendation against the Medicare.gov Plan Finder tool or consult a SHIP counselor at no cost. Whether government oversight and market forces have been successful at limiting agency problems in this market remains an open empirical question.

3 Conceptual Framework for Broker Advice

This section presents the sketch for a simple conceptual framework of the interaction between seniors and brokers. A senior enrolling in Medicare does not understand the different insurance plans and calls a broker for a plan recommendation. The broker is paid by the private insurance company if they enroll the senior in a plan. The broker earns a different commission for each plan and is paid nothing if the senior does not enroll. The senior either accepts the recommendation and enrolls in the plan (the broker closes the sale) or chooses to incur a search cost and call another broker. Online Appendix A proves every result. Section 6 adds structure and estimates the model.

3.1 Setup: Plans, Prices, and Broker Pay

A senior enrolling in Medicare chooses between two insurance plans, $j \in \{L, H\}$, with annual premiums $\rho_L < \rho_H$. The senior seeks a recommendation from broker b. The broker recommends one plan $j \in \{L, H\}$, not the entire menu of plan options. We also assume the broker focuses on persuading the senior that their recommendation is correct and does not explain the alternative plan. The senior then compares the recommended plan against the outside option value v_s, or the continuation value of search. The broker maximizes expected utility, or the probability of making the sale times pay from selling plan j. Selling plan j pays the broker a commission $W_j > 0$. The broker is paid W_j only if the senior enrolls in plan j.

3.2 The Consumer's Problem

The senior chooses the plan that maximizes their perceived utility. We define all utility in monetary terms. Focusing on the difference between the two plans, the net harm from plan L is defined as

$$N \equiv U_c(H) - U_c(L) = \Delta - (\rho_H - \rho_L), \tag{1}$$

where ρ_j is the plan premium and Δ refers to the fit gap, or the gap between the plan's attributes and the senior's preferences (e.g. lost utility from under-insurance). We assume that at the point

of sale the senior perceives plan premium ρ_j exactly and a fraction $\phi \in [0,1]$ of the plan fit gap Δ. We define ϕ as the friction parameter, or a reduced-form object that accounts for the weight the senior puts on premiums versus plan attributes. At $\phi = 1$ the senior perceives the fit gap in full (the full-information benchmark). At $\phi = 0$ the senior perceives none of the fit gap (a pure credence good). ϕ can be thought of as information frictions, a complexity parameter that prevents seniors from fully understanding insurance attributes, or consumer biases such as overweighting plan premiums relative to plan coverage.[11] For a consumer with friction ϕ, the perceived net harm of plan L is

$$\hat{N} \equiv \hat{U}_c(H) - \hat{U}_c(L) = \phi\Delta - (\rho_H - \rho_L). \tag{2}$$

For a senior whose preferences best match plan H, we assume that

$$\Delta > \rho_H - \rho_L, \tag{3}$$

that is, the harm in monetary terms from under-insurance Δ is greater than the savings from the lower plan premium, so the net harm N in (1) is strictly positive. We assume the senior is non-strategic and does not infer anything about the broker's pay or financial incentives from the broker's recommendation.

After the senior receives a recommendation from the broker, the senior compares the recommended plan against the outside option value, or the net continuation value of additional search. The outside option value is worth v_s, a constant dollar amount that varies with senior-specific search costs, but does not vary with the broker recommendation. $s \in \{0,1\}$ indexes the perceived competition treatment arm. We model the perceived competition treatment as a signal about lower search costs: $s = 1$ on treated calls and the treatment is the signal that $v_1 > v_0$. In the comparative statics below we treat the continuation value as a continuous variable v.

We assume a logistic taste shock on the probability that the senior accepts the recommendation or continues search. The probability that a broker recommendation in treatment arm s ends in a completed sale is then

$$P_{r,s} = \Lambda\Big(\hat{U}_c(r) - v_s\Big), \qquad \Lambda(t) = \frac{1}{1+e^{-t}}, \tag{4}$$

where Λ is the standard logistic cumulative distribution function and $P_{r,s}$ is the sale close probability (or close probability for short). Since Λ is strictly increasing, the plan that maximizes the senior's perceived utility $\hat{U}_c(r)$ is the easier sale to close. We invert (4) to convert the probabilities back to the latent dollar index. Taking the difference between the two plans gives us

$$\log\frac{P_H}{1-P_H} - \log\frac{P_L}{1-P_L} = \hat{N}. \tag{5}$$

[11] This framework is related to models where consumers overweight premiums relative to coverage (Abaluck and Gruber, 2011), attend to salient attributes (Bordalo et al., 2013), or do not see shrouded attributes (Gabaix and Laibson, 2006).

Thus $\hat{N}$, the perceived net harm from plan L, determines the gap between the two close probabilities. This implies $P_H > P_L \iff \hat{N} > 0 \iff \hat{U}_c(H) > \hat{U}_c(L)$: intuitively, the product the senior perceives to have the greater utility is the easier sale for the broker to close.

3.3 The Broker's Problem

The broker observes which plan best matches the senior's preferences and is paid W_j if the senior accepts the recommendation and enrolls in the plan and zero otherwise. Brokers choose to recommend plan j that maximizes expected utility

$$U_b(j) \;=\; P_j \;\times\; \Big(W_j \;-\; \kappa_{b,j} N\Big) \tag{6}$$

We slightly abuse notation and use the same symbol P_j to define both the consumer's true sale close probability and the broker's perceived probability that the senior accepts the recommendation versus continuing to search. W_j is broker pay for selling plan j, and net harm N is scaled by the parameter $\kappa_{b,j}$. $\kappa_{b,j} N$ is the suitability cost; κ is the weight the broker places on the senior's net harm. The cost is incurred only when the broker sells the plan that they do not believe matches the senior's preferences, or the "misfit plan". For a senior whose preferences match plan H the suitability cost is zero when the broker recommends plan H ($\kappa_{b,L} = \kappa_b$ and $\kappa_{b,H} = 0$). κ can be thought of as paternalism or guilt from recommending a plan that does not match the senior's preferences, lost continuation value when a poorly matched enrollee disenrolls from the plan (in which case the broker is no longer paid), regulatory liability, or the reputation costs of a poor recommendation. Figure A.5 shows the contributors to κ self-reported by brokers. The broker trades off their own pay versus senior plan suitability.[12]

3.4 Predictions under misaligned incentives

The audit observes broker recommendations under two decisions where incentives are misaligned, that is, where the plan that matches the senior's preferences pays the broker less. Table 2 describes the two decisions, both of which follow the framework and differ only in the fit gap Δ and in which plan maximizes senior utility. Predictions 1–3 generate predictions for Decision 1, where plan H matches the senior's preferences and plan L pays the broker more. Prediction 4 generates a prediction for Decision 2. Online Appendix A.6 proves Predictions 2–4.

Prediction 1 (Decision 1: broker suitability cost cutoff)**.** *Brokers with a high enough suitability cost recommend the plan that best matches the senior's preferences, even though it pays them less. If no broker bore a suitability cost, every broker would recommend L.*

[12]Note that the broker does not directly observe ϕ, but acts on a belief about it. Nothing in the framework requires that belief to be correct, so the predictions below are signed by the broker's belief about which plan is the easier sale rather than by the senior's actual perception.

Table 2: One framework, two decisions

	Decision 1: Medicare product	Decision 2: Medigap plan
Plans L, H	MA; Traditional Medicare + Medigap	Two Medigap plans with identical coverage
Plan that maximizes $U_c(j)$	plan H	plan L
Plan fit gap Δ	$\Delta > 0$	$\Delta = 0$ (coverage is identical)
Suitability cost	charged on a sale of plan L	neither plan is the misfit; not charged
What the senior loses	net harm $N = \Delta - (\rho_H - \rho_L) > 0$	markup $m = \rho_H - \rho_L > 0$
Broker commission (W_j)	$W_L > W_H$	$W_H > W_L$ (%-based commission $W_j = z\rho_j$)
Higher broker commission	plan L	plan H
Prediction tested	1, 2 and 3	4

Notes: W_j is the present value of plan j's multi-year commission stream and z is the Medigap commission percentage (Section 2). In each decision the plan that pays the broker more is the one that is less suitable for the senior. Condition (3) and the net harm N hold in the Decision-1 parameterization only; in Decision 2 both plans fit, so no sale carries the suitability cost and the cutoff (7) does not apply.

For Decision 1, the senior's preferences match plan H, so the suitability cost is zero when the broker recommends plan H, $\kappa_{b,H} = 0$. The expected payoffs are $U_b(H) = P_H W_H$ and $U_b(L) = P_L\Big(W_L - \kappa_b N\Big)$. The broker recommends L if and only if $U_b(L) \geq U_b(H)$. Both sides of that comparison are expected payoffs, so dividing by $P_L > 0$ leaves the inequality unchanged and transforms the sale probabilities into the ratio $R \equiv \frac{P_H}{P_L}$. We refer to R as the sale-probability ratio, the probability of a successful sale of plan H relative to plan L. R_0 is the sale-probability ratio on control calls (when $s = 0$) and R_1 the sale-probability ratio when $s = 1$.

The broker recommendation rule is a cutoff on the weight κ_b:

$$\text{Broker recommends } L \quad \Longleftrightarrow \quad \kappa_b \leq \kappa^* \equiv \frac{W_L - R\,W_H}{N}. \tag{7}$$

Online Appendix A.5 shows the full derivation. Note that the sale-probability ratio R is the only channel through which the senior's friction ϕ enters the broker problem. Equation (5) shows that $R < 1$ precisely when $\hat{N} < 0$.

Every broker whose weight on the senior's net harm exceeds the cutoff, $\kappa_b > \kappa^*$, recommends plan H, the plan that matches the senior's preferences. Multiplying through by N states the same rule in dollars: the broker recommends H when the suitability cost $\kappa_b N$ exceeds the ratio-discounted pay advantage of plan L, $W_L - RW_H$, which is the form Section 6 estimates. The weight varies from call to call, so the weight κ_b of the broker on a given call is a draw from a continuous cumulative distribution function F with density F'. Section 6 lets the location of F vary with caller characteristics and takes F logistic. A logistic F places some mass at $\kappa_b < 0$, which describes a broker who would rather close the misfit sale. By (7), σ, the share of brokers who recommend L to a senior who is a fit for H, is the mass below the cutoff,

$$\sigma = F\Big(\kappa^*\Big), \qquad \text{and by competition treatment arm,} \qquad \sigma_s \equiv F\Big(\kappa_s^*\Big). \tag{8}$$

Note that σ is also the Decision 1 outcome, or the share of calls on which the broker recommends plan L to a senior whose preferences best match plan H.

Prediction 2 (Decision 1: recommendations respond to pay)**.** *Lowering broker pay W_L lowers σ, the share of brokers that recommend L, for every value of ϕ.*

Prediction 2 follows from differentiating (7), which gives $\partial\sigma/\partial W_L = F'(\kappa^*)/N > 0$; Online Appendix A.6 contains the full proof. The intuition is that raising W_L raises the cutoff κ^* by $1/N$, and the brokers who switch to recommending L are those whose weight κ_b lies between the old and new cutoffs, a mass of $F'(\kappa^*)$ per unit of shift. Both terms are strictly positive: F' is a density and $N > 0$ by condition (3). Experiment A observes broker recommendations when $W_L = 0$.

Prediction 3 (Decision 1: impact of perceived competition)**.** *In Decision 1, where $\Delta > 0$, the direction of the perceived competition treatment's effect on σ, the share of brokers recommending L, depends on the broker's perception of the consumer friction ϕ. In particular, the treatment raises σ when $\phi < \phi^\dagger$, lowers σ when $\phi > \phi^\dagger$, and leaves σ unchanged when $\phi = \phi^\dagger$.*

From Equation (2) the senior perceives a gain from the low-price plan, $\hat{N} < 0$, exactly when

$$\phi \;<\; \phi^\dagger \;\equiv\; \frac{\rho_H - \rho_L}{\Delta}. \tag{9}$$

We call $\phi^\dagger$ the complexity threshold: the share of the fit gap above which the plan that maximizes the senior's perceived utility $\hat{U}_c$ is also the plan that maximizes true utility U_c. Condition (3) places $\phi^\dagger$ strictly between zero and one.

The continuation value v A better outside option lowers the close probability of every recommendation, but not in the same proportion. Differentiating Equation (4) gives $-\partial \log P_r/\partial v = 1 - P_r$: each recommendation loses its sales at a proportional rate equal to its own walk-away probability. Thus the harder sale loses the larger share. Differencing the two rates gives the response of the sale-probability ratio,

$$\frac{\partial \log R}{\partial v} \;=\; (1 - P_L) - (1 - P_H) \;=\; P_H - P_L, \tag{10}$$

so relative sale probability tilts toward whichever recommendation was already easier to close.[13] Note that a rise in the continuation value leaves $\hat{N}$ unchanged: it changes how willing the senior is to accept a recommendation, but not which plan the senior prefers.

The continuation value of search enters the cutoff (7) through the sale-probability ratio R. Since W_H, R, N and $F'(\kappa^*)$ are each strictly positive, the sign is that of $P_L - P_H$, which by (5) and

[13] Lemma 2 in Online Appendix A.3 derives the three close-probability derivatives in full, and Remark 1 shows that the logistic form is a normalization rather than a restriction.

(9) is the sign of $\phi^\dagger - \phi$. When $\phi < \phi^\dagger$, a rise in the continuation value raises the share of brokers recommending plan L. When $\phi > \phi^\dagger$, it lowers that share. When $\phi = \phi^\dagger$, the share does not change. Experiment B raises the continuation value from v_0 to v_1. The broker response depends on the broker's beliefs about the consumer's frictions, or where ϕ is relative to $\phi^\dagger$.

$$\begin{aligned} \frac{\partial \sigma}{\partial v} &= \frac{W_H\, R}{N}\, F'(\kappa^*)\,\Big(P_L - P_H\Big), \\ \operatorname{sign} \frac{\partial \sigma}{\partial v} &= \operatorname{sign}\Big(P_L - P_H\Big) \;=\; \operatorname{sign}\Big(\phi^\dagger - \phi\Big). \end{aligned} \tag{11}$$

Prediction 4 (Perceived competition when plans only vary in price)**.** *In Decision 2, when* $\Delta = 0$ *and* $W_j = z\rho_j$*, the broker responds to the perceived competition treatment by recommending the lower-price plan.*

A rise in the continuation value lowers the share of calls on which the higher-priced plan is recommended, and with it the average markup of the plans brokers recommend, for every $\phi \in [0, 1]$. Prediction 4 follows the same logic as Prediction 3, but with $\Delta = 0$. The proof is shown in Online Appendix A.6. Because the two decisions order commissions oppositely, the same movement toward the easier sale raises the broker's expected pay in Decision 1 and lowers it in Decision 2; Remark 4 works through which accounts of broker motivation that joint pattern rules out.

4 Experimental Design

Most of our evidence comes from a large-scale audit study in which senior secret shoppers request Medicare plan recommendations from randomly assigned insurance brokers. We supplement the audit data with two surveys. The first is a nationally representative panel of 2,015 current Medicare beneficiaries fielded through the Understanding America Study (Appendix B.4). The second is a survey of 157 currently licensed Medicare insurance brokers (Appendix B.5).

4.1 Audit Logistics

We randomly assign 109 secret shoppers, most of them age 65 or older, to request a Medicare plan recommendation from 554 randomly assigned insurance brokers on 637 phone calls. Data collection began in September 2024 and concluded in March 2026. This section describes the audit study logistics and protocols, including who the shoppers are, which brokers they reach, and where the consumer lives. Additional details are included in Appendix B.1.

Secret Shopper Sampling We hire and train seniors from two "secret shopping" firms.[14] Shoppers are never told the purpose of the study. Before they are assigned insurance brokers to contact, seniors must first pass an assessment demonstrating familiarity with the call script and a basic understanding of Medicare. We also collect the senior's real age, gender, race, and education level and include these characteristics in our baseline controls. Due to the individual-level random assignment of both brokers and seniors, any differences in recommendation quality across shopper characteristics can be interpreted causally.

Broker Sampling We sample from the two channels seniors actually use to find a Medicare broker. The first consists of two major Medicare call centers, which together enrolled over one million seniors in 2024 and each employ more than 1,000 brokers. Both are named in a 2025 Department of Justice complaint under the False Claims Act (*United States ex rel. Shea v. eHealth, et al.,* No. 21-cv-11777), making the population particularly policy-relevant. The second is the nation's largest online directory of Medicare insurance brokers, which listed over 18,000 brokers as of June 2025. We match the audited brokers to a licensure database constructed from 47 state insurance department registries. This gives us the broker's tenure and the set of carriers the broker is currently appointed to sell. Table A1 in Appendix B.1 reports summary statistics for the seniors and the brokers, and Table A2 compares broker characteristics across the two firm types (call-center versus non-call-center brokers).

Geographic Context Medicare plan availability and price vary by zip code. We select zip codes in California and Texas, two states with reasonably large senior populations. We also target rural areas with a normal amount of Medigap price dispersion and with median household income and education levels that are not too far from the national average. We choose rural areas since these areas tend to have fewer provider networks to choose from and fewer MA plans in general, creating a higher bar for MA recommendations.[15] This makes MA a worse fit for our audit study senior and narrows the MA choice set a broker can steer the senior into.

4.1.1 Measuring Broker Quality

To best measure broker quality, we focus our audit study on a scenario where broker incentives and consumer preferences are misaligned. On each of the two decisions a senior makes, we assign the consumer profile so that the plan that pays the broker more is the plan that serves the senior worse. This scenario creates a conflict of interest between brokers and consumers, and is the exact

[14] A secret shopping firm hires people to pose as regular customers in order to evaluate a business's service quality or its compliance with company policies and regulations. We use two firms due to the different strengths each firm had in reaching a different portion of our broker sample. This is further described in Appendix B.1.

[15] While MA plans guarantee a minimum level of coverage, beneficiaries in rural areas may have to travel farther for in-network care, making network restrictions potentially more binding (Park et al., 2021).

scenario policymakers at CMS and the DOJ have raised concerns about.[16]

The Medigap-fit senior Our main scenario assigns the shopper preferences that make them a textbook case for Traditional Medicare plus Medigap, built from the reasons listed on Medicare.gov for not choosing MA. The assigned preferences include wanting the flexibility to seek care in multiple states and having providers in multiple states (MA has network restrictions), not wanting to seek prior authorization for medical procedures (MA requires prior authorization), wanting to see specialists without a referral (MA requires referrals), and not expressing a preference for dental, vision, and hearing coverage (which MA has and Medigap does not).[17] We call a shopper with these preferences a "Medigap-fit" senior. Enrolling this senior in MA nonetheless pays the broker about $400 more than a Medigap enrollment.

How common is a Medigap-fit consumer? The exact share of Medigap-fit consumers is difficult to estimate given the detailed data on insurance preference, current and future health risks, and financial data needed to determine which Medicare plan is most suitable for a beneficiary.[18] Our best proxy is to use broker beliefs when they are not influenced by enrollment incentives. In our Medicare broker survey, described further in Appendix B.5, we directly ask currently licensed Medicare insurance brokers what share of seniors is an unambiguous fit for each plan. Brokers report that 51% are a clear fit for MA (versus a population average of 55%), 34% are a clear fit for Traditional Medicare plus Medigap (versus 24%), and 15% are not a clear fit for either plan.

Given the institutional setting, we believe these broker survey and population estimates are reasonable proxies for the share of Medicare beneficiaries who are a strong fit for Medigap for the following reasons. First, unlike individuals enrolled in MA, individuals enrolled in Medigap are free to switch to an MA plan annually without medical underwriting during a 2-month Annual Enrollment Period. This means individuals who believe they are a poor fit for Medigap can switch to MA, suggesting beneficiaries who remain in Medigap are satisfied with the service. Second, brokers are not financially incentivized to sell Medigap over MA, so it is unlikely that individuals are incorrectly steered into Medigap. This is in line with concerns raised by the DOJ and CMS; all concerns focus on steering into MA, none reflect steering into Medigap. Finally, Medigap plans are more expensive than MA plans. Since existing studies have documented a bias towards lower-premium plans with higher out of pocket costs, we expect that the bias would run against enrolling in Medigap, once again making the true enrollment a lower bound on Medigap-fit consumers in

[16] https://www.justice.gov/opa/pr/united-states-files-false-claims-act-complaint-against-three-national-health-insurance

[17] We verify these characteristics in interviews with multiple insurance brokers, against the Medicare.gov plan search tool, and against publicly available plan choice tools including the Fidelity Medicare tool and the AI-backed tool Healthpilot. Appendix Figure A.2 lists the assigned preferences beside the plan attributes they map to.

[18] This would include consumer preference and behavior data related to current and future travel plans, current and future health risks, and network restrictions, specialists, and prior authorization.

the population (Abaluck and Gruber, 2011; Handel and Kolstad, 2015).

Outcome 1: Recommend MA Our first outcome is an indicator for whether the broker recommends MA over Traditional Medicare plus Medigap. To estimate the consumer surplus loss from an MA recommendation for a Medigap-fit consumer, in a separate survey we ask brokers to report the maximum additional monthly premium a senior with our audit study characteristics would pay for various MA and Medigap plan attributes. Assuming the attributes are additively separable, we sum the components to calculate the compensating variation.[19] Appendix Figure A.7 reports the decomposition.

Using brokers' own beliefs, we estimate the net loss in consumer surplus for a Medigap-fit senior in our audit study to be about $5,574 a year. Two additional checks support this interpretation. First, we compare our broker survey estimates to network restriction costs estimated by Higuera et al. (2018), which value the plan attribute at $84–$390 a year. At the upper end of this range, this one MA restriction alone would imply that an MA enrollment results in a net loss in consumer surplus for the average consumer. Second, in the same broker survey, we also estimate the net loss for an *average* senior who has MA to be $2,711 (Appendix Figure A.11). The broker valuations match stated broker beliefs: in interviews, several brokers told us that "anyone who can afford to should have Traditional Medicare plus Medigap." Focus groups conducted by the Commonwealth Fund also show brokers are more likely to choose Traditional Medicare plus Medigap over MA for themselves (Leonard et al., 2023).

Outcome 2: Medigap Markup Our second outcome is the annual premium difference between the Medigap plan recommended by the broker and the low-cost plan with identical coverage available to the senior. This difference is the monetary cost of a dominated plan recommendation and requires no assumption about consumer preferences. The conflict of interest is direct: most brokers earn a percentage of the annual Medigap premium the senior pays, but each additional dollar of premium is a dollar of lost consumer surplus.

Misaligned Recommendation Rate We combine the two outcomes into a single measure, which equals one if the broker recommends the plan that pays the broker more even though it fits the consumer worse: that is, if the broker recommends MA, or if every Medigap plan the broker recommends carries a markup of at least $360 per year.[20] Table 3 collects the primary and secondary quality measures. We define the secondary measures in Appendix B.1.

[19] Appendix C.1 states the full accounting methodology, elicitation details, and the discounting and mortality assumptions for lifetime calculations.

[20] We pre-registered a threshold of $240 per year ($20 per month). After reviewing the data we judged $20 per month too harsh and use $30 per month, or $360 per year, which remains an economically meaningful threshold and sits closer to the mean Medigap markup of $377 per year.

Table 3: Summary of Measures of Broker Quality

Variable	Description	Definition
Panel A: Primary Outcomes		
Misaligned Recommendation Rate	Indicator = 1 if broker recommends a plan that pays them more, even if it is a poor fit for the consumer If broker recommends MA *or* only recommends Medigap plans with a markup $\geq$ \$360	$\mathbb{1}\{\text{Recommend MA} \vee \text{Medigap Markup} \geq \$360\}$
Recommend MA	Indicator = 1 if broker recommends MA over Traditional Medicare + Medigap (TM + MG)	$\mathbb{1}\{\text{Recommend MA}\}$
Medigap Markup	Premium difference between broker-recommended plan and the low-cost identical plan	$P_{\text{rec}} - P_{\text{min}}$
Panel B: Additional Outcomes		
Medigap (%) Markup	Percentage overcharge relative to the low-cost identical plan available to the consumer	$\frac{P_{\text{rec}} - P_{\text{min}}}{P_{\text{min}}}$
Broker Markup	Premium difference between recommended plan and cheapest plan from insurance company the broker is *licensed* to sell	$P_{\text{rec}} - P_{\text{min licensed}}$

Notes: This table defines the primary and secondary outcomes of interest. "Cheapest available" refers to the lowest-cost coverage-identical plan in the consumer's zip code, whereas "cheapest licensed" restricts the choice set to plans the broker is actively appointed to sell according to the CA and TX state licensure databases.

4.2 Experimental Variation

Experiment A: Removing the Commission To evaluate whether broker recommendations are driven by incentives, we take advantage of a natural experiment in which all national Medicare Advantage insurance companies stop paying commissions for MA enrollments in our audit region. This has become a common strategy for insurance companies that misprice their MA plans: because regulation allows MA plans to be pulled from the market only at certain points of the year, insurance companies slow enrollments by stopping MA commissions to brokers.[21] Consumers can still enroll in the MA plans, for example through Medicare.gov, but brokers are no longer paid the $\sim$\$750 commission for the enrollment.

Figure 3 shows the full sequence of commission changes. The key date is September 1, 2025. Before September 1, national carriers paid MA commissions in our California zip code. After September 1, the only commissionable MA plans there came from Imperial Health, a California-only carrier that is not appointed with either national call center, so a call-center broker earned nothing from an MA enrollment.[22] Non-call-center brokers could still get appointed with Imperial

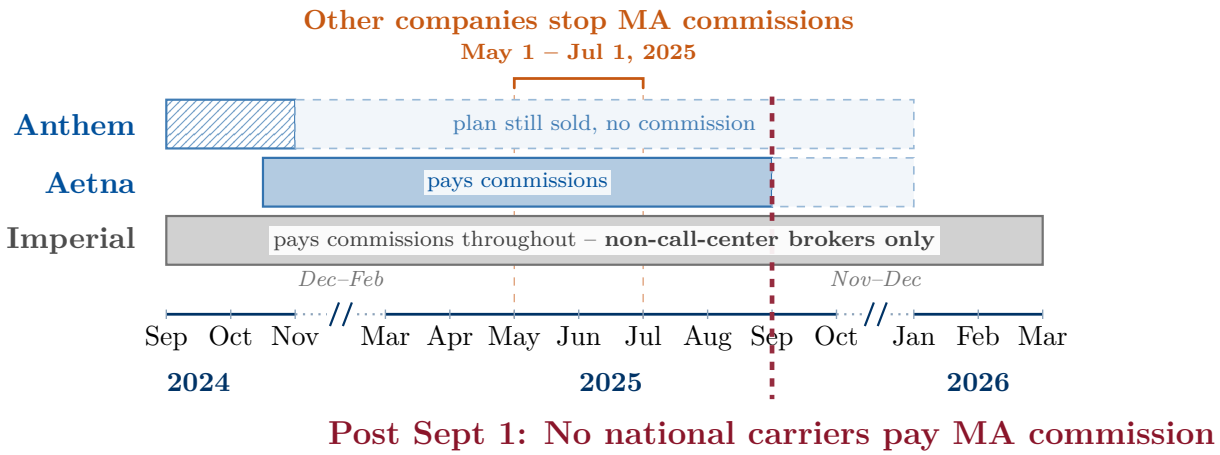
[21]The event is discussed on several Medicare broker blogs. See https://www.medicaremarketinsights.com/p/non-commissionable-plans-united-healthcare; https://www.insurance-forums.com/community/threads/california-not-paying-mapd-commission-is-this-a-national-trend.115640/.

[22]We confirm the appointments in the California Health Insurance License database. Both call centers also removed the non-commissionable plans from their internal search tools, though the plans were still available on

Health, so the commission removal binds only for call-center brokers. We therefore test whether removing the incentive to recommend MA reduces MA recommendations by comparing California call-center calls before and after September 1.

We collect audit data before and after September 1, holding consumer preferences fixed.[23] One unintended consequence of the natural experiment is that the change prevented us from finishing data collection for the treatment arm for California call centers. As a result, our treatment and control groups for Experiment B, described below, are not balanced on call center status.

Figure 3: MA Commission Changes Timeline



Experiment B: Signaling Perceived Competition Experiment A asks whether misaligned recommendations respond to the broker's pay. Experiment B asks whether a treatment designed to increase perceived competition can discipline advice.

The perceived competition treatment, Experiment B, instructs shoppers to inform the broker at the beginning of the call that they plan to get a few quotes and talk to multiple brokers before enrolling in a plan. Shoppers in the control group are instructed to say they want a recommendation from the broker to better understand their options, or to say nothing at all; Appendix B.3 gives the exact wording of both variations. We refer to this treatment as the perceived competition treatment, or the competition treatment for short, in the rest of the paper.

The treatment takes advantage of competitive pressure already in the broker market and makes it more salient. Collecting data from 47 state departments of insurance, we estimate that there were over 1 million currently licensed health insurance brokers in the US at the beginning of 2025.[24]

Medicare.gov. Additional details are in Appendix B.2.

[23]We allocate the post-September call center calls to California, where the commission removal binds, so that the pre-post comparison is adequately powered. Appendix B.2 explains why a paired Texas call center control series was infeasible.

[24]Collecting data from the California and Texas state department of insurance agencies, we estimate that there

In Texas alone, approximately 14,000 resident health insurance brokers held a Medicare product certification at the beginning of the study. This is likely a lower-bound estimate due to limited enforcement and only applies to Texas residents. Almost all brokers are (or can become) licensed in multiple states and can be easily contacted over the phone or online. Overall, the numbers suggest a large supply of brokers. However, in our Understanding America Study (UAS) survey, 66% of current Medicare beneficiaries spoke to only one broker before enrolling in Medicare.

The treatment does not change market structure or manipulate broker supply. Instead, it manipulates broker beliefs about the probability that this call ends in a sale by making consumer retention risks salient in an already competitive market. Competition acts on a broker through the risk that the senior ends the call without enrolling and buys from someone else: a senior can reach another licensed broker through any of hundreds of search results or toll-free numbers, and in interviews brokers frequently complained about competitors stealing their clients. The treatment makes that possibility salient on the call. In doing so, the treatment also tests the effectiveness of advice consumers are routinely given in intermediated markets: to seek a second opinion.[25]

We stratify randomization by state and firm type (call-center versus non-call-center brokers) to ensure balance, and assign brokers to treatment or control status before any calls are made.[26] The calls are then randomly assigned to shoppers at the individual level. We implement the treatment and control conditions in waves (at different points in time) to minimize shopper mistakes.[27] Due to the unexpected timing of the natural experiment, we are unbalanced on call center status and state. Table A3 in Appendix B.3 reports the balance table and Table A4 compares the broker and caller characteristics of compliers versus non-compliers.

Baseline controls Every reduced-form regression on the audit data shares one set of controls, which we call the baseline controls. The broker controls are the broker's gender, an indicator for high tenure status (licensed as a health insurance broker for over 4 years), and an indicator for having many insurance carrier appointments interacted with non-call-center status.[28] Caller controls include caller gender, real age and age squared, race, college education status, and caller experience (the number of audit calls the caller has previously completed).[29] We also include a

were over 200,000 and over 400,000 actively licensed health insurance brokers in California and Texas, respectively, at the beginning of the audit study.

[25]Consumers shopping for a mortgage, for example, are frequently advised to bring competing quotes to a broker.

[26]The individual-level random assignment applies to non-call-center brokers. For call-center brokers we use the call center's random matching of seniors to brokers.

[27]We originally assigned both treatment and control condition calls simultaneously, but noticed problems with treatment compliance and switched to implementing the treatment in waves.

[28]We use 4 years as the cutoff for high tenure since health insurance licenses are granted in 2-3 year cycles in most states, so observing a tenure of over 4 years indicates the broker has renewed their license at least once. We only allow appointments to vary for non-call-center brokers since broker appointments are set by the call center and therefore do not vary significantly within the firm or within call center status.

[29]Race is an indicator for whether or not the caller is white. College education status is an indicator for whether the caller holds an associate's degree or higher. Both use the shopper's self-reported data.

call center fixed effect, a fixed effect for the caller's assigned state of residence, and an indicator for the Medicare Annual Enrollment Period (AEP) interacted with call center status.[30]

Regressions on the MA outcome also include an indicator for calls placed after the September 1, 2025 commission removal interacted with call center status. This is when the commission change was made salient to call-center brokers. For non-call-center brokers the incentive and plan menu change became salient on October 1, 2025, the carrier exit announcement date. We control for calls placed after October 1 interacted with non-call-center status. Regressions on a Medigap outcome also include a plan letter (N or G) fixed effect. We also cluster standard errors at the caller level. Table footnotes state any departure from the baseline set of controls.

5 Results

5.1 Baseline Quality of Broker Advice

In total, across all treatment arms, we generate over 1,692 recommendations from 637 phone calls to Medicare insurance brokers. Since some seniors in our UAS survey do call multiple brokers, we pool the results in our control group (no perceived competition) and treatment group (perceived competition treatment), to reflect the average recommendation quality seniors might receive. To measure broker recommendations when incentives are misaligned, we limit the level of baseline Misaligned Recommendation and MA Recommendation results to calls where brokers are paid the full commission for MA enrollments.[31] Table 4 summarizes the baseline broker recommendations for our main outcomes of interest.

Brokers depart from the welfare-maximizing benchmark on both decisions. For Outcome 1, whether the broker recommends MA or Traditional Medicare plus Medigap, we see that brokers steered the consumer toward an MA plan in 32% (SE 2) of calls. While MA plans often have lower premiums, they impose network restrictions, require prior authorization, and are not recommended for seniors who travel often, violating the consumer's stated preferences assigned in the script. We do not find evidence that the Medigap markups in Decision 2 are intentionally inflated to make Medigap look worse and to rationalize MA recommendations. We include this result in Appendix C.5.

For Outcome 2, the Medigap Markup, we find that brokers recommend a plan that is on average $377 per year (SE 9) more expensive than the optimal available option, representing a 29% markup

[30]Call centers recruit new brokers and hire brokers on a temporary basis during AEP, the highest-volume enrollment period of the year, so the composition of brokers answering call center calls tends to change during those months. The interaction absorbs that change.

[31]We also limit Medigap markup observations to calls where the resident state of the caller is one where the full MA commission is paid. Changing the sample to include calls where MA commissions are not paid does not materially change the Medigap markup.

Table 4: Baseline Broker Recommendations

Outcome Variable	**Mean**	**SD**	**Obs**
Panel A: Primary Outcomes			
Misaligned Recommendation (%)	52%	50	458
Recommend MA (%)	32%	47	458
Medigap Markup ($ Yearly)	$ 377.10	280.65	1,055
Panel B: Secondary Outcomes			
Medigap Pct Markup (% Yearly)	29%	23	1,055
Medigap Broker Markup ($ Yearly)	$ 168.25	281.65	1,002
Avg Medigap Premium ($ Yearly)	$ 1,764	383.04	1,055

Notes: Table 4 shows the baseline broker recommendations. Recall that for the senior preferences we assigned, the welfare-maximizing choice is Traditional Medicare plus Medigap for Outcome 1 (Recommend MA %) and the lowest-cost Medigap plan available to the senior for Outcome 2 (the Medigap Markup). Panel A shows our primary outcomes of interest. Recommend MA (%) and the Misaligned Recommendation rate are computed on the 458 calls on which the broker earns the full MA commission, excluding the no-commission call-center calls and the post-carrier-exit calls of Experiment A. Panel B shows the secondary outcomes.

over the low-cost identical plan available to the senior. This excess premium represents a pure wealth transfer from the consumer to the broker (and insurance company) for a plan with identical coverage. Brokers recommend a zero markup Medigap plan (the lowest-cost plan with identical coverage available to the senior) in 6% of 1,055 baseline Medigap recommendations. Brokers recommend the low-cost Medigap plan within the choice set of insurance companies they are *currently* licensed to sell on 19% of Medigap recommendations with an average broker markup of $168 a year. Finally, brokers recommend a high-markup Medigap plan (or a Medigap plan with a markup of at least $360 a year) on 41% of calls.

Combining Outcome 1 and Outcome 2, we say that brokers provide a misaligned recommendation if they either recommend MA or recommend a high-markup Medigap plan. 52% (SE 2) of brokers provide misaligned recommendations and steer seniors toward higher-commission plans, even when the plan is less suitable for the senior.

Several precautions in our design suggest these estimates may be lower bounds on the level of broker steering. First, given the basic training we provide on the Medicare market to ensure our shoppers can correctly identify recommendations, our senior shoppers may appear more sophisticated than the average senior. Second, to ensure the Medigap recommendation, which comes second, is not contaminated by the initial Traditional Medicare versus MA recommendation, we instruct the shoppers to open the call with an overview of the two topics of conversation (MA plans and Medigap plans). This overview may signal that the shopper has a basic understanding of Medicare and increase perceived consumer sophistication. Finally, we try to limit the number of calls during AEP, when brokers are especially busy and may be more likely to make mistakes.[32]

[32]AEP is also when individuals usually change MA plans, meaning MA might be more likely to be on the broker's

Outcome 1: Medicare Advantage In Section 4.1.1, we show that brokers estimate that 1 year of having an MA plan would cost a Medigap-fit senior approximately $5,574 using the plan attribute valuations brokers report in our survey (Figure A.7). Since seniors cannot enroll in Medigap outside of guaranteed-issue periods without medical underwriting, we assume that the annual loss recurs. We hold the senior in the MA plan and accumulate the loss from age 65 through the life table's terminal age of 100, weighting each year by survival and discounting at a real rate of $r = 3\%$. Appendix C.1 states the accounting identity, the elicitation, and the discounting and mortality assumptions.

We elicit the valuations from brokers rather than recover them from observed plan choices, which existing research has shown to be distorted by consumer choice frictions. Two potential concerns arise with a broker-elicited measure, and both work against our results. The first concerns broker motives: a broker inclined to rationalize their own MA steering would understate the Medigap-distinguishing protections or overstate MA's ancillary benefits, shrinking our measured consumer surplus loss toward zero. The second concerns broker competence. If brokers do not understand the different Medicare products, a concern we show in Section 5.5 to be valid, the elicited measures would be noisy. Restricting the sample to the brokers who score highest on the knowledge test would increase our MA consumer surplus loss estimates (Appendix C.1); we use the full-sample estimate to be conservative.

Over a 36-year retirement, using the broker beliefs, we estimate that an accepted broker MA recommendation would cost a Medigap-fit senior roughly $5,574 a year or $79,756 in mortality-adjusted present value dollars. For the average senior who may not be as strong a fit for Medigap as our audit study senior, the loss is $2,711 a year or $38,790 in mortality-adjusted present value dollars. Figure A.12(a) plots the lifetime loss accumulation from an MA enrollment for a Medigap-fit senior. Given the assumptions needed to estimate the welfare loss for MA, we place greater weight on the consumer surplus estimates from the dominated Medigap recommendations below.

Outcome 2: Medigap Markup To value the Medigap markup we project the year-1 markup forward over the enrollment horizon of our hypothetical 65-year-old senior with an average (10%) premium growth rate. Appendix C.2 states the full methodology.

We estimate that a dominated Medigap recommendation in our audit sample costs the senior approximately $12,000–$24,000 of forgone wealth over their lifetime under typical nominal premium growth rates of 10% and 15% a year. Table A8 reports the full set of estimates and Figure A.12(b) shows the monetary loss from the dominated plan choice over time.

What Predicts Advice Quality – Seniors vs Brokers We then analyze whether senior or broker characteristics are more predictive of recommendation quality. Doing so helps us pinpoint

mind, so incorrect MA recommendations may be more common during this time.

the nature of the market failure. While we do not explicitly vary consumer characteristics to test for discrimination, seniors are randomly assigned to call the insurance brokers, so we can interpret any correlation between recommendation quality and senior characteristics causally. Isolating these drivers is also important for designing effective policy interventions.

Table A7 in Appendix C shows the coefficients from regressions of our two primary measures of broker quality on senior and broker characteristics. Note that the caller characteristics are rarely, if ever, statistically significant. Broker characteristics, on the other hand, tend to be correlated with advice quality. Call-center brokers are more likely to recommend MA on Outcome 1, suggesting firm-level differences such as in training or incentives may be important. Brokers with higher tenure also tend to provide higher-quality advice on both outcomes. This suggests broker experience or selection may be important. We discuss this heterogeneity further in Section 5.5.

5.2 What seniors value and learn on calls

That advice quality varies with broker characteristics and not with consumer characteristics emphasizes the importance of understanding the supply-side of the market. In this section, we evaluate whether the demand-side feedback might help discipline the market.

To better understand which call attributes seniors value, we ask our secret shoppers to provide an insurance broker rating after every call within the audit study and test whether the ratings are correlated with our two quality measures.[33] Shoppers are asked the following question:

> *"On a scale from **1 (definitely would not recommend)** to **5 (definitely would recommend)**, how likely are you to recommend this insurance broker to a friend or family member selecting insurance in this state?"*

Shoppers give brokers a rating of 4 (likely to recommend) or 5 (definitely would recommend) on 79% of calls. Recall that the seniors in our audit study conduct multiple broker calls (specifically, an average of 6 calls per shopper) and must pass a multiple-choice quiz demonstrating a basic understanding of the Medicare market before making any calls, meaning our auditors may be more sophisticated and possess more Medicare knowledge than the average senior. Even these positively selected shoppers, who make multiple phone calls, rate the vast majority of brokers highly.

Table 5 regresses the probability of a highly rated broker call on our measures of broker recommendation quality and call length. Each column reports a separate regression of an indicator for a high rating (a rating of 4 or 5 out of 5), measured in percentage points, on one of our objec-

[33]We focus on the overall recommendation rating. However, seniors were also asked to provide ratings for broker knowledge and broker friendliness. The three ratings are highly correlated. The overall recommendation rating was added to the shopper response form at a later date. For the 20 calls where the response to the overall recommendation rating is missing, we take the average of the broker knowledge and broker friendliness ratings.

tive measures of broker recommendation quality. We standardize each measure so each coefficient reports the change in the probability of a high rating, in percentage points, from a one standard deviation increase in that measure.

The results highlight the disconnect between our objective measures of broker recommendation quality and the subjective consumer ratings. None of the four measures of broker quality spanning Decision 1 and Decision 2 shifts the probability of a high rating by an amount statistically distinguishable from zero. Call length does, however, shift the broker rating. The results in Column (5) show that a call that is one standard deviation, or 14 minutes, longer raises the probability the broker receives a high rating by 6.16 percentage points. The results suggest that seniors' ratings do not vary with objective quality metrics such as the plan recommendation, and instead reward noisy proxies for quality such as time spent on the call.[34] Appendix C.3 shows how the correlation between senior ratings and recommendation quality evolves as callers gain experience with the market. The evidence is consistent with limited caller learning.

Table 5: Broker Rating by Recommendation Quality and Call Characteristics

	(1) Pr(High Rating) (4 or 5)	(2) Pr(High Rating) (4 or 5)	(3) Pr(High Rating) (4 or 5)	(4) Pr(High Rating) (4 or 5)	(5) Pr(High Rating) (4 or 5)
Recommend MA (per 1 SD)	-2.37				
	(2.07)				
Medigap Markup (per 1 SD)		-0.76			
		(1.61)			
Medigap Broker Markup (per 1 SD)			-0.57		
			(1.57)		
Medigap Markup % (per 1 SD)				-0.70	
				(1.61)	
Call Length (per 1 SD)					6.16***
					(1.91)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055	637
R^2	0.05	0.08	0.08	0.08	0.06

Notes: Table 5 shows the relationship between the senior's rating of the broker and our objective measures of broker recommendation quality and call length. A high rating is a 4 (likely to recommend) or 5 (definitely would recommend) on the recommendation question in Section 5.2. Each column reports a separate OLS regression of an indicator for a high rating, in percentage points, on the row variable. The row variable is standardized over each regression's estimation sample, so each coefficient shows the impact of a one standard deviation increase. Columns (1) and (5) are estimated at the call level. Columns (2)–(4) are estimated at the Medigap plan quote level, with up to one quote per plan letter and state on each call. All columns include baseline controls, controls for the call-center and non-call-center commission change, and a control for perceived competition assignment. Columns (2)–(4) also include the Medigap plan-letter fixed effect. Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Consumer feedback through ratings, referrals, and regulatory action (e.g. reporting broker steering) is often the primary mechanism through which consumers might discipline low-quality intermediary advice. Seniors who rely on brokers because they have an imperfect understanding of the Medicare market do not have a benchmark against which to judge recommendation quality. This inability to recognize and punish low-quality advice, even after repeat-exposure to the market as in Appendix Figure A.13, may allow agency problems to persist. Our ratings are only a proxy

[34]The result provides empirical support for the model assumption that consumers cannot evaluate plan fit.

and do not reflect senior satisfaction with the recommendation *after* enrollment. However, the initial broker ratings are nonetheless informative given limited feedback and opportunities for consumer learning in this market.[35]

Another reason consumers may be unable to discern Medicare plan fit and identify low-quality advice is that brokers may not be educating beneficiaries or not providing them with accurate information about Medicare plans during these phone calls. To test this within the audit study, as part of the post-call survey we instruct our secret shoppers to ask and report whether the broker informs the shopper that Medigap plans are standardized or that all Medigap plans within a letter are the same. Since all shoppers are instructed to ask about Medigap plans regardless of the plan recommended in Decision 1, one might expect experts looking to educate consumers on the different Medicare plans to bring up this important fact; however, shoppers report being informed that Medigap plans are standardized on only 56% of calls.

To further understand why our secret shoppers may have difficulty discerning Medicare plan fit, we use text analysis to analyze the call transcripts from the Medicare plan consultations. We use a large language model to analyze 497 transcripts (78% of the audit calls) to better understand the content of the phone calls (Appendix C.8).

Seniors may not learn with greater exposure to the broker market if brokers do not educate the seniors or if brokers make inaccurate or contradictory remarks that are difficult to verify. Figure A.16 shows the justifications brokers give for recommending MA and Traditional Medicare (TM) + Medigap (MG) in Decision 1. Recommendation reasons do vary along predictable product attributes: brokers cite additional benefits and lower monthly premiums as reasons for recommending MA on 66% and 75% of calls, respectively, versus 9% and 34% of calls where Medigap is recommended. Each of the seven reasons, however, is cited for *both* recommendations, highlighting that the same reasons can be used to justify opposite recommendations when consumers do not understand the products. Thus, a senior who cannot verify the plan attributes learns little from the call.[36]

The LLM analysis further highlights that calls focus disproportionately on current costs and health status and few calls discuss future risks and uncertainty (Figure A.15 in Appendix C.8). Brokers highlight a $0 monthly premium on 57% of calls and discuss the trade-off between low

[35]A Medicare Advantage enrollee can change plans at most once a year and most do not change unless circumstances require it, and Medigap enrollees change plans even less often. This limits occasions where beneficiaries are forced to update their assessment of both the quality of their Medicare plan and of the broker who advised them.

[36]For example, "predictable healthcare costs" is cited on 67% of calls recommending TM + MG and still on 23% of calls recommending MA. A broker recommending TM + MG might emphasize that an MG plan is better for someone who wants fewer unexpected bills and predictable costs since MG covers a large portion (and sometimes all) of OOP costs. Conversely, a broker recommending MA might emphasize that someone who wants fewer unexpected bills and predictable costs should have an MA plan because MA plans have a maximum OOP limit while some MG plans do not have an annual OOP limit. Both statements are factually correct, but the second statement is misleading since the beneficiaries with MA plans tend to pay more in OOP costs.

premiums and low out-of-pocket costs on 59% of calls. Brokers also ask about current health status on 67% of calls. In contrast, few calls discuss future health risks and uncertainty: brokers ask about expected future health risks on 6% of calls, personal health history on 4%, and family health history on 1% of calls. The transcripts suggest that brokers are not educating consumers on the long-term health and financial implications of choosing between Medicare Advantage and Medigap. The initial enrollment period is a unique window when beneficiaries can sign up for any Medicare plan without medical underwriting. Consumers who are not fully informed about the long-term trade-offs inherent in choosing a multi-year insurance contract risk being locked into an unsuitable plan, especially if their health deteriorates.

5.3 Experiment A: Changing Incentives

Because seniors cannot distinguish suitable from unsuitable advice, little in the market punishes a broker who follows their own financial incentive. We now ask how much of the advice those incentives account for. To do so, we exploit a natural experiment where both national Medicare Advantage carriers stop paying broker commissions to reduce enrollment in mispriced plans.

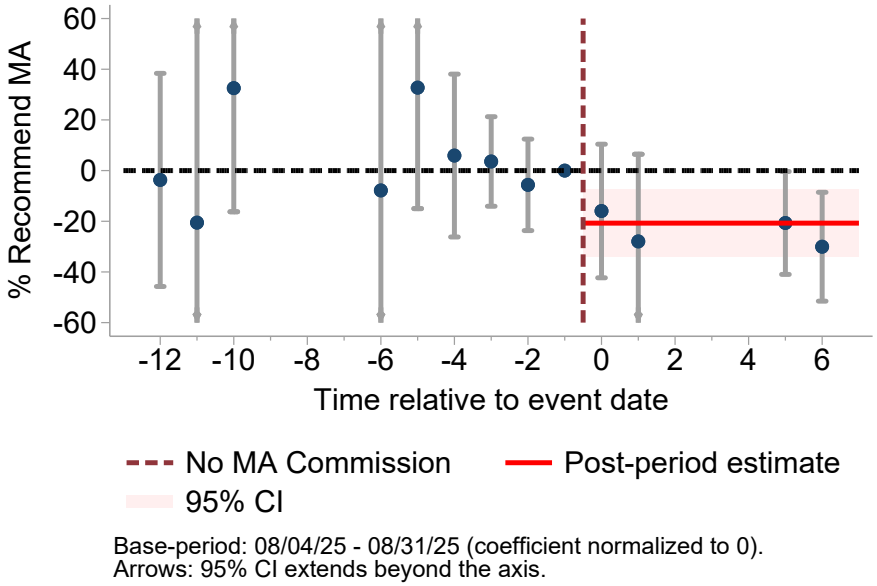
To trace how MA recommendations evolve in calendar time around the commission removal, we estimate an interrupted time-series specification. Let t_i denote the calendar date of call i, and let $t^\star$ = September 1, 2025 be the event date, when all call center carriers stopped paying MA commissions.[37] We aggregate calendar dates into four-week bins ($d = 28$ days), placed so that bin 0 begins at $t^\star$: call i falls in bin $k_i = \lfloor (t_i - t^\star)/d \rfloor$, so pre-event calls have $k_i < 0$ and post-event calls have $k_i \geq 0$. We estimate

$$\text{Rec MA}_i \;=\; \alpha \;+\; \sum_{k \neq -1} \beta_k \, \mathbb{1}\{k_i = k\} \;+\; \gamma' \mathbf{X}_i \;+\; \varepsilon_i, \tag{12}$$

where the sum runs over the bins observed in the sample and bin -1 (the four weeks ending August 31, 2025) is the omitted reference period, so each β_k measures the difference in the MA recommendation rate between bin k and the four weeks immediately before the commission removal. The sample is restricted to the no perceived competition arm (that is, to calls where the shopper was not assigned to inform the broker that they were talking to competitors); this ensures the variation in the outcome is not contaminated by the experimentally introduced perceived competition. $\mathbf{X}_i$ is the vector of baseline controls, with the September 1, 2025 indicator omitted because the event it marks is what the bins trace. Figure 4 plots the estimated coefficients $\widehat{\beta}_k$.

[37] Recall that there are differences in the timing of when the incentive removal became salient to call-center versus non-call-center brokers. For call-center brokers, the plans were immediately removed from the internal plan search systems as soon as the company stopped paying commissions. For non-call-center brokers, who do not always have separate plan search platforms and often use Medicare.gov to check which plans are available, the plan commission removal was not salient until October 1, when Anthem and Aetna also announced plans to leave the MA market in the county of interest beginning January 1, 2026. Since non-call-center brokers could still contract with a carrier

Figure 4: MA Recommendation Rates Over Time



Notes: Figure 4 plots the estimated coefficients $\widehat{\beta}_k$ from equation (12) with 95% confidence intervals: the difference in the MA recommendation rate (in percentage points) between each four-week calendar bin and the omitted reference bin, the four weeks ending August 31, 2025. The sample is the no perceived competition arm of the audit. The vertical line marks September 1, 2025, when all call center carriers stopped paying MA commissions; the horizontal line shows the pooled estimate from equation (13). Baseline controls. Standard errors are clustered at the caller level.

To measure the pooled impact of the MA commission change, we estimate:

$$\text{Rec MA}_i \;=\; \alpha \;+\; \beta_1\,\text{Call Center}_i \;+\; \beta_2\,\text{No MA Commission}_i \;+\; \gamma'\mathbf{X}_i \;+\; \varepsilon_i, \tag{13}$$

where $\text{No MA Commission}_i \equiv \mathbb{1}\{\text{Call Center}\}_i \times \mathbb{1}\{\text{CA}\}_i \times \mathbb{1}\{t_i \geq \text{September 1, 2025}\}$ and CA_i indicates that the shopper's assigned primary state is California. The indicator equals one exactly when the broker answering call i earns no commission from an MA enrollment: call center carrier appointments are set at the company level, and on September 1, 2025 every carrier available to call center brokers stopped paying MA commissions in our California county, while non-call-center brokers could still get appointed with additional companies that continued to pay. MA plans themselves remained available for consumers to enroll in throughout (Section 4). The coefficient of interest, β_2, measures the change in the MA recommendation rate for call center brokers once the MA commission is removed; because every post-September-1 call in the estimation sample is a call center California call, β_2 is identified from the before–after comparison for call center calls, conditional on covariates. $\mathbf{X}_i$ contains the controls from equation (12) other than the call center indicator, which now enters separately.[38]

that continued to pay MA commissions, Imperial Health, the commission removal applies only to call center brokers.

[38]The AEP × Call Center indicator is identified off the 2024 and 2025 enrollment windows; the specification cannot include a smooth calendar-time trend or a stand-alone post-September-1 indicator, either of which would be collinear with the treatment in this sample.

Table 6: Natural Experiment: MA Incentives Removed

	(1) Recommend MA (%)
No MA Commission	-20.73***
	(6.78)
Counterfactual Mean (No Commission Change)	44.33
Controls	Yes
Observations	384
R^2	0.14

Notes: Table 6 reports OLS estimates of Equation (13) on calls in the no perceived competition arm. The outcome is the share of calls on which the broker recommends MA over TM + MG. The coefficient on "No MA Commission" is the change in that share for California call-center brokers after carriers stopped paying MA commissions on September 1, 2025. The counterfactual mean is the share of calls on which these brokers would have recommended MA over the same months if the commission had not been removed. All columns include baseline controls. Standard errors are clustered at the caller level. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$.

Table 6 shows that removing MA commissions lowers the share of calls on which a California call center broker recommends MA by 21 percentage points (SE 7). Absent the change, these brokers would have recommended MA at a counterfactual rate of 44% of calls over the same months. The commission removal therefore lowers the MA recommendation rate by 47% relative to the counterfactual mean. Crucially, the non-commissionable plans and every other MA plan remained in the market despite the commission change. Consumers could still enroll in the Anthem and Aetna plans through the end of 2025 and in the Imperial Health MA plans throughout 2025 and 2026, and a broker who checked Medicare.gov would still see all three MA plans available. Administrative enrollment data from the audit county provide suggestive evidence that a portion of MA enrollments may have been driven by commissions (Appendix D).

The second important takeaway from the natural experiment is that removing commissions does not *completely* eliminate MA recommendations even when recommending MA 1) is unsuitable for the senior, and 2) does not earn the broker a commission. We find that call center brokers still recommend MA on 24% of calls, even after commissions are removed. This suggests MA recommendations are partially driven by broker information frictions such as limited product knowledge, limited training, or a genuine belief that MA is the best-fit plan for seniors in this scenario. We discuss this further in Section 5.5.

5.4 Experiment B: Changing Perceived Competition

Experiment A shows that misaligned recommendations respond to the broker's pay, but that commissions do not explain every MA recommendation. Experiment B tests whether competitive pressure can discipline brokers and improve advice quality. Our main regression specification is:

$$Y_{is} = \alpha + \beta_1 \mathbb{1}(\text{Perceived Comp}_{is}) + \beta_2 \text{Call Center}_{is} + \gamma' \mathbf{X}_{is} + \epsilon_{is} \tag{14}$$

where Y_{is} is the recommendation outcome provided by broker i to senior s. Perceived Comp_{is} is an indicator variable for whether the senior s was assigned the competition line for the call with broker i. Call Center_{is} is an indicator variable for broker firm (that is, whether broker i is reached through a call center phone number). Finally, $\mathbf{X}_{is}$ is the vector of baseline controls. Additional robustness checks, including specifications that control for caller experience and calendar time and the instrumental-variable estimates, are in Appendix C.

Our main coefficient of interest is β_1, which measures the change in broker recommendation quality in response to the perception of increased competition. Our main specification reports the Intent to Treat (ITT) in Table 7, where we regress the main outcomes of interest on treatment assignment. The IV results are similar in magnitude and are shown in Appendix C.7.

Table 7: Impact of Perceived Competition Assignment on Advice Quality

	(1) Recommend MA (%)	(2) Medigap Markup ($)	(3) Medigap Broker Markup ($)	(4) Medigap % Markup
Perceived Competition Assignment	11.35**	-84.15***	-83.99***	-7.07***
	(4.93)	(20.01)	(17.30)	(1.65)
Generalized Lee Bounds		[-168, -26]	[-154, -15]	[-17, -2]
Control Mean	28.13	407.25	217.69	31.47
Controls	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055
R^2	0.16	0.16	0.16	0.23

Notes: Table 7 reports the effect of perceived competition assignment on our main measures of broker recommendation quality. All columns include baseline controls. Column (1) also includes the September 1, 2025 commission removal indicator interacted with call-center status and the October 1, 2025 carrier exit announcement indicator interacted with non-call-center status. Columns (2)–(4) also include the Medigap plan-letter fixed effect. To account for selection on the Medigap outcomes, we report 95% confidence regions for the generalized Lee bounds (Semenova, 2025). Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Focusing on Outcome 1, column (1) shows that the competition treatment *increases* the rate of MA recommendations by a statistically significant 11 p.p. (SE 5). Recall that we assign the senior with preferences that match Traditional Medicare plus Medigap, according to Medicare.gov, but brokers are paid more for enrolling seniors in MA.

The Medigap outcome, however, implies the opposite result. Columns (2) to (4) of Table 7 show that across all the ways to measure the Medigap markup, the competition treatment causes brokers to recommend lower-cost Medigap plans. Since Medigap products are standardized and homogeneous within plan letter, these recommendations save the consumer money and *improve* consumer welfare. Since brokers are paid a percent of the Medigap premium, this implies brokers respond in a way that *lowers* their pay. On this decision the treatment lowers the mean recommended markup by $84.15 a year (SE 20.01). This amounts to a dollar-for-dollar increase in consumer surplus since the plans have identical coverage.

Together, the results show that increasing perceived competition causes brokers to recommend

the *lower-premium plan*, even when it reduces consumer welfare. Furthermore, the Medigap results suggest that brokers do possess the ability to provide high-quality recommendations, but supply them strategically, withholding the cheaper coverage-identical plan from consumers who give no sign that they will compare quotes.

Initially, the results may seem counterintuitive; we often expect competition to increase efficiency and improve consumer outcomes and one might expect the threat of external verification to lead brokers to improve their recommendation quality. The results also highlight that competitive pressure and consumer search may have unintended consequences in complex markets with asymmetric information where quality is difficult to verify. However the results are consistent with the conceptual framework in Section 3; a broker who expects a sale to be less certain responds to competitive pressure by recommending the plan they believe the senior is more likely to accept, which depends on the utility the consumer perceives. The results show that in complex markets where consumers lack complete information on product quality and intermediary incentives, competition may actually *reduce* consumer surplus, as in Outcome 1.

Comparing the two decisions also rules out the most natural alternative explanation: that the signal pushes brokers toward the plan that pays more. The signal raises MA recommendations, which pay brokers more, but lowers the markup of the recommended Medigap plan, which pays brokers less. A broker who pushed whichever plan pays the larger commission would have moved toward MA in Decision 1 and toward the higher-priced plan in Decision 2. Instead, when a sale becomes less certain, brokers recommend the lower-premium plan to close it, even when that plan does not fit the consumer (Decision 1) and even when it pays them less (Decision 2).

Finally, we test whether the result is an artifact of caller learning or the timing of the treatment waves by controlling for caller experience and calendar time trends. Appendix Table A9 shows that the results are robust to these specifications.

Discussion of Alternative Explanations The perceived competition treatment can signal several attributes of the caller at once: that the consumer is sophisticated, that they are price sensitive, that their search costs are low, or simply that the sale is now less likely to close. One limitation of the treatment is that we are unable to precisely identify the mechanisms; that is, we cannot fully separate a change in what the broker infers about the consumer (sophistication, education, income, etc.) from a change in competitive pressure that impacts the broker's incentive to make the sale. Below, we discuss the leading alternative explanations in turn.

The first potential confound is that planning to talk to multiple brokers and get multiple quotes signals that the consumer is informed or more educated. However, one would expect brokers to give *better* advice to shoppers they believe to be sophisticated. This should improve recommendation quality on every margin and we would expect to see *fewer* MA recommendations for our Medigap-fit

senior. We find the opposite: sophistication cannot account for the MA result in Decision 1.

Another credible alternative explanation is that the perceived competition treatment signals price sensitivity. That is, a consumer who searches and solicits multiple quotes may be revealing that they are income-constrained or care mainly about price.[39] This explanation rationalizes the results of both Outcome 1 (MA) and Outcome 2 (Medigap); a broker accommodating a price-sensitivity preference would steer consumers toward the cheapest premium in both markets and would think they are improving consumer welfare, even when there are no other changes to income and shoppers are explicitly instructed to say they can afford any plan.

Recall that this would require brokers to rationalize a level of price sensitivity where an average premium savings of $120/month ($1,440/year) is enough to overcome an estimated consumer surplus loss of $5,574/year for a Medigap-fit consumer or $2,711/year for the average senior. These numbers are based on the stated beliefs of brokers themselves.[40] Because brokers who answer more of the survey's knowledge-check questions correctly report larger consumer surplus losses (Appendix C.1), we view these valuations as lower bounds.

We also test for price sensitivity within the audit study calls using a large language model (LLM) analysis. For each call where we have a transcript, we ask the LLM to rate the shopper's perceived price sensitivity on a one-to-five scale.[41] We might expect the LLM to rate the callers who deliver the perceived competition line as more price sensitive. Columns (1)–(3) of Table 8 show that the rating does not move: assignment to treatment changes the price sensitivity rating by -0.04 points (SE 0.05) against a control mean of 2.87 and a control standard deviation of 0.57. Column (4) shows that the impact of the perceived competition treatment on the MA outcome remains, even when limited to the subset of calls with call transcripts.

Finally, we directly test whether the treatment signals greater price sensitivity in our broker survey. We include an experiment where we hold consumer type fixed and show each broker two identical Medigap-fit prospective client vignettes that differ only in whether they deliver the perceived competition line. We also randomize the perceived competition line across the two vignettes. Figure 5 shows the maximum willingness to pay for a Medigap plan reported by the broker split by perceived competition signal. Recall that if the MA recommendation is driven by genuine concerns over the price of the Medigap premium, the difference in WTP would have to be larger than the average Medigap premium, or at least $120/month. We do not find large differences in WTP in response to the signal.

[39] A similar interpretation is that brokers know consumers have a bias towards lower premiums and that brokers are catering to rather than correcting these biases as in Anagol et al. (2017) and Mullainathan et al. (2012).

[40] We expect the survey population to be representative. If anything, the survey population may have a lower opportunity cost of time and be negatively selected.

[41] Ratings are available for 497 of the 637 calls. Coverage is 82% among calls assigned to control and 72% among calls assigned to treatment; the 10.8 percentage point gap (SE 5.3) falls to 4.1 percentage points (SE 5.2) once we control for call center status, since call center calls are 21.9 percentage points more likely to be transcribed.

Table 8: Perceived Competition and Caller Price Sensitivity

	(1) Caller Price Sensitivity (1-5)	(2) Caller Price Sensitivity (1-5)	(3) Caller Price Sensitivity (1-5)	(4) Recommend MA (%)
Perceived Competition Assignment	-0.04	0.08	0.00	12.34**
	(0.05)	(0.06)	(0.04)	(5.46)
Control Mean	2.87	2.87	2.87	29.43
Controls	No	Yes	Yes	Yes
Caller Experience	None	Linear	Fixed Effects	Linear
Observations	497	497	497	497
R^2	0.00	0.45	0.74	0.17

Notes: Table 8 reports the effect of perceived competition assignment on the caller's price sensitivity rated by a large language model using the call transcripts (Appendix C.8). In columns (1)–(3) the outcome is a one-to-five price sensitivity rating where five is the highest level of price sensitivity. In column (4) the outcome is an indicator for an MA recommendation, in percentage points. Column (1) includes no controls. Columns (2) and (4) include baseline controls. Column (3) replaces the linear caller experience control with caller experience fixed effects (Appendix C.6). The sample is the 497 calls with a transcript rating from 98 callers. Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

5.5 Evidence of Broker Information Frictions

This section presents evidence that brokers may not have perfect information on both of the two main Medicare plan types: MA and Medigap. These broker information frictions motivate our two-type broker model described in Section 6.[42]

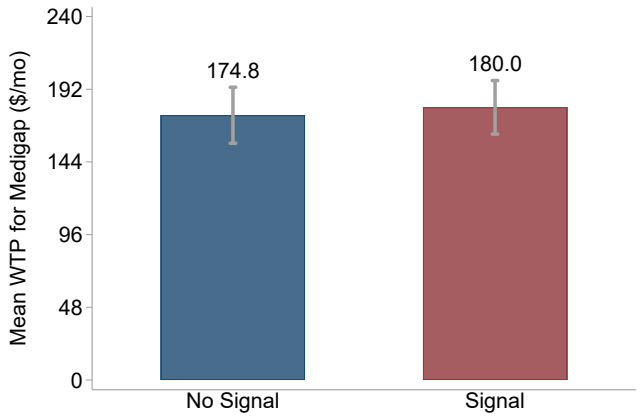
First, we show evidence from the audit call transcripts and broker survey. Brokers say they only sell either MA or Medigap on 131 of 497 transcribed calls. The MA-only brokers are concentrated at call centers: 32 of the 33 calls where brokers say they only sell MA come from a call-center broker in a sample where call centers make up 48% of calls.[43] This share is likely a lower bound since a broker who is only knowledgeable about one product is unlikely to volunteer this information. We see the same pattern in our survey of currently licensed Medicare insurance brokers: 18% of brokers only sell one of the two Medicare products (15% only sell MA and 3% only sell Medigap).[44]

[42] Anagol et al. (2017) find similar evidence of intermediary information frictions among life insurance agents in India and Linnainmaa et al. (2021) find that financial advisors also show biases in their own investment accounts by holding high-fee funds, chasing returns, and being under-diversified.

[43] Whether a broker sells one product or both is the endogenous choice of the firm and the broker. Both audit states license MA and Medigap under a single health line of authority (LOA). No publicly recorded certification differentiates brokers who sell MA or Medigap. Insurance company appointments are attached to the LOA rather than a specific product (Tex. Ins. Code §§4001.201–4001.206). This means a broker appointed to sell health insurance for an insurance company can sell any of that carrier's health products, including both MA and Medigap plans. Federal regulation requires agents selling MA to be trained and tested annually. However, these product-specific requirements are not publicly recorded and it is unclear whether they are enforced. Medigap is regulated at the state level and does not have additional training requirements in most states.

[44] Brokers believe single-product specialization is far more common than their own licensure implies: on average they believe 28% of brokers sell only MA and 22% sell only Medigap, together half the market and roughly three times the 18% implied by their own licensure. The two sources also disagree about which of the two specializations is more common, and we do not adjudicate between them. The argument here turns on the share of seniors who reach a single-product broker, not on which product that broker carries.

Figure 5: WTP for Medigap by Perceived Competition Treatment



Notes: Figure 5 shows the average WTP for a Medigap plan ($/mo) by competition signal, or by perceived competition treatment arm, from our broker survey. Grey bars are ±1 SE. Figure 5 shows the results from the within-broker randomization. Figure A.4 in Appendix B.5 shows the results from the across-broker randomization. The results are consistent across both specifications.

The second piece of evidence is the result from the natural experiment that a large share of brokers continue to recommend MA when it is not a good fit for the senior and when it does not pay a commission (Section 5.3). We create a proxy for broker knowledge from the coded indicator for whether or not the broker stated on the call that Medigap plans are standardized. While this measure is endogenous, callers are explicitly told to ask about Medigap regardless of the initial recommendation, giving all brokers a chance to state this important fact. Including baseline controls, brokers without the knowledge proxy flag recommend MA on 37% of calls (versus 22% for brokers with the knowledge proxy). The results are shown in Appendix Figure A.6.

A third piece of evidence, also highlighted in Section 5.1, is that high-tenure brokers tend to provide better advice.[45] Table A11 in Appendix C.8 further highlights the broker knowledge disparities by showing that high-tenure brokers are more likely to ask key diagnostic questions needed to provide a high-quality recommendation than low-tenure brokers. The content of the transcripts is endogenous, meaning we cannot distinguish between a broker who provides a bad recommendation because they do not *know* which diagnostic questions to ask and a broker who *chooses* not to ask the questions so they can provide a bad recommendation. Nonetheless, the differences by broker tenure are consistent with broker learning or positive selection over time.

High-tenure brokers may provide better advice for two reasons: learning and selection. Brokers may learn which plan fits a senior as they gain experience. Alternatively, brokers who provide low-quality advice may be more likely to leave the market, therefore the high-tenure brokers may be positively selected. We cannot separate the two explanations with our current audit data. All calls take place within 16 months, and only 52 of the 554 audited brokers are called more than once, so we do not observe how a given broker's advice changes as the broker gains experience. We show

[45]High-tenure brokers are brokers actively licensed for more than four years, as specified in the baseline controls.

two under-powered tests in Appendix C.4 where we attempt to disentangle the two explanations. Finally, we plan to track our audit study brokers in publicly available insurance licensure data at the end of 2027 to see if brokers who provided low-quality advice are more likely to leave the market. Since licenses renew in 2 year cycles, it is still too soon to test for these differences.

Finally, the results from our broker survey show that many Medicare brokers demonstrate important gaps in Medicare knowledge. In our survey we ask brokers to answer three multiple-choice knowledge-check questions phrased as hypothetical questions beneficiaries might ask them. These three questions are built from facts every licensed Medicare broker should know and that are clearly stated on Medicare.gov.[46] Performance is strikingly low for a licensed professional population: only 61% correctly identify that switching from MA to Medigap requires medical underwriting, 48% know that Traditional Medicare has no out-of-pocket cap, and only 39% know that same-letter Medigap plans are standardized.

6 A Structural Model of Broker Advice

The two experiments establish that broker recommendations respond to financial incentives and the probability of losing a sale (perceived competition) and provide evidence consistent with broker information frictions. In this section we use a structural model to quantify the relative importance of these forces and to estimate consumer surplus under different counterfactual policies. The estimated model is based on the conceptual framework in Section 3, but adds a misinformed broker type to represent brokers who, whether due to asymmetric training or information frictions, do not know that the Medigap plan matches the audited senior's preferences.

The model is deliberately parsimonious: its purpose is to isolate the supply-side factors governing the broker's recommendation decision and to recover the parameters needed to simulate counterfactual policies. We model the broker recommendation decision for our two main audit study outcomes: Decision 1 (Medicare Advantage versus Medigap) and Decision 2 (which Medigap plan to recommend). Broker pay is calibrated from the regulations in Table 1. Commissions and suitability costs are both measured in present-value (PV) dollars over the broker's commission horizon.[47] Baseline sale probabilities and the net harm of an unsuitable recommendation are calibrated from the broker survey. The remaining parameters are estimated from the audit study; these include commission sensitivity β, the Decision 1 suitability cost location $\bar{K}_1$, the misinformed broker share $\bar{\lambda}$, the sale-probability ratio on treated calls R_1, the perceived competition treatment deflator η, the Decision 2 suitability cost weight κ_2, and the appointment cost a_{appt}.

[46] Figure A.9 shows the results. The full questions and answers are shown in Appendix Figure A.8.

[47] Commission streams are discounted at $\delta = 0.95$ per year over $T = 6$ years. A Medigap stream pays 1 in year 1 and 1/2 in each renewal year, so it is worth $\mathcal{M}_{\text{MG}} = 3.15$ times its year-1 payment (Appendix E). The senior's losses, the net harm N and the Medigap markup m_j, are annual amounts that recur at their full value in every year of the same horizon, so their present value is $\mathcal{A} = \sum_{t=1}^{T} \delta^{t-1} = 5.30$ times the annual amount.

6.1 Setup

By design, every audited senior is a "Medigap-fit" senior, or a senior whose assigned preferences match the attributes of Medigap plans (Section 4.1.1). On call i, broker b makes the two recommendations observed in the audit study. As in Section 3.3, the broker trades off pay against a *perceived* suitability cost of recommending the "misfit plan," or the plan the broker believes does not match the senior's preferences (Decision 1), or against the suitability cost of recommending a plan that overcharges the senior for identical coverage (Decision 2).

In Section 3, we use plan j to index both the Decision 1 and Decision 2 recommendations to highlight the symmetry in the perceived competition result. In this section they are estimated using different data and are kept separate: $r \in \{\text{MA}, \text{MG}\}$ indexes the Decision 1 plan recommendation and j indexes the Decision 2 Medigap recommendation. The numeric subscripts on the suitability cost refer to the decision (e.g. Decision 1 vs Decision 2). K_1 and κ_1 belong to Decision 1 and K_2 and κ_2 to Decision 2, the second subscript on $K_{1,i}$ indexes call i, and the subscripts on $K_{2,ij}$ index call i and plan j. The subscript on the sale-probability ratio R_s instead indexes the perceived competition arm. Table 9 lists the model parameters and the source of identifying variation.

Broker types Each broker b has a fixed latent type $\tau_b \in \{I, M\}$ that determines which product the broker believes matches the audited senior's preferences. The informed type (I) knows that Medigap matches the audited senior's preferences and believes MA is the misfit plan. The misinformed type (M) believes MA matches and Medigap is the misfit plan. The broker type is determined by information frictions: a broker of the misinformed type may not be fully informed or fully trained to sell both products, or may face inertia or other frictions that make it difficult to sell the other Medicare product.[48] Section 5.5 presents reduced-form evidence of the two broker types. The probability that broker b is misinformed is

$$\lambda(x_b) = \Lambda\Big(-(\gamma_{ncc} + \gamma_{cc}\, cc_b + \gamma_{\text{ten}}\, \text{ten}_b)\Big),$$

where $x_b = (cc_b, \text{ten}_b)$ consists of two broker characteristics fixed before the call (cc_b indicates that the broker works at a call center, and ten_b is the log of one plus the broker's tenure in years of licensure, standardized across audited brokers). The index in parentheses is the log-odds that the broker is informed: γ_{ncc} is that log-odds for a non-call-center broker of average log tenure, γ_{cc} is the call-center shift, and γ_{ten} is the tenure coefficient. We report the misinformed share $\bar{\lambda}$, or the average of $\lambda(x_b)$ over the audited brokers, and $\bar{\lambda}_{cc}$ and $\bar{\lambda}_{ncc}$, its averages over the call-center and the non-call-center brokers.

[48] The misinformed type can be thought of as "MA-default brokers," who recommend MA on almost every call.

Decision 1 On call i, broker b earns the present-value commission $W_{r,i}$ if the senior enrolls in the recommended plan $r \in \{\text{MA}, \text{MG}\}$, which the broker expects to happen with probability P_{r,s_i}, where $s_i \in \{0, 1\}$ is the perceived competition arm of call i. As in Section 3.3, the broker also incurs a suitability cost $K_{1,i}$, but only on the sale of the plan the broker believes is the misfit. The expected payoffs of the two recommendations are

	Recommend MA	Recommend Medigap
Informed (I)	$P_{\text{MA},s_i}\big(W_{\text{MA},i} - K_{1,i}\big)$	$P_{\text{MG},s_i}\, W_{\text{MG},i}$
Misinformed (M)	$P_{\text{MA},s_i}\, W_{\text{MA},i}$	$P_{\text{MG},s_i}\big(W_{\text{MG},i} - K_{1,i}\big)$

(15)

and the broker recommends the plan with the higher expected payoff. The two types face the same commissions and sale probabilities; they differ only in which sale carries the suitability cost.

In both decisions the suitability cost is a weight times the present value of the senior's loss. In Decision 1, $K_{1,i} = \kappa_{1,i}\mathcal{A}N$, where N is the annual net harm in Equation (1) that the broker believes the misfit sale does, $\mathcal{A}N$ is its present value over the commission horizon, and $\kappa_{1,i}$ is the weight the broker on call i places on that present value (κ_b in Section 3.3, now indexed by call because some brokers are observed on multiple calls and the suitability cost weight is redrawn on each call). In Decision 2, the suitability cost of recommending plan j is $K_{2,ij} = \kappa_2 \mathcal{A} m_j$, where m_j is the plan's annual Medigap markup (Equation (18)) and κ_2 is the weight the broker places on the markup's present value. The costs $K_{1,i}$ and $K_{2,ij}$ are in PV dollars, like the commissions, so the weights $\kappa_{1,i}$ and κ_2 have no units. In the main specification both types perceive the same net harm, $N^{\text{MA}} = N^{\text{MG}} = N$, calibrated from the broker survey, so both types draw $K_{1,i}$ from the same distribution.[49]

The perceived competition arm is assigned call by call. Since the broker compares two recommendations, only the ratio of the two sale probabilities determines which product pays more in expectation: $R_s \equiv P_{\text{MG},s}/P_{\text{MA},s}$ for $s \in \{0, 1\}$, with R_0 on control calls, calibrated from the broker survey, and R_1 on treated calls, estimated from the perceived competition treatment. On call i the broker faces R_{s_i}.[50] Comparing the two payoffs in each row of (15), and dividing the informed type's comparison by P_{MA,s_i} and the misinformed type's by P_{MG,s_i}, gives each type a cutoff rule

[49]The harm enters the broker's problem only through the product $\kappa_{1,i}\mathcal{A}N$, so the audit identifies the distribution of the dollar cost $K_{1,i}$ and not $\kappa_{1,i}$ and N separately; N enters only the consumer surplus calculations (Online Appendix A.10). Table A14 in Appendix E.2 reports estimates with a different perceived harm for the misinformed type.

[50]The audit records recommendations, not enrollments, so the data cannot identify the sale probability levels; the model needs only their ratio (Section 6.2).

on the suitability cost:

$$\begin{aligned} I \text{ (Informed) Recommends MA iff:} \quad & K_{1,i} \;\le\; W_{\mathrm{MA},i} - R_{s_i}\, W_{\mathrm{MG},i}. \\ M \text{ (Misinformed) Recommends MA iff:} \quad & K_{1,i} \ge \Big(W_{\mathrm{MG},i} - W_{\mathrm{MA},i}/R_{s_i}\Big). \end{aligned} \tag{16}$$

For the informed broker type I, the right-hand side of Equation (16) is the "expected net pay gap from recommending MA," or the MA commission net of the Medigap commission, discounted by the relative sale-close probability. The informed type I only recommends MA if the suitability cost (incurred on an MA recommendation) is less than or equal to the expected net pay gap. Conversely, the misinformed type M incurs the suitability cost on a Medigap sale and recommends MA iff the suitability cost is larger than the expected net pay gap from recommending Medigap.

The suitability cost varies across calls: $K_{1,i}$ is drawn independently on each call from a logistic distribution with location $\bar{K}_{1,i}$ and scale $1/\beta$. The location depends on the characteristics of the caller who placed call i:

$$\bar{K}_{1,i} \;=\; \bar{K}_1 \,\exp\Big(\delta'(x_i - \bar{x})\Big),$$

where x_i contains the caller's college education, race (white), gender (female), age, and experience (the number of audit calls the caller had completed before call i), and $\bar{x}$ is the mean of x_i over calls. The vector $\delta = (\delta_{\text{college}}, \delta_{\text{white}}, \delta_{\text{female}}, \delta_{\text{age}}, \delta_{\text{exper}})$ holds one coefficient for each characteristic, with age measured in decades and experience in tens of prior calls. $\bar{K}_1$ is therefore the location when each caller characteristic is at its mean over calls ($x_i = \bar{x}$). The location is common to call-center and non-call-center brokers. Call-center status enters the model through the broker type probability $\lambda(x_b)$. We show that adding a call-center shift to the location does not improve the fit of the model (Appendix E). These are the caller controls of the reduced-form analysis without the square of age. They absorb part of the correlation among the recommendations that different brokers give the same caller (Appendix E).

The share of calls on which the cost falls below any cutoff c is therefore $\Pr(K_{1,i} \le c) \;=\; \Lambda\Big(\beta(c - \bar{K}_{1,i})\Big)$ where $\Lambda(t) = 1/(1+e^{-t})$. As in Equation (8), each type's probability of recommending MA is the mass of this distribution on the MA side of that type's cutoff in (16):

$$\begin{aligned} \Pr(r_i{=}\text{MA} \mid I) &= \Lambda\Big(\beta\Big[\Big(W_{\mathrm{MA},i} - R_{s_i} W_{\mathrm{MG},i}\Big) - \bar{K}_{1,i}\Big]\Big), \\ \Pr(r_i{=}\text{MA} \mid M) &= \Lambda\Big(\beta\Big[\bar{K}_{1,i} - \Big(W_{\mathrm{MG},i} - W_{\mathrm{MA},i}/R_{s_i}\Big)\Big]\Big). \end{aligned} \tag{17}$$

Equation (17) is a binary logit, the same one a random-utility model with logistic utility shocks would produce. Here the random term has an economic meaning: it is the variation in the suitability cost across calls, and it enters only the payoff of the plan the broker's type believes is the misfit. All commissions and suitability costs are measured in present-value thousands of

dollars. β is the commission sensitivity, or log-odds of an MA recommendation per present-value \$1,000 of the expected pay gap in brackets.[51] We assume that commission sensitivity β is common across brokers (including informed, misinformed, call-center, and non-call-center brokers). The location $\bar{K}_{1,i}$ is the commission gap at which an informed broker type is as likely to recommend MA as Medigap. Equation (17) is a logit with slope β on the commission gap and intercept $-\beta\bar{K}_{1,i}$, so the dollar location is minus the intercept divided by the slope.

Decision 2 Since broker type is a belief about which product (MA versus Medigap) matches the senior's preferences, we assume that broker type does not enter broker utility for Decision 2. Broker b recommends a Medigap plan j from the senior's choice set according to:

$$u_{ij} = \beta\Big[e^{-\eta\, s_i m_j}\, z\mathcal{M}_{\text{MG}}\big(\rho_{\min,i} + m_j\big) \;-\; K_{2,ij} \;-\; a_{\text{appt}}\mathbb{1}\{j \notin \mathcal{P}_i\}\Big] \;+\; \varepsilon_{ij}, \qquad K_{2,ij} = \kappa_2 \mathcal{A} m_j, \tag{18}$$

where ε_{ij} is i.i.d. extreme value and the bracketed term multiplied by β is the present-value commission dollars of plan j net of the Decision 2 suitability cost $K_{2,ij}$ and appointment cost a_{appt}.[52] Medigap commissions are a percentage z of the plan premium, so a higher-markup Medigap plan pays the broker more. The Decision 2 suitability cost penalizes the Medigap markup: the broker places the weight κ_2 on the present value of the premium the senior pays above the low-cost coverage-identical plan. We assume separate suitability cost weights for Decision 1 and Decision 2 (κ_1, κ_2).

The appointment cost a_{appt} reflects the cost of getting licensed to sell plans outside of $\mathcal{P}_i$, the portfolio of insurance carriers the broker on call i is already appointed to sell. The cost may include licensing fees, which can be waived by insurance companies, time costs, or general effort costs.[53] Note that in this setting appointments are not a binding constraint on the broker's choice set due to "just-in-time" appointments: brokers can still recommend plans outside their choice set and do so within our audit study on 29% of Medigap recommendations.[54]

To account for the broker response to the perceived competition treatment $s_i = 1$, the competition treatment deflator, or "the deflator", discounts the expected Medigap commission on treated calls $e^{-\eta s_i m_j}$. The deflator accounts for broker beliefs that the sale probability on a treated call could decrease (positive η) or increase (negative η) per dollar of markup m_j.

[51] We report the scale of the logistic distribution of $K_{1,i}$ in dollars as $\mu_K \equiv \$1{,}000/\beta$ (its standard deviation is $\pi\mu_K/\sqrt{3}$): a \$1,000 change in either commission gap shifts the log-odds of an MA recommendation by $\$1{,}000/\mu_K$.

[52] Since all beneficiaries who enroll in Medigap must also enroll in Part D or pay a penalty, the calibrated Medigap commission also includes the present value commission stream from a Part D enrollment. The commission is suppressed in Equation (18) since the value is the same for every Medigap plan recommendation (Appendix E).

[53] Appendix E explains why κ_2 and κ_1 are distinct objects, and why the appointment cost is a cost the broker can avoid rather than a binding constraint.

[54] Just-in-time (JIT) appointments allow brokers to obtain appointments right as they are about to enroll a beneficiary in an insurance company's plan.

We assume the same commission sensitivity term β multiplies both Decision 1 and Decision 2. The assumption implies a present-value dollar of commission converts to the same utility in both decisions. The Medigap recommendation in our audit study data identifies only the net markup slope $\alpha \equiv \beta(z\mathcal{M}_{\text{MG}} - \kappa_2\mathcal{A})$, the coefficient on the markup m_j in Equation (18), and the scaled appointment cost from the broker's recommendation given the consumer's choice set. Separating the costs relies on the assumption that the commission sensitivity β is the same in Decision 1 and Decision 2.[55] The Decision 2 model fit does not depend on this assumption. The values of K_2 and a_{appt} in Table 10 are converted at β, and $e^{-0.5\eta}$ is estimated given β, so all three rely on the assumption. Appendix E.3 identifies which counterfactuals in Table A17 also rely on the assumption.

6.2 Identification and Estimation

Table 9 summarizes each parameter of the estimated model and its source of identifying variation in the audit study.

We estimate the model by maximum likelihood. By assumption, broker type is a fixed attribute across calls. Let $\mathcal{C}_b$ be the calls broker b answered and $\mathcal{D}_2(i)$ the broker's Medigap plan recommendations on call i. Broker b's contribution to the likelihood is therefore:

$$\mathcal{L}_b \;=\; \Big[\lambda(x_b)\prod_{i\in\mathcal{C}_b}\Pr(r_i \mid M) \;+\; \big(1-\lambda(x_b)\big)\prod_{i\in\mathcal{C}_b}\Pr(r_i \mid I)\Big] \;\times\; \prod_{i\in\mathcal{C}_b}\prod_{\ell\in\mathcal{D}_2(i)}\Pr\big(j_{i\ell}\big)^{w_{i\ell}}, \tag{19}$$

where $\Pr(r_i \mid \tau)$ is the Decision 1 probability in (17) evaluated at the observed recommendation, $\lambda(x_b)$ is the probability that broker b is misinformed (Section 6.1), $\Pr(j_{i\ell})$ is the Decision 2 conditional-logit probability from (18) given the Medigap recommendation plan $j_{i\ell}$ and the senior's choice set. $w_{i\ell} = 1/|\mathcal{D}_2(i)|$ re-weights the recommendations so each call counts once in Decision 2 even if the call has multiple Medigap recommendations.[56] We estimate 14 parameters $\Theta = (\beta, \bar{K}_1, \delta, \gamma_{ncc}, \gamma_{cc}, \gamma_{\text{ten}}, R_1, \eta, \kappa_2, a_{\text{appt}})$ by maximum likelihood, where δ contains the five caller-characteristic coefficients. Standard errors and intervals are estimated from a caller-cluster bootstrap over 1000 resamples. See Appendix E for details.

Misinformed broker share Approximately 17% of brokers in our audit study are estimated to be misinformed. Consistent with the reduced-form results, the share is heavily concentrated

[55]The plan-choice data cannot test this assumption. In our model we assume that the Medigap commission is a fixed percentage of the premium. This means that within a senior's choice set every dollar of markup raises the commission by the same amount and the data identify α, but not its two components. The only payment that varies separately from the premium is a carrier bonus, which we observe only for plans the broker is already appointed to sell and which is paid once rather than as a commission stream.

[56]Decision 2 sits outside the bracket because (18) does not vary with broker type. A call where the broker recommends MA in Decision 1 but also recommends a Medigap plan in Decision 2 still contributes a Decision 2 factor (Appendix E).

Table 9: Parameters of estimated model and identifying variation

Parameter	Economic interpretation	Identifying variation
Estimated		
β	Commission sensitivity	Variation in broker pay (including cross-time variation and Experiment A). Experiment A: $W_{\text{MA}} \to 0$ (call centers), $W_{\text{MA}} \to \frac{1}{3}W_{\text{MA}}$ with perceived competition treatment (non-call-center brokers)
$\bar{\lambda}$	Misinformed broker share	Share that recommend MA at $W_{\text{MA}} = 0$ (call centers); non-call-center brokers extrapolated from $\frac{1}{3}W_{\text{MA}}$ using β and logistic shape; repeat calls within broker
$\bar{K}_1$	Decision 1 suitability cost location	Level of MA recommendations unexplained by commission gap given β and $\bar{\lambda}$
R_1	Sale-probability ratio when $s_i = 1$	Experiment B, Decision 1 given calibrated R_0
η	Treatment discount (when $s_i = 1$) on sale probability per \$1,000 of annual Medigap markup	Experiment B, Decision 2
κ_2	Suitability cost weight on the PV of the Medigap markup	Within-menu Medigap plan choices
a_{appt}	Cost of obtaining appointment	Implied cost from plans chosen outside broker's portfolio
Calibrated		
Discount factor	Annual discount factor applied to commission streams and suitability costs	Conventional value in the literature: 0.95
W_{MA}, W_{MG}	Present value commission stream	CMS commission schedule, discounted at the discount factor over 6 years
$N^{\text{MA}}, N^{\text{MG}}$	Annual net harm from misfit sale	Broker survey; $N^{\text{MA}} = N^{\text{MG}}$ in main model
$\mathcal{A}$	PV of \$1 of annual loss over the commission horizon	Calibrated: discount factor of 0.95 per year over a 6-year time horizon
R_0	Sale-probability ratio when $s_i = 0$	Broker survey

among call center brokers: 39% of call-center brokers are estimated to be misinformed versus 6% of non-call-center brokers.[57] Note that the non-call-center share rests on an extrapolation from the one-third-pay calls; Table A14 re-estimates the model with a calibrated share of misinformed non-call-center brokers and finds similar results.[58]

[57] We assume a two-type model given the reduced-form evidence of broker information frictions in Section 5.5. We also test for a one-broker-type model in which the suitability cost is drawn independently on each call. The one-type model is rejected with likelihood ratio 56.2 (Table A14).

[58] This exercise uses the share of non-call-center brokers who report that they are only licensed to sell MA in the broker survey (13.3%). The data narrowly reject the calibrated restriction at the 5% level (likelihood ratio 7.01, $p = 0.008$). This version of the calibrated model changes the gains in consumer surplus from pay parity from \$377 to \$352 per senior per year and moves no other policy in Figure 6 by more than \$26.

Table 10: Structural Model Parameters in Dollar-Equivalent Terms

Parameter	Description	Estimate	95% interval
Decision 1: Medicare Advantage or Medigap			
$\mu_K = \$1{,}000/\beta$	Scale of the suitability cost (PV \$)	1,107	[633, 2,545]
$\bar{K}_1$	Location of the suitability cost (PV \$)	3,696	[2,834, 8,638]
Broker types			
$\bar{\lambda}$	Misinformed share, all brokers	0.17	[0.11, 0.22]
$\bar{\lambda}_{cc}$	Misinformed share, call-center brokers	0.39	[0.29, 0.49]
$\bar{\lambda}_{ncc}$	Misinformed share, non-call-center brokers	0.06	[0.00, 0.11]
Signal of talking to multiple brokers (treated calls)			
R_1	Sale-probability ratio, Medigap to MA[a]	0.00	[0.00, 0.30]
$e^{-0.5\eta}$	Relative acceptance of a plan \$500/yr above the cheapest	0.72	[0.32, 0.91]
Decision 2: Medigap plan			
K_2	Suitability cost per \$1,000/yr of markup (PV \$)	2,571	[1,723, 5,364]
a_{appt}	Appointment cost (PV \$)	1,966	[1,106, 4,650]

Notes: Table 10 reports the dollar equivalents of the maximum likelihood estimates of the main specification. Appendix Table A12 reports every estimated parameter on the scale it is estimated. Dollar parameters are in present-value (PV) commission dollars. Each dollar parameter is the corresponding estimated parameter divided by the commission sensitivity β. $e^{-0.5\eta}$ is the treatment deflator at a markup of \$500 per year: on a treated call, this is the broker's expected probability of selling a plan \$500/yr above the cheapest coverage-identical plan relative to the same plan on a control call. We use \$500 because it is close to the median markup across the plans on the audit menus (\$480). $K_2 = \kappa_2 \mathcal{A} \times \$1{,}000 = 1{,}000\,(z\mathcal{M}_{\text{MG}} - \alpha/\beta)$ is the PV suitability cost of recommending a plan \$1,000/yr above the cheapest coverage-identical plan, so it does not depend on the annuity factor $\mathcal{A}$. Note that the dollar values of $\bar{K}_1$ and K_2 are not directly comparable. $\bar{K}_1$ is the location of the cost of an entire unsuitable Decision 1 recommendation, while K_2 is the cost of \$1,000 a year of markup. Commission streams and the markup are valued over 6 years at a discount factor of 0.95 per year. Each 95% interval runs from the 2.5th to the 97.5th percentile of the parameter across 1000 caller-cluster bootstrap refits (Appendix E). [a]The estimate of R_1 is zero to numerical precision, the lower limit of its range. Its interval is the 95% confidence interval from inverting the likelihood-ratio test (Appendix E): the values of R_1 that the test does not reject at the 5% level. Appendix E.1 shows that the model reproduces the perceived competition treatment's effect on Decision 1 with R_1 in this interval. Appendix Table A13 reports the same parameters, and the percent change in $\bar{K}_1$ for each caller characteristic, with intervals from caller-cluster and broker-cluster bootstrap refits.

The Decision 1 suitability cost The estimated location of the dollar suitability cost is $\bar{K}_1 =$ \$3,696 (Table 10).[59] The estimates imply there are costs that discipline brokers against recommending the higher-commission MA plans; on average a broker is indifferent between recommending Medigap and MA when MA pays \$3,892 more in PV-commission dollars, or 2.8 times the \$1,367 sale-probability-ratio-discounted commission gap the informed broker type faces on those calls.

The Decision 2 suitability cost A Medigap plan that is \$1,000 a year more expensive than the cheapest coverage-identical plan pays the broker \$630 more in PV commission, but carries a suitability cost of $K_2 = \kappa_2 \mathcal{A} \times \$1{,}000 = \$2{,}571$ (Table 10), or 4.1 times the extra commission. Equivalently, the implied suitability cost weight is $\kappa_2 = 0.49$: the broker behaves as if they incur 0.49 PV dollars of cost for each PV dollar of markup the senior pays.

[59]Adding a separate location for call-center brokers does not improve the fit (likelihood-ratio statistic 0.92, $p = 0.34$; Table A14 and Appendix E).

Appointment cost The estimated appointment cost, or the cost of becoming licensed to sell plans for a specific insurance company, is $a_{\text{appt}} = \$1{,}966$ of PV-commission.

Decomposing broker advice Table A16 in Appendix E.3 reports how the share of brokers recommending MA to a Medigap-fit senior changes when each force in the model is switched off, holding the others at their estimated values.

Together, we estimate that 38%–49% of MA recommendations are driven by commissions and that information frictions account for 41% of recommendations. In the estimated model, brokers recommend MA on 34% of commission-paying calls. When brokers are not paid for either product the share of MA recommendations falls by 13 percentage points. Removing only the MA commission, the change carriers made for call-center brokers in Experiment A, lowers the share by 17 percentage points. Under zero MA commissions, changing every misinformed broker to the informed type would reduce the share of MA recommendations by 14 percentage points. Note that this does not account for the role incentives may have played in shaping which products brokers are trained to sell. Table A17 in Appendix E.3 reports the same exercise for the mean recommended Medigap markup.

The suitability cost receives a large weight in both decisions. Removing the suitability cost from both decisions and holding commissions, broker types, and choice sets constant raises the share of brokers recommending Medicare Advantage to a Medigap-fit senior from 34% to 83% of calls. Similarly, the mean recommended Medigap markup would increase from \$386 to \$681 a year.

7 Counterfactuals and Welfare

The following counterfactuals assume the broker recommendations are accepted by the senior. A Medigap-fit senior who follows the recommendation of a broker in our audit study loses \$2,154 a year in expectation, or \$30,811 in present-value wealth over their lifetime, from the broker recommendation. In Decision 1 the senior loses \$1,899 in expectation per year when the broker recommends Medicare Advantage to the Medigap-fit senior. In Decision 2 the senior loses \$255 in expectation per year when the broker recommends a Medigap plan with an average markup of \$386, multiplied by the probability that the broker recommends a Medigap plan.

Each counterfactual policy changes one feature of the supply-side environment or the senior's search behavior and recalculates the broker recommendation probabilities implied by the model under this change. We value each averted MA recommendation at the net annual harm elicited from the broker survey less the expected Medigap markup the senior pays when recommended Medigap instead. We report every result per Medigap-fit senior per year. Once again, this assumes

the recommendation is accepted. Additional details and a discussion of the partial-equilibrium assumption are included in Appendix E.3.

7.1 Policy Simulations

We examine two sets of policies: 1) market changes that governments or regulators can impose; and 2) changes to consumer search (e.g. announcing plans to seek recommendations from multiple brokers or consulting only highly-rated brokers). We also consider a third counterfactual that we refer to as "consumer education." The idea behind these counterfactual simulations is that brokers may have two roles: to advise and to educate consumers. These counterfactuals estimate consumer surplus gains from recommendations to consumers who understand a greater portion of the Medigap insurance attributes (or who have a smaller friction ϕ in Section 3). We focus on two changes to broker commissions: equalizing the MA and Medigap commissions, which we call pay parity, and capping the Medigap markup at 25% above the low-cost identical plan. Appendix E.3 shows the full set of counterfactual policies.

Supply-side policies improve broker advice Figure 6 shows the annual consumer surplus gain for a Medigap-fit senior under six policy counterfactuals. Changing the market rules recovers the largest share of consumer surplus. We estimate that pay parity raises consumer surplus by $377 per senior per year, recovering 18% of lost consumer surplus, and a markup cap set at 25% of the cheapest coverage-identical premium is worth $128 (6% of lost consumer surplus).

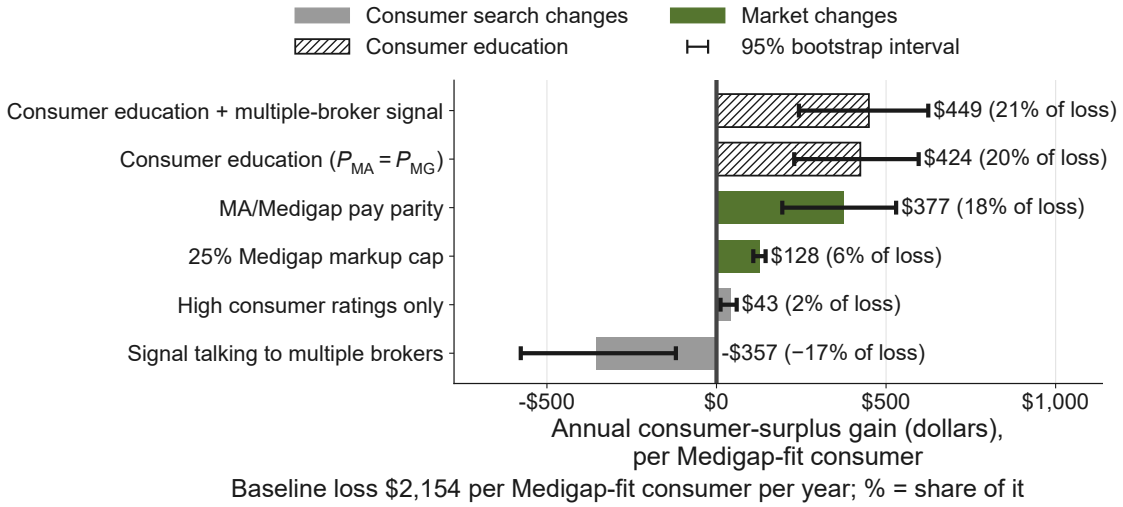


Figure 6: Annual Consumer Surplus Gain by Policy

Notes: Figure 6 shows the change in annual consumer surplus per Medigap-fit senior under each counterfactual policy, relative to a baseline loss of $2,154 a year. Grey bars show changes to consumer search, black hatched bars show changes to consumer education, and green bars show changes to the market that can be implemented by governments or regulators. Every bar is measured against the audit's observed mix of treated and control calls. Confidence intervals show 95% caller-cluster bootstrap percentile intervals from the refits (Appendix E.3).

Limitations of consumers as a disciplining mechanism Two counterfactuals in Figure 6 show changes in consumer search and two show changes in consumer education. In the two consumer search changes consumers solicit advice only from brokers with high senior ratings and always signal plans to talk to multiple brokers. The consumer education counterfactuals show the consumer surplus gains if brokers believe a senior is equally likely to accept a recommendation of MA or a recommendation of Medigap, $P_{\text{MA}} = P_{\text{MG}}$ or $R = 1$. The higher perceived Medigap sale-close probability may be driven, for example, by policies educating seniors on the different MA and Medigap plan attributes so seniors put more weight on the plan attributes and less weight on plan premiums. This could also be driven by perceptions that competitors will educate seniors on plan attributes.[60]

The two consumer search policies recover less of the lost consumer surplus than any supply-side policy described above. The "high consumer ratings" only counterfactual restricts the pool of brokers to brokers who receive a high rating, or a rating of a 4 or 5 (Section 5.2). The counterfactual recovers \$43 per senior per year, or 2% of the loss. Since shopper ratings carry almost no information about broker or recommendation quality the screen recovers little consumer surplus.

The "signal talking to multiple brokers" counterfactual, where all seniors tell brokers they plan to solicit multiple recommendations, has a *net negative* impact on consumer surplus. Moving every call from the control arm to the treated arm raises the share of brokers recommending MA from 31% to 41% of calls and lowers the average Medigap markup by \$56 a year. Valuing each additional Medicare Advantage recommendation at the survey-elicited net annual harm less the expected Medigap markup the senior would otherwise pay, the net impact is a change in consumer surplus of −\$494 per Medigap-fit senior per year. Figure 6 shows a smaller loss of \$357 because it measures the same policy against the audit's observed mix of treated and control calls, in which some calls already carry the signal.[61] The treatment moves the recommendation toward the attribute the senior can directly observe and verify, the premium, and away from the coverage attributes the senior cannot. The net impact is a reduction in consumer surplus.

As an illustrative benchmark rather than a policy, we show that the "consumer education" counterfactual recovers 20% of lost consumer surplus. The counterfactual, which simulates recommendations for consumers who better perceive a greater portion of insurance attributes and have lower frictions ϕ, lowers the share of brokers recommending MA from 34% to 26% of calls, recovering \$424 of consumer surplus per senior per year. By construction, the counterfactual changes the broker's perception of the senior's acceptance probability for Decision 1 only, not of a high markup Medigap plan. The gain comes almost entirely from brokers of the informed type, who

[60] $R = 1$ is the complexity threshold of Section 3.4 or the sale-close probabilities when the senior perceives a share $\phi^{\dagger} = 0.25$ of the coverage harm. The interpretation of this perceived harm share is that at this share the perceived harm from under-insurance is exactly equal to the premium the senior saves when enrolling in MA.

[61] Appendix E.3 reports the valuation that uses the reduced-form treatment effects in place of the model's fitted effects, and the band that the boundary on the treated sale-probability ratio places around the point estimate.

steer less once a Medigap sale is as likely to close as an MA sale. Consumer education cannot remove the steering of misinformed brokers, who believe MA fits the senior and therefore continue to recommend it on almost every call. Adding the multiple-broker signal raises the gain to $449 by shifting the Medigap markup in Decision 2.

7.2 Scaling to the national population

We scale the consumer surplus loss from broker recommendations to the national population two ways: 1) for the seniors who turn 65 each year, and 2) for current Medicare beneficiaries. Both calculations use the following assumptions, described in detail in Appendix E.4. We assume that Medicare beneficiaries fall into one of three plan-fit types: 34% unambiguous Medigap-fit, 51% unambiguous MA-fit, and 15% ambiguous-fit (not a clear fit for either plan). The shares reflect the broker beliefs elicited in our survey. A Medigap-fit senior loses the net annual harm of $5,574 if a broker enrolls them in MA and loses the Medigap markup otherwise. An MA-fit senior loses the over-insurance harm that brokers report in the survey, or $1,243 a year, plus the Medigap markup if the broker enrolls them in Medigap. Brokers report this harm with both plans priced at $0, so it leaves out the $1,855 higher premium the senior pays for Medigap, and the MA-fit senior's loss is conservative. An ambiguous-fit senior loses only the Medigap markup, on the 58% of calls where brokers recommend Medigap for a senior profile designed to be an "ambiguous fit" for both products.[62]

Since the audit study only measures recommendations to a Medigap-fit senior, we use the structural model to predict recommendations to an MA-fit senior. We assume that an informed broker follows the estimated Decision 1 rule, with the suitability cost now incurred by a Medigap recommendation, and that the same share of brokers is misinformed and believes a Medigap plan matches the MA-fit senior's preferences. Finally, we assume that 50% of seniors enroll through a broker. This assumption is conservative; in our UAS survey, 56% of beneficiaries who report neither Medicaid eligibility nor access to an employer plan say a broker helped them enroll in their current Medicare plan (Appendix Table A6).

For the first calculation, we assume that seniors enroll in Medicare when they turn 65. Excluding beneficiaries with full Medicaid benefits, whose cost sharing Medicaid pays, and beneficiaries with employer or union plans, we estimate that 2.81 million of the 4.1 million seniors who turn 65 each year (Fichtner, 2024) face the choices in our audit study, and 1.41 million of them use a broker. Figure 7 reports lost consumer surplus for these new enrollees at baseline and under two market policies. We estimate the total consumer surplus loss to be about $1.25 billion a year, or $17.9 billion in present-value wealth over their lifetime. Of the annual loss, about $1.04 billion

[62]This rate comes from a broker survey question where we ask brokers to provide a plan recommendation for a senior vignette designed to be a fit for either MA or Medigap.

comes from steering seniors into the wrong plan (MA vs. Medigap) and $210 million comes from the Medigap markup. About 83% of lost consumer surplus falls on Medigap-fit seniors.

Our estimates suggest pay parity increases total consumer surplus for these seniors by $151 million a year, and adding the 25% Medigap markup cap increases it by another $124 million. Together, the two policies recover $275 million a year, or 22% of baseline consumer surplus loss. The final bar in Figure 7 adds the federal savings (or taxpayer savings) generated by the two policies. A 2025 Medicare Payment Advisory Commission report estimated that Medicare spends 20% more on beneficiaries enrolled in Medicare Advantage than it would spend on these same beneficiaries in Traditional Medicare (Medicare Payment Advisory Commission, 2025). Net of the additional Medicare spending caused by more comprehensive Medigap coverage (Cabral and Mahoney, 2019), this is approximately $846 per enrollee per year, implying the two policies would lower federal spending by $45 million a year. Appendix E.4 describes the assumptions behind this calculation.

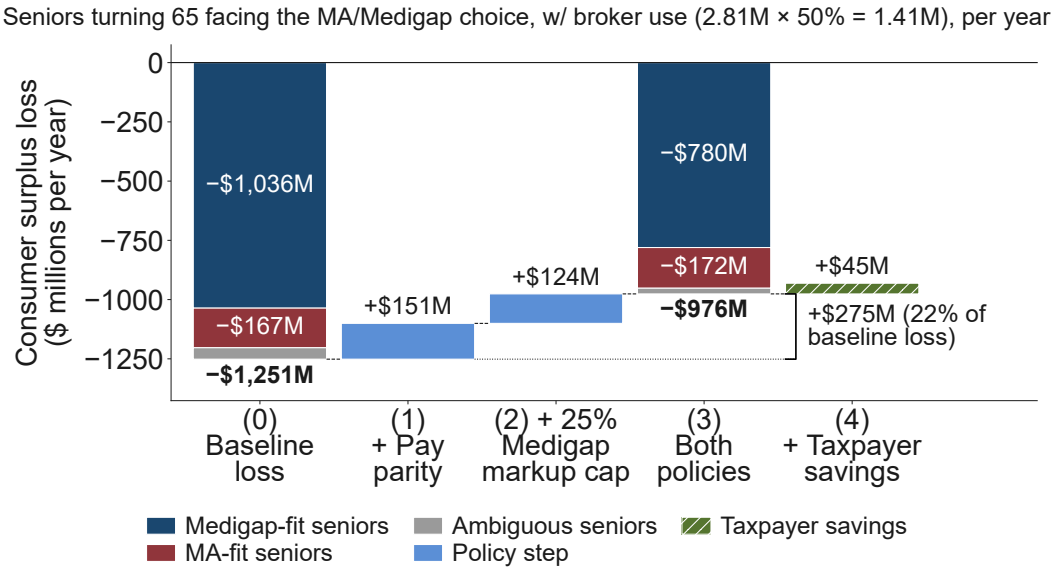


Figure 7: Consumer Surplus Loss Among Seniors Turning 65 Who Face the Medicare Advantage–Medigap Choice and Use a Broker, and the Share Two Market Policies Recover

Notes: Figure 7 shows the annual consumer surplus loss for the 1.41 million seniors who turn 65 each year, choose between Medicare Advantage and Medigap in the individual market, and use a broker (Appendix E.4). Lost consumer surplus is measured relative to each senior enrolling in the plan that matches their plan-fit type from the broker survey, at a $0 Medigap markup. Negative values are losses. Bars (0) and (3) split the loss by plan-fit type. Bar (1) shows the change in consumer surplus from equalizing Medicare Advantage and Medigap commissions. Bar (2) shows the additional change from capping the Medigap markup at 25% of the lowest-priced coverage-identical plan once pay parity is in place. The bracket shows the gain from bar (0) to bar (3). Bar (4) shows the reduction in federal Medicare spending under both policies, valued at $846 per Medicare Advantage enrollee per year. Premiums, commission schedules, choice sets, and broker composition are held fixed.

For the second calculation, we start from current beneficiaries aged 65 and over who are currently enrolled in either Medicare Advantage or a Medigap plan. Once again, we exclude beneficiaries with full Medicaid benefits and beneficiaries with employer or union plans. Assuming the same 50% broker-use share, we estimate that 15.6 million current MA or Medigap beneficiaries used a

broker (Appendix E.4). We use the same plan-fit shares from the broker survey as for seniors turning 65. As a check, these shares predict that 61% of current beneficiaries who used a broker are enrolled in Medicare Advantage, compared with 61% of these beneficiaries in the 2023 Medicare Current Beneficiary Survey (Appendix E.4). Assuming their plans reflect the broker recommendations we observe today, we estimate the total consumer surplus loss for current beneficiaries to be about $13.9 billion a year (Figure 8). Of this, about $10.1 billion falls on Medigap-fit seniors steered into Medicare Advantage. Since a policy would only change the recommendations made to seniors at enrollment we do not report policy counterfactuals for current beneficiaries.

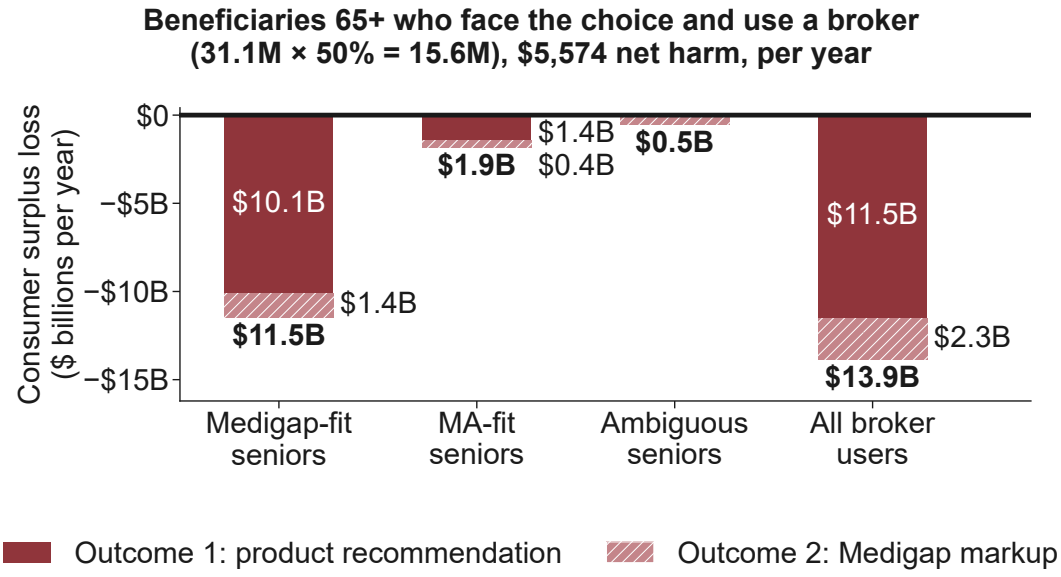


Figure 8: Consumer Surplus Loss Among Current Medicare Beneficiaries Who Face the Medicare Advantage–Medigap Choice and Use a Broker

Notes: Figure 8 shows the annual consumer surplus loss for the 15.6 million beneficiaries aged 65 and over who hold Medicare Advantage or Medigap and used a broker, by plan-fit type and outcome (Appendix E.4). The plan-fit shares come from the broker survey, as in Figure 7. Dark segments are losses from recommending the plan type that does not fit the senior (Decision 1). Light segments are Medigap markups over the cheapest coverage-identical plan (Decision 2). Bold labels are group totals, and losses are plotted as negative values. Premiums, commission schedules, choice sets, and broker composition are held fixed.

8 Conclusion

More than half of the nearly 70 million people on Medicare enroll through an insurance broker, with the broker directly enrolling beneficiaries in the plan they recommend. The initial enrollment decision is difficult to reverse, so the recommendation a senior receives on one phone call can shape their insurance coverage and premiums for years. These stakes make the quality of expert advice consequential, yet the same complexity that creates demand for brokers can also make their advice difficult for consumers to evaluate.

In the first large-scale audit study of Medicare insurance brokers, we show that brokers steer

seniors toward the higher-commission, worse-fit product on 52% of calls when incentives are misaligned. In the Medigap market, these agency problems generate a directly measurable cost: the average recommendation costs a senior more than $377 per year, or a 29% markup over an otherwise identical plan. Compounded over time, a senior who follows this advice at age 65 loses more than $12,000 in lifetime wealth. These are likely conservative estimates given our decision to base our seniors in rural areas and to avoid enrollment periods when brokers are busiest.

Two experiments and a structural model identify two sources of low-quality advice: misaligned incentives and broker information frictions. A natural experiment in which carriers stopped paying MA commissions reduced MA recommendations by 21 p.p. Yet brokers continued to recommend MA even when MA recommendations did not generate commissions and did not match senior preferences. Additional evidence from recommendation differences by broker tenure, call transcripts, and a broker survey is consistent with broker information frictions rather than beliefs that MA matches senior preferences. Using the structural model to quantify the components of advice, we estimate that commissions account for 38%–49% of MA recommendations, and broker information frictions for 41%.

Our perceived competition treatment shows why competition alone may not discipline these distortions. The experimental results show that increasing competitive pressure by signaling that a consumer plans to solicit multiple recommendations reduces Medigap markups, but also increases steering toward Medicare Advantage. Competitive pressure therefore disciplines the attribute consumers can verify, the premium, and not the attributes they cannot, such as plan coverage and network restrictions. Consistent with this interpretation, senior ratings of brokers are uncorrelated with our objective measures of recommendation quality, and this relationship does not strengthen with caller experience. Thus, the consumer-search counterfactuals show little benefit from consumer-directed mechanisms: screening brokers using high ratings recovers only a small fraction of the consumer surplus loss, while encouraging all consumers to solicit multiple recommendations lowers consumer surplus.

The findings illustrate a broader challenge in designing markets with complex choices: when consumers cannot evaluate the products themselves or expert advice, policies that rely on consumer choice and competition may fail to address conflicts of interest created by misaligned incentives. Instead, policies should target the root causes of low-quality advice directly, by aligning compensation and ensuring that intermediaries have sufficient information and training to evaluate *all* the products they are licensed to sell.

One implication concerns the growing use of AI-based decision support in advice markets. Our results suggest why such technologies may be especially valuable in settings like Medicare. If programmed correctly, AI technology can help align incentives and reduce the information frictions that generate poor advice even when financial incentives are aligned. At the same time, AI does

not by itself resolve an agency problem when the intermediary's incentives remain misaligned. If technology makes brokers better informed, it may instead increase the effectiveness of incentive reform by giving brokers the information needed to act on aligned incentives. Thus, the gains from AI in expert advice markets may depend not only on the quality and incentives of the technology itself, but also on the incentives faced by the experts using this technology.

Overall, our results highlight the importance of well-designed markets and incentives, especially in complex markets with information frictions and limited feedback. More broadly, consumer information frictions can reinforce supply-side frictions: when consumers have difficulty evaluating product quality or expert advice, competitive pressure may provide little discipline and firms may have weaker incentives to invest in training and expertise when markets do not reward high-quality advice. The same information frictions that create demand for expert intermediaries can therefore weaken the market forces that would discipline agency problems, allowing distortions to persist even in competitive markets.

References

Abaluck, Jason and Jonathan Gruber, "Choice Inconsistencies among the Elderly: Evidence from Plan Choice in the Medicare Part D Program," *American Economic Review*, June 2011, *101* (4), 1180–1210.

Afendulis, Christopher C., Anna D. Sinaiko, and Richard G. Frank, "Dominated Choices and Medicare Advantage Enrollment," *Journal of Economic Behavior & Organization*, 2015, *119*, 72–83.

Agarwal, Sumit, John C. Driscoll, Xavier Gabaix, and David Laibson, "The Age of Reason: Financial Decisions over the Life Cycle and Implications for Regulation," *Brookings Papers on Economic Activity*, 2009, *40* (2), 51–117.

Aizawa, Naoki and You Suk Kim, "Advertising and Risk Selection in Health Insurance Markets," *American Economic Review*, 2018, *108* (3), 828–867.

Ali, Riaz, Aimee Cicchiello, Morgan Hanger, Lesley Hellow, Ken Williams, and Gretchen Jacobson, "How Agents Influence Medicare Beneficiaries' Plan Choices," Technical Report, The Commonwealth Fund 2021.

Alsan, Marcella, Owen Garrick, and Grant Graziani, "Does Diversity Matter for Health? Experimental Evidence from Oakland," *American Economic Review*, December 2019, *109* (12), 4071–4111.

Ameriks, John, Andrew Caplin, Minjoon Lee, Matthew D. Shapiro, and Christopher Tonetti, "Cognitive Decline, Limited Awareness, Imperfect Agency, and Financial Well-Being," *American Economic Review: Insights*, March 2023, *5* (1), 125–140.

Anagol, Santosh, Shawn Cole, and Shayak Sarkar, "Understanding the Advice of Commissions-Motivated Agents: Evidence from the Indian Life Insurance Market," *The Review of Economics and Statistics*, March 2017, *99* (1), 1–15.

Andrews, Donald W. K., "Inconsistency of the Bootstrap When a Parameter Is on the Boundary of the Parameter Space," *Econometrica*, March 2000, *68* (2), 399–405.

Arias, Elizabeth, Jiaquan Xu, and Kenneth D. Kochanek, "United States Life Tables, 2022," National Vital Statistics Reports vol. 74, no. 2, National Center for Health Statistics, Hyattsville, MD April 2025.

Bagnoli, Mark and Ted Bergstrom, "Log-Concave Probability and Its Applications," *Economic Theory*, 2005, *26* (2), 445–469.

Balafoutas, Loukas and Rudolf Kerschbamer, "Credence Goods in the Literature: What the Past Fifteen Years Have Taught Us about Fraud, Incentives, and the Role of Institutions," *Journal of Behavioral and Experimental Finance*, June 2020, *26*, 100285.

Bekker-Grob, Esther W. De, Mandy Ryan, and Karen Gerard, "Discrete Choice Experiments in Health Economics: A Review of the Literature," *Health Economics*, February 2012, *21* (2), 145–172.

Bertrand, Marianne and Adair Morse, "Information Disclosure, Cognitive Biases, and Payday Borrowing," *The Journal of Finance*, 2011, *66* (6), 1865–1893.

_ **and Sendhil Mullainathan**, "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination," *American Economic Review*, September 2004, *94* (4), 991–1013.

Bhargava, Saurabh, George Loewenstein, and Justin Sydnor, "Choose to Lose: Health Plan Choices from a Menu with Dominated Option," *The Quarterly Journal of Economics*, August 2017, *132* (3), 1319–1372.

Bhattacharya, Utpal, Andreas Hackethal, Simon Kaesler, Benjamin Loos, and Steffen Meyer, "Is Unbiased Financial Advice to Retail Investors Sufficient? Answers from a Large Field Study," *The Review of Financial Studies*, April 2012, *25* (4), 975–1032.

Biniek, Jeannie Fuglesten, Nolan Sroczynski, and Tricia Neuman, "Use of Prior Authorization in Medicare Advantage Exceeded 46 Million Requests in 2022," August 2024.

Boccuti, Cristina, Gretchen Jacobson, Kendal Orgera, and Tricia Neuman, "Medigap May Be Elusive for Medicare Beneficiaries with Pre-Existing Conditions," 2018.

Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer, "Salience and Consumer Choice," *Journal of Political Economy*, 2013, *121* (5), 803–843.

Brot-Goldberg, Zarek C., Amitabh Chandra, Benjamin R. Handel, and Jonathan T. Kolstad, "What Does a Deductible Do? The Impact of Cost-Sharing on Health Care Prices, Quantities, and Spending Dynamics*," *The Quarterly Journal of Economics*, August 2017, *132* (3), 1261–1318.

_ **, Timothy Layton, Boris Vabson, and Adelina Yanyue Wang**, "The Behavioral Foundations of Default Effects: Theory and Evidence from Medicare Part D," *American Economic Review*, 2023, *113* (10), 2718–2758.

Bucher-Koenen, Tabea, Andreas Hackethal, Johannes Koenen, and Christine Laudenbach, "Gender Differences in Financial Advice," *American Economic Review*, December 2025, *115* (12), 4218–4252.

Cabral, Marika and Neale Mahoney, "Externalities and Taxation of Supplemental Insurance: A Study of Medicare and Medigap," *American Economic Journal: Applied Economics*, April 2019, *11* (2), 37–73.

_ **, Michael Geruso, and Neale Mahoney**, "Do Larger Health Insurance Subsidies Benefit Patients or Producers? Evidence from Medicare Advantage," *American Economic Review*, 2018, *108* (8), 2048–2087.

Carroll, Gabriel, "Robust Incentives for Information Acquisition," *Journal of Economic Theory*, May 2019, *181*, 382–420.

Centers for Medicare & Medicaid Services, "Agent Broker Compensation," https://www.cms.gov/medicare/health-drug-plans/managed-care-marketing/medicare-marketing-guidelines/agent-broker-compensation 2026.

Centers for Medicare and Medicaid Services, "Medicare and Medicaid Programs; Contract Year 2022 Policy and Technical Changes to the Medicare Advantage Program, Medicare Prescription Drug Benefit Program, Medicaid Program, Medicare Cost Plan Program, and Programs of All-Inclusive Care for the Elderly (CMS-4190-F2)," *Federal Register*, January 2021, *86* (11), 5864–6135.

_ , "Medicare Program; Contract Year 2024 Policy and Technical Changes to the Medicare Advantage Program, Medicare Prescription Drug Benefit Program, Medicare Cost Plan Program, and Programs of All-Inclusive Care for the Elderly (CMS-4201-F)," *Federal Register*, April 2023, *88*, 22120–22345.

_ , "CMS Fast Facts, April 2025 Version," Technical Report, Centers for Medicare and Medicaid Services April 2025.

_ , "Contract Year 2026 Agent and Broker Compensation Rates, Referral/Finder's Fees, Submissions, and Training and Testing Requirements," HPMS Memorandum, Centers for Medicare and Medicaid Services, Baltimore, MD June 2025.

_ , "Medicare Program; Contract Year 2027 and Certain Contract Year 2026 Policy and Technical Changes to the Medicare Advantage Program, Medicare Prescription Drug Benefit Program, and Medicare Cost Plan Program (CMS-4208-F3 and CMS-4212-F)," *Federal Register*, April 2026, *91*, 17384–17602.

CMS, "2026 Annual Report of the Boards of Trustees of the Federal Hospital Insurance and Federal Supplementary Medical Insurance Trust Funds," Technical Report, Centers for Medicare & Medicaid Services, Washington, DC June 2026.

Cubanski, Juliette and Anthony Damico, "Medicare Part D Enrollment, Premiums, and Cost Sharing in 2026," June 2026.

Curto, Vilsa E., "Pricing Regulations in Individual Health Insurance: Evidence from Medigap," *Journal of Health Economics*, September 2023, *91*, 102785.

Dulleck, Uwe and Rudolf Kerschbamer, "On Doctors, Mechanics, and Computer Specialists: The Economics of Credence Goods," *Journal of Economic Literature*, February 2006, *44* (1), 5–42.

_ , _ , and Matthias Sutter, "The Economics of Credence Goods: An Experiment on the Role of Liability, Verifiability, Reputation, and Competition," *American Economic Review*, April 2011, *101* (2), 526–555.

Egan, Mark, "Brokers versus Retail Investors: Conflicting Interests and Dominated Products," *The Journal of Finance*, 2019, *74* (3), 1217–1260.

Ericson, Keith M. Marzilli, "Consumer Inertia and Firm Pricing in the Medicare Part D Prescription Drug Insurance Exchange," *American Economic Journal: Economic Policy*, 2014, *6* (1), 38–64.

Farber, Henry S., "Why You Can't Find a Taxi in the Rain and Other Labor Supply Lessons from Cab Drivers*," *The Quarterly Journal of Economics*, November 2015, *130* (4), 1975–2026.

Fichtner, Jason J., "The Peak 65 Zone Is Here: Creating a New Framework for America's Retirement Security," White Paper, Alliance for Lifetime Income, Retirement Income Institute January 2024.

Freed, Meredith, Alex Cottrill, Jeannie Fuglesten Biniek, and Tricia Neuman, "What Do People with Medicare Think About the Role of Marketing, Shopping for Medicare Options, and Their Coverage?," Technical Report, KFF September 2023.

_ , Nancy Ochieng, Juliette Cubanski, and Tricia Neuman, "Key Facts About Medigap Enrollment and Premiums for Medicare Beneficiaries," Technical Report, KFF October 2024.

Gabaix, X. and D. Laibson, "Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets," *The Quarterly Journal of Economics*, May 2006, *121* (2), 505–540.

Gaffney, Adam, Stephanie Woolhandler, and David U. Himmelstein, "Less Care at Higher Cost-The Medicare Advantage Paradox," *JAMA internal medicine*, August 2024, *184* (8), 865–866.

Gennaioli, Nicola, Andrei Shleifer, and Robert Vishny, "Money Doctors," *The Journal of Finance*, 2015, *70* (1), 91–114.

Gneezy, Uri, John A. List, and Michael K. Price, "Toward an Understanding of Why People Discriminate: Evidence from a Series of Natural Field Experiments," NBER Working Paper 17855, National Bureau of Economic Research 2012.

Green, Paul E. and V. Srinivasan, "Conjoint Analysis in Marketing: New Developments with Implications for Research and Practice," *Journal of Marketing*, October 1990, *54* (4), 3–19.

Gruber, Jonathan, Benjamin R. Handel, Samuel H. Kina, and Jonathan T. Kolstad, "Managing Intelligence: Skilled Experts and AI in Markets for Complex Products," April 2020.

Gunja, Munira Z., Gretchen Jacobson, Faith Leonard, and Reginald D. Williams II, "Health Care Affordability for Older Adults: How the U.S. Compares to Other Countries," Technical Report, The Commonwealth Fund 2024.

Handel, Benjamin and Jonathan Kolstad, "Health Insurance for "Humans": Information Frictions, Plan Choice, and Consumer Welfare," *American Economic Review*, August 2015, *105* (8), 2449–2500.

_ **and Joshua Schwartzstein**, "Frictions or Mental Gaps: What's Behind the Information We (Don't) Use and When Do We Care?," *Journal of Economic Perspectives*, February 2018, *32* (1), 155–178.

Handel, Benjamin R, "Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts," *American Economic Review*, December 2013, *103* (7), 2643–2682.

Heiss, Florian, Daniel L. McFadden, Joachim K. Winter, Amelie C. Wuppermann, and Bo Zhou, "Inattention and Switching Costs as Sources of Inertia in Medicare Part D," *American Economic Review*, 2021, *111* (9), 2737–2781.

Higuera, Lucas, Caroline S. Carlin, and Bryan Dowd, "Narrow Provider Networks and Willingness to Pay for Continuity of Care and Network Breadth," *Journal of Health Economics*, July 2018, *60*, 90–97.

Horowitz, Joel L., "Bootstrap Methods in Econometrics," *Annual Review of Economics*, August 2019, *11* (1), 193–224.

Hortaçsu, Ali and Chad Syverson, "Product Differentiation, Search Costs, and Competition in the Mutual Fund Industry: A Case Study of S&P 500 Index Funds*," *The Quarterly Journal of Economics*, May 2004, *119* (2), 403–456.

Inderst, Roman and Marco Ottaviani, "Misselling through Agents," *American Economic Review*, May 2009, *99* (3), 883–908.

_ **and** _ , "Financial Advice," *Journal of Economic Literature*, June 2012, *50* (2), 494–512.

Jacobson, Gretchen A. and David Blumenthal, "Medicare Advantage Enrollment Growth: Implications for the US Health Care System," *JAMA*, June 2022, *327* (24), 2393–2394.

Karakoç, Gülen, Marco Pagnozzi, Salvatore Piccolo, and Giovanni Walter Puopolo, "Information Acquisition and Financial Advice," *Economic Modelling*, December 2024, *141*, 106891.

Kessler, Judd B., Corinne Low, and Colin D. Sullivan, "Incentivized Resume Rating: Eliciting Employer Preferences without Deception," *American Economic Review*, November 2019, *109* (11), 3713–3744.

Ketcham, Jonathan D., Claudio Lucarelli, and Christopher A. Powers, "Paying Attention or Paying Too Much in Medicare Part D," *American Economic Review*, January 2015, *105* (1), 204–33.

Kézdi, Gábor and Robert J. Willis, "Expectations, Aging, and Cognitive Decline," in David A. Wise, ed., *Discoveries in the Economics of Aging*, Chicago: University of Chicago Press, 2014, pp. 305–340.

Kim, Hugh Hoikwang, Raimond Maurer, and Olivia S. Mitchell, "How Financial Literacy Shapes the Demand for Financial Advice at Older Ages," *The Journal of the Economics of Ageing*, 2021, *20*, 100329.

Kline, Patrick, Evan K Rose, and Christopher R Walters, "Systemic Discrimination Among Large U.S. Employers*," *The Quarterly Journal of Economics*, November 2022, *137* (4), 1963–2036.

Leonard, Faith, Gretchen Jacobson, Lauren A. Haynes, and Sara R. Collins, "Traditional Medicare or Medicare Advantage: How Older Americans Choose and Why," *The Commonwealth Fund*, 2022.

_ , _ , **Michael Perry, Sean Dryden, and Naomi Mulligan Kolb**, "The Challenges of Choosing Medicare Coverage: Views from Insurance Brokers and Agents," Technical Report, The Commonwealth Fund February 2023.

Linnainmaa, Juhani T., Brian T. Melzer, and Alessandro Previtero, "The Misguided Beliefs of Financial Advisors," *The Journal of Finance*, 2021, *76* (2), 587–621.

Lusardi, Annamaria, Olivia S. Mitchell, and Vilsa Curto, "Financial Literacy and Financial Sophistication in the Older Population," *Journal of Pension Economics & Finance*, 2014, *13* (4), 347–366.

Malmendier, Ulrike and Devin Shanthikumar, "Are Small Investors Naive about Incentives?," *Journal of Financial Economics*, August 2007, *85* (2), 457–489.

Mazzonna, Fabrizio and Franco Peracchi, "Are Older People Aware of Their Cognitive Decline? Misperception and Financial Decision-Making," *Journal of Political Economy*, June 2024, *132* (6), 1793–1830.

McGuire, Thomas G and Mark V Pauly, "Physician Response to Fee Changes with Multiple Payers," *Journal of Health Economics*, 1991, *10* (4), 385–410.

Medicare Payment Advisory Commission, "Report to the Congress: Medicare Payment Policy — Chapter 11: The Medicare Advantage Program Status Report," Technical Report, MedPAC March 2025.

Meyers, David J., Jay Shroff, Jeffrey Marr, Em Balkan, Andrew M. Ryan, and Amal N. Trivedi, "Trends in Broker Enrollment and Spending in Medicare Advantage," *JAMA Internal Medicine*, July 2026, *186* (7), 902.

Mullainathan, Sendhil, Markus Noeth, and Antoinette Schoar, "The Market for Financial Advice: An Audit Study," Technical Report w17929, National Bureau of Economic Research, Cambridge, MA March 2012.

National Association of Insurance Commissioners, "Model Regulation to Implement the NAIC Medicare Supplement Insurance Minimum Standards Model Act," Model Regulation 651, National Association of Insurance Commissioners October 2008.

Ochieng, Nancy, Gabrielle Clerveau, Juliette Cubanski, and Tricia Neuman Published, "A Snapshot of Sources of Coverage Among Medicare Beneficiaries," December 2023.

_ , Juliette Cubanski, and Anthony Damico, "Medicare Households Spend More on Health Care Than Other Households," Technical Report, KFF March 2024.

Park, Sungchul, David J. Meyers, and Brent A. Langellier, "Rural Enrollees In Medicare Advantage Have Substantial Rates Of Switching To Traditional Medicare," *Health Affairs*, March 2021, *40* (3), 469–477.

Pitchik, Carolyn and Andrew Schotter, "Honesty in a Model of Strategic Information Transmission," *The American Economic Review*, 1987, *77* (5), 1032–1036.

Robles-Garcia, Claudia, "Competition and Incentives in Mortgage Markets: The Role of Brokers," *Unpublished working paper*, 2019.

Self, Steven G. and Kung-Yee Liang, "Asymptotic Properties of Maximum Likelihood Estimators and Likelihood Ratio Tests under Nonstandard Conditions," *Journal of the American Statistical Association*, June 1987, *82* (398), 605–610.

Semenova, Vira, "Generalized Lee Bounds," *Journal of Econometrics*, September 2025, *251*, 106055.

United States Office of Management and Budget, "Rescission and Reinstatement of Circular A-4 and Circular A-94," Technical Report M-25-15, Office of Management and Budget, Executive Office of the President, Washington, DC 2025.

USC Schaeffer Center, "Improving Access to Medigap When Beneficiaries Leave Medicare Advantage," April 2024.

Vabson, Boris, Andrew L. Hicks, and Michael E. Chernew, "Medicare Advantage Denies 17 Percent Of Initial Claims; Most Denials Are Reversed, But Provider Payouts Dip 7 Percent," *Health Affairs*, June 2025, *44* (6), 702–706.

Wolinsky, Asher, "Competition in a Market for Informed Experts' Services," *The RAND Journal of Economics*, 1993, *24* (3), 380–398.

Appendix

A Medicare Market Details

Medicare Supplement (Medigap) Plans Traditional Medicare covers 80% of the cost of most outpatient appointments and procedures. Beneficiaries are responsible for the remaining 20%. There is no limit to the 20% coinsurance expenses (that is, no maximum out-of-pocket limit, or MOOP). In exchange for a fixed monthly premium, Medigap plans help fill these "gaps" by paying for a portion of these cost-sharing expenses. Figure A.1 shows the standardized Medigap plans by plan letter.

Figure A.1: Standard Medicare Supplement (Medigap) Plans

Medigap Benefit	Plan A	Plan B		Plan D		Plan G*	Plan K	Plan L	Plan M	Plan N
Part A coinsurance and hospital costs up to an additional 365 days after Medicare benefits are used up	✓	✓		✓		✓	✓	✓	✓	✓
Part B coinsurance or copayment	✓	✓		✓		✓	50%	75%	✓	✓***
Blood (first 3 pints)	✓	✓		✓		✓	50%	75%	✓	✓
Part A hospice care coinsurance or copayment	✓	✓		✓		✓	50%	75%	✓	✓
Skilled nursing facility care coinsurance	×	×		✓		✓	50%	75%	✓	✓
Part A deductible	×	✓		✓		✓	50%	75%	50%	✓
Part B deductible	×	×		×		×	×	×	×	×
Part B excess charge	×	×		×		✓	×	×	×	×
Foreign travel exchange (up to plan limits)	×	×		80%		80%	×	×	80%	80%
Out-of-pocket limit**	N/A	N/A		N/A		N/A	$6,940 in 2023	$3,470 in 2023	N/A	N/A

Source: Medicare.gov

Broker disclosure requirements Brokers must disclose that they do not represent all insurers and all Medicare plans and tell the senior that information on all available options is available from Medicare.gov, 1-800-MEDICARE, or the free State Health Insurance Assistance Program (SHIP) (42 C.F.R. § 422.2267(e)(41); CMS, 2023). Brokers are also required to review each client's doctors, drugs, and costs before enrollment (CMS, 2025b). Note that these are the rules in place during our audit, which ran from September 2024 to March 2026, some of which were amended in Contract Year 2027. For example, brokers are no longer required to mention SHIP resources and starting in Contract Year 2027 call recordings must be stored for up to 6 years instead of the original 10 years (CMS, 2026).

B Study Details

Figure A.2: Medicare Plan Characteristics vs Senior Preferences

Medicare Advantage (MA) Pros	Senior Preferences
Low cost: ▪Plans usually less than $100/month ▪Some low-income beneficiaries quality for $0 premium plans	▪Main priority is having good insurance coverage ▪Willing to spend more money for good coverage ▪No SSI or food stamps *(does not quality for extra discounts)*
Drug Plan: Most plans include Medicare drug coverage	▪Not a priority - no regular prescriptions (only multivitamins)
May include dental, vision, and hearing	▪Not a priority - have pretty good vision and hearing
Medicare Advantage (MA) Cons	**Senior Preferences**
Prior authorization: May need approval from your plan before it covers certain services or supplies.	▪Do not want to deal with prior authorization
Seeing specialists: May need a referral before seeing a specialist.	▪Want to be able to see a specialist if needed
Network restriction: in many cases, you can only use doctors and hospitals in the plan's network (for non-emergency or non-urgent care).	▪Do not want to worry about network restrictions ▪Live in a rural area area *(more limited networks)*
Travel: maybe harder to get coverage outside of where you live due to network restrictions	▪Spend 4-5 months of the year in a rural area of a different state

Notes: Figure A.2 lists the preferences and characteristics assigned to the "Medigap-fit" shopper beside the plan attributes on which Medicare Advantage and Traditional Medicare plus Medigap differ. Each assigned preference corresponds to a reason Medicare.gov gives for not choosing Medicare Advantage.

Table A1: Broker and Senior Characteristics

	Obs.	Mean	SD	Q1	Q3
Broker Characteristics					
Call-Center Brokers	637	0.40	0.49	0.00	1.00
Broker - Female	637	0.43	0.50	0.00	1.00
# Company Appointments (Group)	624	8.42	4.29	5.00	12.00
Broker Tenure (yrs)	634	9.40	8.10	3.78	13.11
High Tenure (4+)	637	0.70	0.46	0.00	1.00
Consumer Characteristics					
Caller - Female	637	0.83	0.38	1.00	1.00
Caller - Real Age	637	71.18	6.28	67.00	74.00
Caller Race - White	637	0.81	0.39	1.00	1.00
College Degree	637	0.59	0.49	0.00	1.00

B.1 Audit Study Logistics

Shopper Recruitment and Screening We hire and train seniors from two secret shopping firms. Secret shoppers are all senior citizens and most are over 65 years old and therefore Medicare-eligible. We use two firms because each could reach a different part of our broker sample: one firm could mask shopper phone numbers, which suited it to contacting call center brokers, while the

Table A2: Call Center v Non-Call-Center Brokers

Broker Characteristics	All Mean (SE)	Call-Center Brokers Mean (SE)	Non-Call-Center Brokers Mean (SE)	Difference Δ	Reg P-value (cntrl)
# States Registered In	23.4	32.5	17.4	15.1	0.00
	(0.6)	(0.7)	(0.6)		
# Insurance Comps Appointed To (Group)	8.4	12.8	5.4	7.4	0.00
	(0.2)	(0.1)	(0.1)		
Broker Tenure (yrs)	9.4	5.8	11.8	-6.0	0.00
	(0.3)	(0.4)	(0.4)		
High Tenure	70.0%	51.8%	82.4%	-30.6	0.00
	(1.8)	(3.1)	(2.0)		
% Broker Female	43.3%	42.8%	43.7%	-0.9	0.77
	(2.0)	(3.1)	(2.5)		
N (Medigap Recs)	1,055	373	682		
N (Calls)	637	257	380		

Notes: Table A2 compares broker characteristics for brokers reached through a call center and brokers reached through the online broker directory. The first three columns report means with standard errors in parentheses, and the Difference column is the call center mean less the non-call-center mean. The final column reports the *p*-value on the call center indicator from a controlled regression of the row characteristic.

other could pre-schedule calls, which suited it to contacting brokers drawn from the large online directory.

Shoppers are never told the purpose of the study. Before a shopper is assigned any broker to contact, the shopper must pass an assessment demonstrating familiarity with the call script and a basic understanding of Medicare. The content of that training is described under the Caller Training section below.

We also collect demographic information on each shopper's real age, gender, race, and education level, and we include these consumer characteristics in our baseline controls. Potential discrimination is not our primary treatment of interest, but we use the natural variation in shopper characteristics to test whether recommendation quality differs across seniors. Because we randomly assign both brokers and seniors at the individual call level, any correlation between recommendation quality and shopper characteristics carries a causal interpretation. We do not enforce balance on gender and race, given the cost of recruiting a balanced sample.

Caller Training We also provide the seniors with some basic facts about the Medicare market that many Medicare-eligible seniors may already know (for example, the definition of Traditional Medicare and Medicare Advantage, what a Red White and Blue card is, who is eligible for Medicare). We also have shoppers pass a comprehension check quiz to ensure they are familiar with the script and to ensure the shoppers would be able to accurately document the broker recommendations. The goal of the pre-call assessment and script was to keep the content of the calls as consistent as possible, aside from our experimentally assigned treatments.

Broker Sampling and the Licensure Database We sample brokers from two sources commonly used by seniors. First, we sample from two major Medicare call centers. Combined, the two call centers in our study enrolled over one million seniors in Medicare plans in 2024, a figure taken from the public financial records of the companies. Each employs over 1,000 brokers and runs extensive television, mail, and internet advertising campaigns. Second, we sample from the nation's largest directory of Medicare insurance brokers, which included over 18,000 brokers as of June 2025. We stratify the sampling to ensure a mix of brokers from both sources in both the treatment and control groups. As noted in the text, unforeseen changes in broker commissions forced us to halt data collection for call center brokers prematurely, which left a smaller call center treatment group.

We sample brokers licensed in both California and Texas, the two states in which we assign our shoppers residency. Because the majority of brokers are licensed in more than one state, we are not concerned that this procedure yields a selected sample. Between 33% and 37% of California brokers in the large broker directory are also licensed in Texas, and between 18% and 19% of Texas brokers in the directory are also licensed in California, depending on the exact date the broker data were pulled.

To understand which broker characteristics correlate with advice quality, we construct a broker licensure database from insurance licensure data covering 47 states and match our audit brokers to it. The database gives us broker tenure, that is, how long the broker has held a health insurance license; firm of employment, including call center versus non-call-center status; and the number of companies the broker is licensed to sell. In several states, including both California and Texas, an insurance broker must be appointed with a company before selling that company's plan. Appointments are filed with the state Department of Insurance, which allows us to observe a broker's current appointments. Both California and Texas also allow just-in-time appointments, so a broker can become appointed quickly before making a sale.

Geographic Choice We intentionally choose two states with different rules on when seniors can change Medigap plans. California has a "Birthday Rule" that lets an individual with a Medigap plan switch each year around their birthday, without medical underwriting, to another plan with the same or fewer benefits; Texas has no such rule. Birthday Rules are spreading: as of December 2025, 10 states had one, and more plan to adopt it. In principle, the rule makes it easier to keep the senior in the cheapest plan: if the plan's premium rises, which it can do once a year, a broker can move the senior to a cheaper plan. In practice, the Medigap markups brokers recommend are similar for the California and Texas plans, so the rule does not appear to change the initial recommendation. We pool the two states and include state fixed effects as a control.

Since Medicare plan coverage varies by zip code, we intentionally select zip codes in counties that (1) are not in urban areas, (2) have a normal amount of Medigap price dispersion, and (3) have

demographic characteristics that do not deviate significantly from national averages on income and education.

Focusing on less urban areas, especially for an audit consumer who has providers in the rural areas in more than one state, raises the potential consumer surplus loss from an MA plan since rural areas tend to have fewer provider networks to choose from and result in longer travel and greater difficulty accessing care. It also means the zip codes we choose offer fewer MA plans than urban areas do, which further limits the choice set for a consumer who seeks flexible care.

Call Script and Survey Instruments Each broker is asked to make two decisions, which correspond to the two main decisions faced by seniors enrolling in Medicare. The first decision asks brokers to choose between Medicare Advantage (MA) and Traditional Medicare plus a Medicare Supplement (Medigap) plan. The second decision asks brokers to recommend a Medigap plan for the provided zip code. These two decisions correspond to our two main outcomes of interest.

The calls proceed as follows. First, the senior explains that they have just become Medicare-eligible and are interested in discussing both MA and Medigap plans. We intentionally ask seniors to highlight that they want to discuss both Medicare plan types at the beginning of the call so the broker is prepared to provide two recommendations. This also reduces the risk that brokers interpret the discussion of a different plan as a rejection of their plan recommendation. Seniors then provide their assigned characteristics and coverage preferences and ask the broker whether they would recommend MA or TM + Medigap. Whether or not the broker recommends MA at this point in the conversation is our first primary outcome (Outcome 1).[63]

Seniors are specifically instructed *not* to explicitly accept or reject the first plan recommendation before redirecting the conversation to ask about Medigap plans. Seniors ask the broker for a Medigap plan recommendation for the two most popular Medigap plans, Plan N and Plan G. These Medigap plan recommendations are our second main outcome of interest. We also intentionally design the scenario so the senior travels back and forth between two states, CA and TX, for months at a time. This allows seniors to ask about the Medigap plan premiums in each state, a design which gives us up to four Medigap plan recommendations per call.

The shopper collects the name of the insurance company that provides the Medigap plan and the quote of the monthly premium provided by the broker. To ensure that we use consistent and accurate pricing, we match the Medigap plan recommended by the broker to the Medigap search tool on Medicare.gov to determine the true premium available to the senior for the recommended

[63]As specified in the pre-analysis plan, responses where seniors select "Other" and describe the recommendation as ambiguous or explain that the broker said it was the senior's decision are coded as not recommending MA to be conservative. After reviewing the data, we found this to be appropriate since all responses where the seniors select "Other" correspond to cases where the broker initially talked about MA, but ultimately recommended Traditional Medicare plus Medigap.

plan without improperly applied discounts.[64] We compare this premium to the lowest-cost premium available to the senior for a plan with identical coverage to calculate the Medigap markup, or Outcome 2.[65]

One potential confound is that the Medigap plan recommendation comes *after* the initial Medicare Advantage versus Traditional Medicare recommendation. To mitigate this concern, we instruct all seniors to remain intentionally vague about their decision. Specifically, seniors are instructed to not say yes or no to a recommendation. Instead, they thank the broker, say they will think about it more, and move on to the next question. We also have seniors outline the topics of discussion at the beginning of the call so redirecting the conversation to Medicare Supplement is not seen as a rejection of the broker's recommendation. This may signal senior sophistication and a strong understanding of the Medicare market, which may bias the broker recommendations toward better recommendations. For this and other reasons discussed in Section 5.1, we believe our figures may represent an *underestimate* of unsuitable recommendations.

To try to keep calls as standardized as possible, we assign consumer preferences and design a call script for the secret shoppers to follow when contacting brokers. The call scripts are based on several real conversations between Medicare insurance brokers and beneficiaries. The script was developed after interviewing multiple Medicare beneficiaries and currently and previously licensed Medicare insurance brokers.

Secondary Outcome Definitions We report two secondary measures alongside the two primary outcomes. The first is the Medigap Percent Markup, which expresses the Medigap Markup as a share of the annual premium of the cheapest coverage-identical plan available to the consumer. This measure gives us a sense of the overpayment in relative terms and allows us to compare the Medigap markup to the markup found in other markets.

The second is the Medigap Broker Markup, which asks how much of the markup survives once we hold the broker's own choice set fixed. It compares the premium of the recommended Medigap plan to the lowest-premium coverage-identical plan available to the senior that the broker is currently licensed to sell.[66] The measure is coded as missing when the broker does not name a specific Medigap plan on the call, when the broker cannot be identified, and when appointment

[64] As specified in the pre-analysis plan, we use the publicly available Medicare.gov premium to avoid comparisons where brokers apply household discounts or short-term promotional pricing that the senior is not eligible for when quoting the premium in order to make the sale. We use a combination of the insurance company name and the premium to match. Additional details about the plan matching process are provided in the pre-analysis plan.

[65] We code non-standard Medigap plans as missing. Non-standard Medigap plan recommendations are rare and only happen 1% of the time in our sample.

[66] We collect these data from the publicly available state department of insurance licensure databases. California and Texas both require the broker to be appointed with an insurance company before selling that company's plan, and appointments are reported to the state Department of Insurance. Depending on match quality, we match insurance companies at the parent or the subsidiary level.

data are unavailable.[67]

Data Collection We have surveyors fill out an information collection form within three business days of completing the call, although most shoppers fill out the form within 24 hours. If shoppers do not submit the survey instrument within 8 hours they get a message reminding them to submit the form. Data collection requires surveyors to fill out a form recounting the recommendations made on their calls and explanations or statements made by the broker during the call. We also collect a call recording, which is converted to a call transcript, when possible.

B.2 Natural experiment: plan availability and design constraints

When we began data collection, three insurance companies offered MA plans in our California zip code: two national carriers, Anthem and Aetna, and one California-focused carrier, Imperial Health. In October 2024, Anthem announced that it would stop paying MA commissions in several California regions, including our zip code, effective November 1, 2024. Since this was too late to remove the plans from the market for 2025, Anthem plans remained in the market through December 31, 2025.

Between May and July 2025, several large insurance companies, including Anthem Blue Shield of California and UnitedHealthcare, announced similar commission cuts for MA plans in other California counties. These changes drew significant attention from state governments and from Medicare broker blogs and forums.[68] While the changes between May and July 2025 did not directly affect the MA plans in our California audit zip code, the prevalence of commission cuts throughout California may have made the previous Anthem commission change and future cuts more salient to brokers.

Finally, Aetna announced that it would stop paying MA commissions after September 1, 2025. At this point, no national insurance carriers, and as a result no carriers that contracted with call centers, paid broker commissions within our audit study region.

Throughout the audit study window at least one MA plan was always available for enrollment. Prior to September 1, 2025, there was also at least one national carrier that paid broker commissions: Medicare.gov publishes the available plans for the upcoming plan year on October 1 and the Annual Enrollment Period opens October 15, meaning brokers could earn a commission on an Aetna enrollment from October 15, 2024 onward. Anthem's commission cut took effect on November 1, 2024, but Anthem's commissions ran through October 31.

[67] California's appointment records appear less complete than Texas's, and active Medicare brokers are often missing appointments there. Brokers can also become appointed with a company after deciding to make a sale. For both reasons we treat this as a secondary measure.

[68] See www.jrreport.wordandbrown.com/2025/07/01/blue-shield-of-california-cuts-medicare-advantage-broker-commissions; www.beckerspayer.com/payer/unitedhealth-cuts-some-medicare-advantage-broker-commissions.

Both the Anthem and Aetna plans remained available and continued to accept new enrollments through December 31, 2025. This means it was still possible for call center brokers to enroll consumers in those plans through the end of the year: the Anthem plan was available through December 2025 and the Aetna plans from October 1, 2024 through December 2025. MA plans were still available to consumers after January 1, since two of the four Imperial Health Medicare Advantage plans remained available for the entire duration of the audit study. A broker who checked Medicare.gov would have seen all three carriers' plans.

We confirmed that Imperial Health does not work with the national call centers by checking the California Health Insurance License database, which requires companies and individuals to be appointed with any carrier whose products they sell. Neither of the two major call centers we sample from was licensed to sell Imperial Health products at the time of the commission change or in the three months after.

Note that when an MA carrier leaves the market unexpectedly, its enrollees receive a new enrollment window in which they may choose any MA plan without medical underwriting or the risk of denial. This means enrolling in an MA plan that will leave the market in 1-2 months may still be the best choice for the senior. The senior can receive the benefits of this (potentially underpriced) MA plan for a few months, then enroll in whichever Medicare plan they choose without medical underwriting once the plan leaves the market.

Ideally, we would have also collected post-September call-center data in Texas as a control group. However, this was not feasible in the field. Call-center brokers required shoppers to provide a phone number during each call (and often required the number to be verified by text). Each call with the shopper's phone number was then logged in the shared firm-wide system and each additional call raised the risk that a call center would become suspicious.[69] Thus, our pool of roughly one hundred shoppers set a ceiling on the number of call-center calls we could place during the post-MA commission removal window. We allocated all of these calls to California, where the commission removal binds, so that the before and after comparison for California call centers would be adequately powered. Dividing these calls between the two states would have left the analysis underpowered.[70]

B.3 Experiment B: Additional Details

At the beginning of the phone call, shoppers assigned to the perceived competition treatment say one of the following:

[69]We considered using Google Voice numbers, but the secret-shopper firm we partnered with advised against it since, in their experience, these phone numbers were often screened or blocked by similar firms.

[70]The seven Texas call center calls in September occurred before the secret shopper firm had a chance to cancel the calls that had already been assigned.

"I want to get a few quotes from different brokers before making a decision"

"I want to shop around before I choose a plan"

"I want to get a few recommendations from other brokers before I choose a plan"

Shoppers in the control group are instructed to say that they want a recommendation from the broker to better understand their options, or to say nothing at all. The signal is delivered at the start of the call, usually in the first sentence, to make it salient.

The treatment does not vary market structure or manipulate market-level broker supply. It works by lowering the broker's perceived probability of making the sale, which makes the risk of losing the client salient in a market that is already competitive. Finding a Medicare broker is inexpensive for a consumer: an online search returns hundreds of results and multiple toll-free numbers that a beneficiary can call at almost any hour to reach a licensed broker. Brokers are aware of this at baseline. In interviews, several complained about competitors taking their clients, and many Medicare beneficiaries receive cold calls from brokers seeking their business.

Randomization is stratified by state and firm type, and brokers are assigned to treatment or control before any calls are placed. Calls are then randomly assigned to shoppers at the individual level. We implement the treatment and control conditions in waves, at different points in time, to reduce shopper confusion and mistakes: when we originally assigned both conditions simultaneously, we found compliance problems, because our senior shoppers had difficulty remembering which condition applied to a given call. Because the arms were fielded at different times, Appendix C.6 reports specifications that flexibly control for caller experience and calendar time.

Table A3: Competition Assignment - Balance Table of Broker and Consumer Characteristics

	All Mean (SE)	Competition Assigned Mean (SE)	Not Assigned Mean (SE)	Difference Δ	Reg P-value (strata)
% Competition Treated	41.6%	89.7%	9.9%	79.8	.
	(2.0)	(1.9)	(1.5)		
Broker Characteristics					
% Call-Center Brokers	40.3%	22.1%	52.3%	-30.2	.
	(1.9)	(2.6)	(2.6)		
# States Registered In	23.4	20.9	25.1	-4.2	0.61
	(0.6)	(0.9)	(0.7)		
# Insurance Comps Appointed To (Group)	8.4	7.1	9.3	-2.2	0.95
	(0.2)	(0.3)	(0.2)		
Broker Tenure (yrs)	9.4	10.4	8.7	1.7	0.92
	(0.3)	(0.5)	(0.4)		
High Tenure	70.0%	75.5%	66.4%	9.1	0.78
	(1.8)	(2.7)	(2.4)		
Broker Female	43.3%	41.5%	44.5%	-3.0	0.44
	(2.0)	(3.1)	(2.5)		
Caller Characteristics					
Caller Female	83.0%	84.2%	82.3%	1.9	0.63
	(1.5)	(2.3)	(2.0)		
Caller White	81.0%	87.0%	77.1%	9.9	0.26
	(1.6)	(2.1)	(2.1)		
Caller Black	6.9%	5.9%	7.6%	-1.6	0.54
	(1.0)	(1.5)	(1.4)		
Caller Real Age (yrs)	71.2	71.5	71.0	0.5	0.95
	(0.2)	(0.4)	(0.3)		
Caller Above Med Age	50.5%	55.3%	47.4%	7.9	0.53
	(2.0)	(3.1)	(2.6)		
Caller College Degree	58.9%	57.7%	59.6%	-1.9	0.95
	(2.0)	(3.1)	(2.5)		
Caller # Calls	19.9	27.8	14.8	13.0	0.10
	(1.3)	(2.4)	(1.3)		
N (Medigap Recs)	1,055	427	628		
N (Calls)	637	253	384		

Notes: Table A3 shows the balance table of broker and consumer characteristics by competition assignment.

Table A4: Competition Compliers vs Non-Compliers - Broker and Senior Characteristics

	All Mean (SE)	Competition Correct Mean (SE)	Competition Incorrect Mean (SE)	Difference Δ	Reg P-value (strata)
% Competition Treated	41.6%	39.6%	59.4%	-19.8	.
	(2.0)	(2.0)	(6.2)		
Broker Characteristics					
% Call-Center Brokers	40.3%	39.1%	51.6%	-12.5	.
	(1.9)	(2.0)	(6.3)		
# States Registered In	23.4	23.3	24.8	-1.5	0.93
	(0.6)	(0.6)	(1.7)		
# Insurance Comps Appointed To (Group)	8.4	8.3	9.2	-0.9	0.95
	(0.2)	(0.2)	(0.6)		
Broker Tenure (yrs)	9.4	9.6	7.5	2.2	0.11
	(0.3)	(0.3)	(0.7)		
High Tenure	70.0%	70.0%	70.3%	-0.3	0.38
	(1.8)	(1.9)	(5.8)		
Broker Female	43.3%	42.6%	50.0%	-7.4	0.34
	(2.0)	(2.1)	(6.3)		
Caller Characteristics					
Caller Female	83.0%	83.2%	81.3%	2.0	0.94
	(1.5)	(1.6)	(4.9)		
Caller White	81.0%	82.7%	65.6%	17.1	0.11
	(1.6)	(1.6)	(6.0)		
Caller Black	6.9%	6.8%	7.8%	-1.0	0.99
	(1.0)	(1.1)	(3.4)		
Caller Real Age (yrs)	71.2	71.4	69.1	2.3	0.17
	(0.2)	(0.3)	(0.5)		
Caller Above Med Age	50.5%	52.2%	35.9%	16.2	0.16
	(2.0)	(2.1)	(6.0)		
Caller College Degree	58.9%	59.0%	57.8%	1.2	0.92
	(2.0)	(2.1)	(6.2)		
Caller # Calls	19.9	21.3	7.9	13.4	0.19
	(1.3)	(1.4)	(1.4)		
N (Medigap Recs)	1,055	980	75		
N (Calls)	637	573	64		

Notes: Table A4 shows the balance between the compliers and non-compliers on broker and senior characteristics.

B.4 Understanding America Study (UAS) Senior Survey and Additional Results

To evaluate the external validity of our audit study design, we survey a nationally representative panel of 2,015 Medicare beneficiaries through the Understanding America Study (UAS). The goal of the survey is to better understand broker use and Medicare enrollment and to test the external validity of our audit study design. Our survey population includes seniors who are 65 years of age or older and individuals with disabilities who have Medicare. We re-weight the survey responses to reflect the US population. This helps adjust for a slight oversampling of respondents from California, where the Center for Economic and Social Research at the University of Southern California (USC), which administers the study, is based.

Similar to previous surveys, we find that the majority of beneficiaries used a broker to enroll in Medicare. Table A6 reports broker use by Medicare coverage group: among beneficiaries who report neither Medicaid eligibility nor access to an employer plan, the population to which Section 7.2 scales the structural estimates, 56% used a broker. Figure A.3 highlights the different distribution channels that seniors use to search for and match with brokers. 51% of beneficiaries find their broker in existing networks (through word of mouth or prior relationships) and 49% of beneficiaries find their brokers through new networks (advertisements/TV/mailers, cold outreach where the broker reached out to the senior directly, online search, or "other" methods). We take this as evidence that our sampling method of finding brokers through commonly advertised distribution channels is not unusual.

We also ask whether seniors talk to multiple brokers before enrolling in a Medicare plan. Of the seniors who enrolled in Medicare using a broker, 66% of seniors spoke to only one broker before enrolling in a plan and 7% "do not know" how many brokers they spoke with. While this does not directly test whether *telling* a broker you will speak to their competitors is unusual, the survey results suggest that speaking with multiple brokers before enrolling is uncommon, but not implausible.

Limited search is surprising given how easy it is to find a broker. Brokers heavily advertise to seniors via mailers, cold-calls, online advertisements, and even in-person visits.[71] A broker can easily be reached by calling a call center number at almost any hour of the day or by scheduling an automated appointment online. Additionally, most initial Medicare consultation appointments are about 30 minutes. Since few seniors talk to more than one broker even though it is cheap and easy to do so, telling a broker that you plan to contact multiple brokers is realistic but uncommon. This makes it a meaningful signal that could change a broker's recommendation.

Finally, the results from our UAS survey highlight one of the benefits of using our audit

[71]Multiple brokers we spoke to reported competitors showing up at their clients' doorsteps and frequenting community centers to try to find clients.

study evidence over noisy survey evidence. We find that even among our sample of positively selected seniors who participate in a longitudinal research survey panel, at least 40% of our survey respondents answered questions about what Medicare plan they have or how much they pay in premiums incorrectly.[72] Our audit study methodology addresses these limitations by directly asking brokers to recommend plans in a specially designed "textbook" scenario where we assigned senior preferences, rather than relying on self-reported and potentially noisy survey data.

Table A5: Summary of UAS Survey Results

Panel A. Respondents unable to correctly identify current Medicare coverage

Survey Response	**Share**	**Base**
Medicare but does not know the specific type	10%	All respondents (2,015)
Medigap plan letter of "Other" or "Do not know"	32%	Medigap enrollees (719)
Medigap premium $\leq$ \$20/month	14%	Medigap enrollees (705)
Medicare Advantage (MA) premium $\geq$ \$100/month	34%	MA enrollees (1,042)
Fails at least one of the four tests above	$\geq$ 40%	All respondents (2,015)

Panel B. Broker use and brand preferences

Survey Response	**Share**	**Base**
Used a broker to enroll	51%	Answered item (1,900)
Consulted only one broker before enrolling	66%	Broker users (1,011)
Does not know how many brokers were consulted	7%	Broker users (1,011)
Has neither a preferred nor an avoided insurer	61%	Answered both items
Has no insurer they prefer not to work with	88%	Answered item (1,889)
Found broker through friends, family, or a prior relationship, household income $\leq$ \$60,000	58% (SE 2.37)	Broker users
Found broker through friends, family, or a prior relationship, household income $>$ \$60,000	69% (SE 2.58)	Broker users

Notes: Table A5 summarizes our July 2025 Understanding America Study (UAS) survey of Medicare beneficiaries. All shares are re-weighted to match the US population. Each row's base is the set of respondents who answered the question. Panel A rows are the response inconsistencies described in the text, so the last row is a lower bound on the share of respondents who cannot correctly identify their current Medicare coverage. In Panel B, the brand preference rows compare respondents who report a preferred or avoided insurer with respondents who report caring only about what the plan covers. The referral rows report the share of broker users who found their broker through word of mouth or a prior relationship, split at a household income of \$60,000. Standard errors treat the weighted sample as i.i.d.

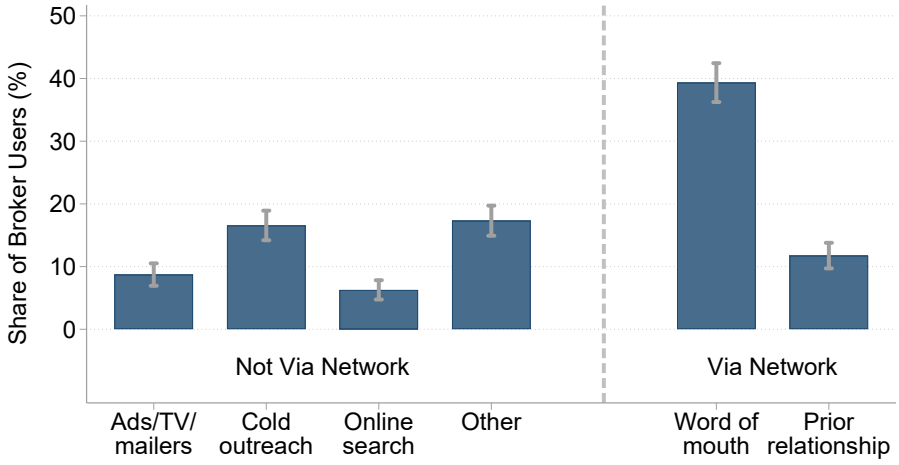
[72]The survey questions were carefully vetted by the Understanding America Study team and tested for clarity before the survey was conducted. We consider the following beneficiaries to be confused about what plan they have: beneficiaries who report having Medigap plans with premiums that are $\leq$\$20 per month (Medigap plans tend not to have monthly premiums this low, meaning the respondent likely has Medicare Advantage), beneficiaries who report having Medicare Advantage plans that are over \$100/month (MA plans are unlikely to have a premium that high), beneficiaries who report having Medigap plan letters that do not exist or who do not know what Medigap plan letter they have, and beneficiaries who answer "do not know" when asked which Medicare plan they have.

Table A6: Broker Use by Medicare Coverage Group (UAS)

Medicare population	Broker use	n	Share of reweighted sample
All	51.3% (SE 1.5)	1,875	100.0%
Medicaid-eligible	57.9% (SE 4.7)	182	12.0%
Employer access	20.8% (SE 3.1)	335	16.0%
Remaining	56.4% (SE 1.8)	1,371	72.8%

Notes: Table A6 reports the share of respondents to our July 2025 Understanding America Study (UAS) survey who say an agent helped them enroll in their current Medicare plan, by Medicare coverage group. "Not sure" counts as no. Groups come from a select-all question on Medicaid eligibility and access to Medicare plans through a current or prior employer. Medicaid eligibility is self-reported and does not separate full-benefit from partial-benefit dual eligibles. The employer row records access to an employer plan, not enrollment in one. 13 respondents report both and appear in both rows. The Remaining row includes the 76 respondents who report military or TRICARE health benefits. Shares use the survey's post-stratification weights, with linearized standard errors in parentheses. The last column is each group's share of the total weight. The sample is the 1,875 respondents who answered the broker question and carry a positive weight.

Figure A.3: Broker Distribution Channel (Broker Users Only)



Notes: Figure A.3 shows the reweighted share of Medicare beneficiaries who found their broker through each distribution channel. Distribution channel options included advertisements/TV/mailers, cold outreach (the broker reached out to the senior directly), online search, word of mouth, prior relationship, or other. Error bars show the 95% confidence intervals. Note that this may underestimate brokers found through distribution channels such as ads/TV, since many seniors who speak to brokers through them believe they are speaking to Medicare representatives and do not realize they are speaking to brokers.

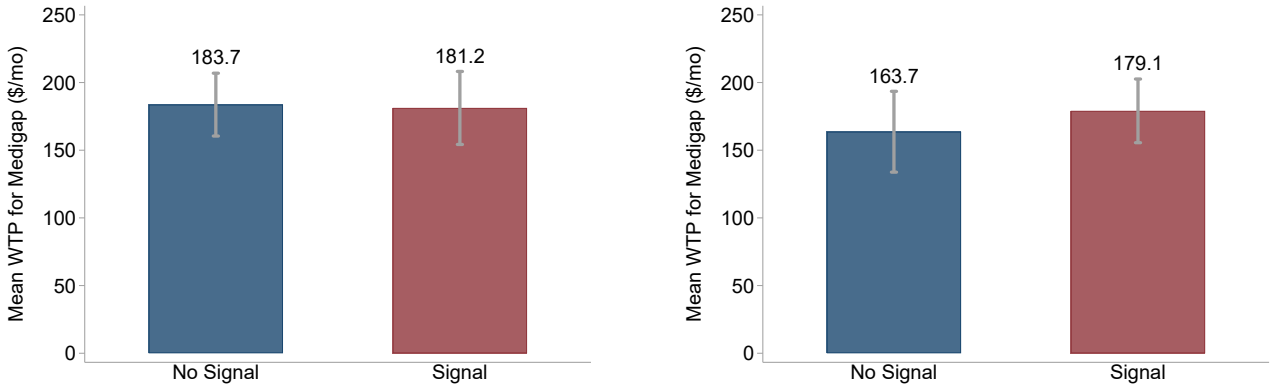
B.5 Broker Survey and Additional Survey Results

We supplement our audit study data with a survey of 157 currently licensed Medicare insurance brokers. To recruit brokers, we hire a Business-to-Business (B2B) market research firm to recruit a sample of licensed Medicare insurance brokers through their partner firms. To distribute the survey, the firm contacts insurance brokerage firms within its existing network of partner firms and asks the insurance brokerage firms to distribute the survey through their networks. The firm's survey portal automatically assigns each respondent a unique ID to prevent duplicate responses.[73]

[73]We also supplement data collection using the online survey platform Prolific and by directly messaging the brokers in our audit study and asking them to participate in our survey, although the majority of the responses are from the B2B provider.

We use the broker survey results to calibrate structural model parameters and as an additional test of the mechanisms contributing to broker advice.

Figure A.4: WTP by Perceived Competition Signal, by Vignette

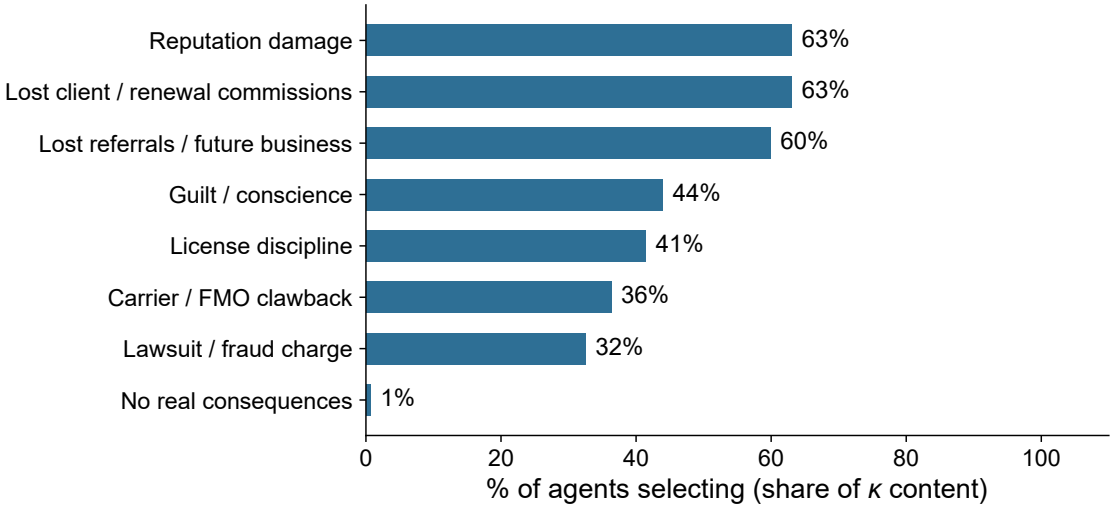


(a) Vignette 1 (b) Vignette 2

Notes: Figure A.4 shows the average maximum WTP for a Medigap plan ($/mo) by competition signal, using across-broker variation. Both Vignette 1 and Vignette 2 describe Medigap-fit seniors similar to the one in the audit study. These figures are the across-broker equivalent of Figure 5. Grey bars are ±1 SE.

Figure A.5 shows the self-reported consequences of a low-quality recommendation reported by currently licensed Medicare brokers in our broker survey. These can be interpreted as the contributors to the weight κ in Section 3.3.

Figure A.5: What a poor-fit recommendation costs the broker



Notes: Figure A.5 shows the share of brokers who select each response to the question "Think about the typical broker in your market. If a broker gives a client a Medicare recommendation that turns out to be a poor fit (e.g., the client overpays or has coverage problems), what are the worst things that can happen to the broker?" Respondents could select any number of options, so shares do not sum to 100 percent. Bar labels show the abbreviated survey options. "Carrier / FMO clawback" stands for "Carrier/FMO action (broker is dropped, or commissions is [*sic*] clawed back)." FMO stands for field marketing organization. "License discipline" refers to disciplinary action by a regulator where the broker's license is suspended or revoked. The figure omits the write-in option "Other (please specify)."

C Additional Results and Robustness Checks

This appendix reports the full regression tables behind two exhibits in Section 5.

Table A7: Predictors of Recommendation Quality

	(1) Misaligned Rec (%)	(2) Rec MA (%)	(3) Medigap Markup	(4) Medigap Broker Markup	(5) Medigap % Markup
Caller - Female	-3.26	-4.68	16.22	26.06	3.77
	(4.73)	(4.41)	(29.80)	(29.27)	(2.49)
Caller - Real Age	4.65	1.60	33.42	21.03	2.12
	(6.45)	(5.97)	(36.37)	(39.44)	(2.68)
Caller - Real Age^2	-0.04	-0.01	-0.25	-0.16	-0.02
	(0.04)	(0.04)	(0.27)	(0.29)	(0.02)
Caller Race - White	2.77	-6.41	34.79	25.02	2.44
	(4.96)	(5.45)	(22.85)	(24.26)	(1.77)
Caller - College Degree	-0.50	9.16*	-29.96	-24.26	-3.29*
	(3.39)	(5.41)	(23.08)	(22.89)	(1.81)
Caller Experience (# Calls)	-0.12*	-0.12	0.76	0.96	0.04
	(0.06)	(0.08)	(0.54)	(0.63)	(0.04)
Call-Center Brokers	0.09	16.80***	-4.65	178.06***	-0.49
	(5.00)	(4.85)	(19.20)	(19.27)	(1.46)
Broker - Female	-1.86	3.23	-35.94**	-32.67	-3.18**
	(3.77)	(3.72)	(17.74)	(21.89)	(1.36)
High # Ins Comps	-16.18***	-9.55**	-61.61***	71.15***	-4.83**
	(3.96)	(3.91)	(20.97)	(21.86)	(1.96)
High Tenure (4+ yrs)	-11.56***	-10.27**	-77.04***	-46.10**	-6.46***
	(4.34)	(4.38)	(19.36)	(21.70)	(1.54)
State/Plan/AEP FE	Yes	Yes	Yes	Yes	Yes
Observations	637	637	1,055	1,002	1,055
R^2	0.11	0.16	0.16	0.16	0.23

Notes: Table A7 shows the full table of recommendation quality by broker and senior covariates. Each column is one regression on the baseline controls, and the reported coefficients are that regression's broker and caller controls. Standard errors, in parentheses, are clustered at the caller level. $^{*}p < 0.10,^{**} p < 0.05,^{***} p < 0.01$.

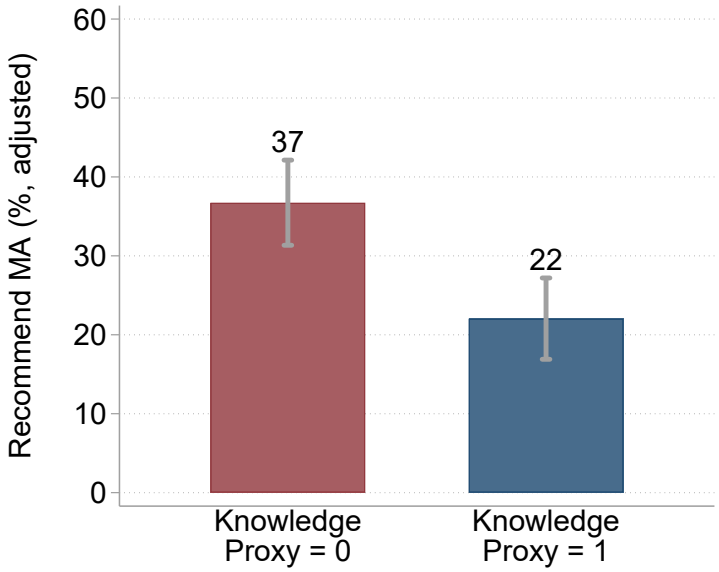
Figure A.6 reports the adjusted margins behind the knowledge-proxy comparison of Section 5.5.

C.1 Calculating lost consumer surplus from an MA Recommendation

This appendix gives the compensating variation behind the annual and lifetime consumer surplus losses reported in Section 5.1 for Outcome 1.

We measure the consumer surplus cost of steering a Medigap-fit senior i into a \$0-premium MA plan as a compensating variation: the monthly dollar transfer that would leave senior i indifferent between the two plans. MA and Traditional Medicare + Medigap (TM+MG) plans differ on a set of well-defined features such as prior authorization requirements, network restrictions, and dental,

Figure A.6: Medicare Advantage Recommendations by Broker Knowledge Proxy



Notes: Figure A.6 shows adjusted predictive margins for the share of calls on which the broker recommends Medicare Advantage, by whether the secret shopper reported that the broker stated Medigap plans within a plan letter are standardized. Estimates come from OLS with the full set of baseline controls. We observe 637 calls. Bars are 95% confidence intervals from standard errors clustered at the caller level. The proxy equals one when the shopper reported that the broker made the statement. It equals zero when the broker said the opposite, never discussed Medigap, or the shopper could not classify the call.

vision, and hearing benefits. We assume that plan attributes are additively separable features and that utility is linear. This allows us to calculate the compensating variation as the sum of the individual plan attributes.[74]

$$\mathrm{CV}_i = \underbrace{S + D + v_i^{\mathrm{dvh}}}_{\text{gains from MA}} - \underbrace{\left(v_i^{\mathrm{net}} + v_i^{\mathrm{auth}} + v_i^{\mathrm{claim}} + v_i^{\mathrm{oop}}\right)}_{\text{Medigap protections lost in MA}} - \underbrace{v_i^{\mathrm{switch}}}_{\text{forgone MA}\to\text{MG switching option}} . \tag{20}$$

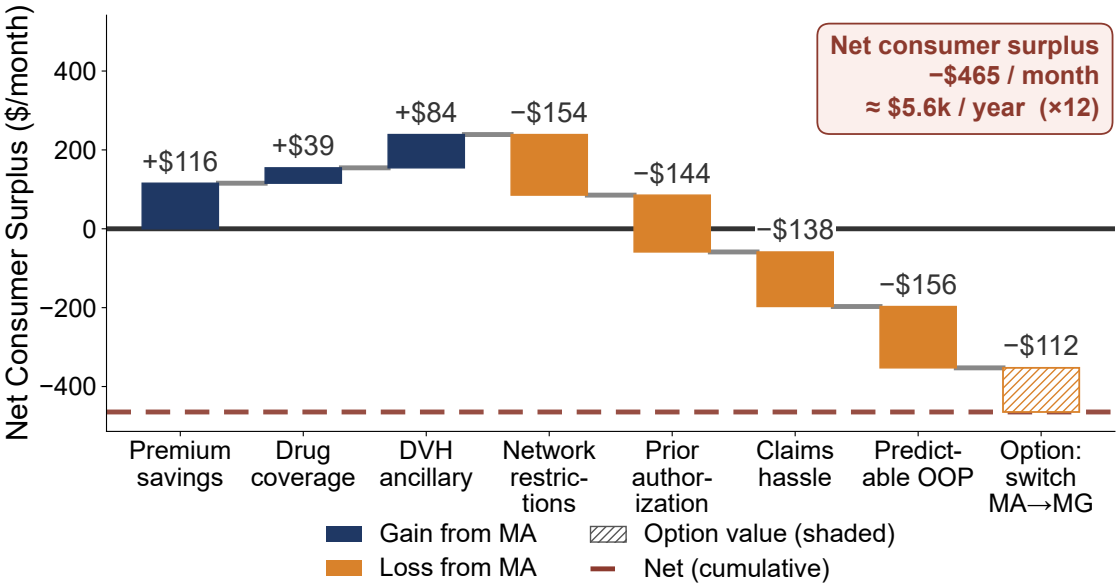
Each v_i^{attrib} is i's willingness to pay (\$/month) for each Medicare plan attribute, including dental/vision/hearing (DVH), network freedom, no prior authorization, freedom from claims-denial hassle costs, and predictable out-of-pocket costs; S is the avoided Medigap premium and D is the money value of MA's bundled drug coverage. To be conservative we set the MA premium and the MAPD drug cost to \$0.[75] A negative CV_i implies the MA plan leaves consumer i worse off. Figure A.7 plots each component of Equation (20).

Given existing evidence of information frictions, switching costs, and consumer inertia resulting in consumer choice that is inconsistent with expected-utility, we elicit the valuations directly as a money-metric from the Medicare broker survey described in Appendix B.5. While it may seem

[74]This additive part-worth structure is methodologically related to conjoint analysis in marketing (Green and Srinivasan, 1990) and to discrete-choice experiments in health economics (De Bekker-Grob et al., 2012).

[75]The average MA-PD premium for Part D coverage was \$7 in 2025 (\$8 in 2026) (Cubanski and Damico, 2026). Setting it to \$0 raises the gains from MA, so it lowers the estimated consumer surplus loss.

Figure A.7: MA Consumer Surplus for a Medigap-fit Senior



Notes: Figure A.7 shows the components of a net annualized consumer surplus loss of −$5,574 from MA restrictions by Medicare plan attribute for a Medigap-fit senior. Measures for each attribute are elicited as a money-metric from a survey of brokers. Premium savings calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study regions. Drug coverage valued at the 2025 national average monthly stand-alone Part D (PDP) premium from Cubanski and Damico (2026). The full methodology is described in Appendix C.1. Figure A.11 reports the same decomposition for the average senior.

counterintuitive to elicit the welfare measure directly from brokers given our evidence of unsuitable broker advice, the survey is designed to elicit the informed expert benchmark without the misaligned incentives present in our audit setting since they are not tied to real enrollments or commissions.[76] Any residual self-serving bias in the valuations would work against our results: a broker inclined to rationalize their own MA steering would understate the Medigap-distinguishing protections or overstate MA's ancillary benefits, shrinking our measured consumer surplus loss toward zero. Thus, our estimate of the consumer surplus loss from an MA recommendation is, if anything, a conservative lower bound on the true consumer surplus cost of an MA recommendation.

The second concern with the broker-elicited measure is broker competence. If brokers do not understand the different Medicare products, a concern we show in Section 5.5 to be somewhat founded, the elicited measures would be noisy and Δ may measure confusion rather than consumer welfare. In Appendix C.1 we show that restricting the sample to the brokers who score highest on the knowledge test would increase our MA consumer surplus loss estimates. We use the full-sample estimate to be conservative.

For each one of the key attributes that distinguish MA from TM+MG, we ask brokers to report the "maximum additional monthly premium ($/month)" that a given senior would pay to have

[76]To incentivize effort brokers are told that the aggregate responses will be shown to real Medicare beneficiaries, but the responses are not tied to potential clients or enrollments, avoiding commission-driven conflicts of interest.

that feature. To account for the lower MA premium cost we include two offsetting attributes: S, the average avoided minimum monthly Medigap premium ($116/mo from our audit data), and D ($39/mo), the 2025 national average monthly stand-alone Part D (PDP) premium (Cubanski and Damico, 2026).[77]

We assume that each plan attribute is additively separable. This assumption seems reasonable given the plan attributes. For example, it seems reasonable that how much a consumer values avoiding prior authorization would not depend on their valuation of network restrictions.

We report an annual consumer surplus loss and a mortality-adjusted present-value consumer surplus loss, both in constant 2025 dollars, for a Medigap-fit senior and for the average senior. Given the difficulty of switching from an MA plan to Traditional Medicare plus Medigap due to the underwriting restrictions of Medigap plans, we hold the senior in the MA plan with annual persistence φ. Our main calculation sets $\varphi = 1$, so the senior stays in the MA plan for the rest of their life. The annual loss is flat in real terms. We accumulate the loss from age 65 through the life table's terminal age of 100, weight each year by survival, and discount at the real rate $r = 3\%$, the U.S. federal benefit–cost default for consumption effects set by the OMB (United States Office of Management and Budget, 2025).[78] The mortality adjustment uses the survival rate of the 2022 NCHS United States Life Tables.[79] The 36-year estimates reported in Section 5.1 apply these weights. Undiscounted, the cumulative 36-year loss is $200,682 for a Medigap-fit senior and $97,604 for the average senior.

MA consumer surplus loss by knowledge test score The broker survey includes three knowledge check questions on fundamental facts about the Medicare market that Medicare brokers should know. We give brokers one point for each correct answer and the total score is out of three points. The knowledge check questions and answers are shown in Figure A.8.

Figure A.10 plots Δ, the point-of-sale under-insurance harm (before premium savings) from an MA recommendation, against the knowledge score. Brokers who miss all three items value the lost protections at $4,807/yr; brokers who get all three right value them at $6,404/yr. The results suggest that the estimated consumer surplus loss from MA is *increasing* in broker knowledge, making the pooled estimate across all knowledge levels a lower bound. The brokers who are more knowledgeable about Medicare report larger consumer surplus loss, so restricting the sample to the most knowledgeable brokers would only increase our consumer surplus loss estimates. We continue

[77] Concretely, the premium savings from Medigap are calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study region.

[78] The 3% figure is a real (inflation-adjusted) discount rate and is the standard rate used by the OMB for similar calculations. At the Medicare Trustees' intermediate inflation assumption of 2.5%, the nominal equivalent is about 5.6%. At $r = 5\%$, the mortality-adjusted present value is $67,181 for a Medigap-fit senior and $32,674 for the average senior.

[79] Specifically, we use Table 1, the NCHS life table for the total population, reported in Arias et al. (2025).

A healthy 67-year-old in a Medicare Advantage plan asks: *"If I try MA now, can I switch to a Medigap plan in a few years if I don't like it?"*

How would you answer this question?

- ○ "Yes — you can switch to any Medigap plan anytime, no health questions."
- ○ "You can apply, but in most states you can be turned down or charged more based on your health."
- ○ "No — switching from MA to Medigap isn't allowed."

A client asks: *"Does Original Medicare by itself — with no Medigap — protect me from very high bills in a catastrophic year?"*

How would you answer this question?

- ○ "Yes — Original Medicare has an annual out-of-pocket maximum."
- ○ "No — Original Medicare alone has no annual out-of-pocket limit."
- ○ "It depends on the state."

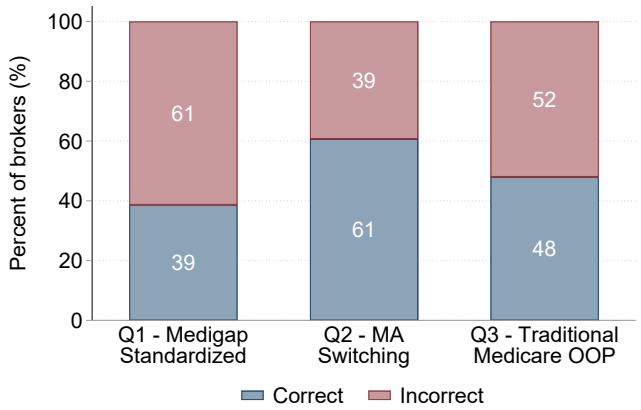
A client comparing Medigap quotes from two different insurance companies asks: *"If it's the same standardized plan letter (like Plan G or Plan N), can one company's plan have better coverage than another company?"*

How would you answer this question?

- ○ "Yes — each company designs its own version, so the benefits can differ quite a bit."
- ○ "No — for a given plan letter the benefits are set by law and identical across companies; they mainly differ on price."
- ○ "Most of it is the same, but each company adds or drops a few benefits."

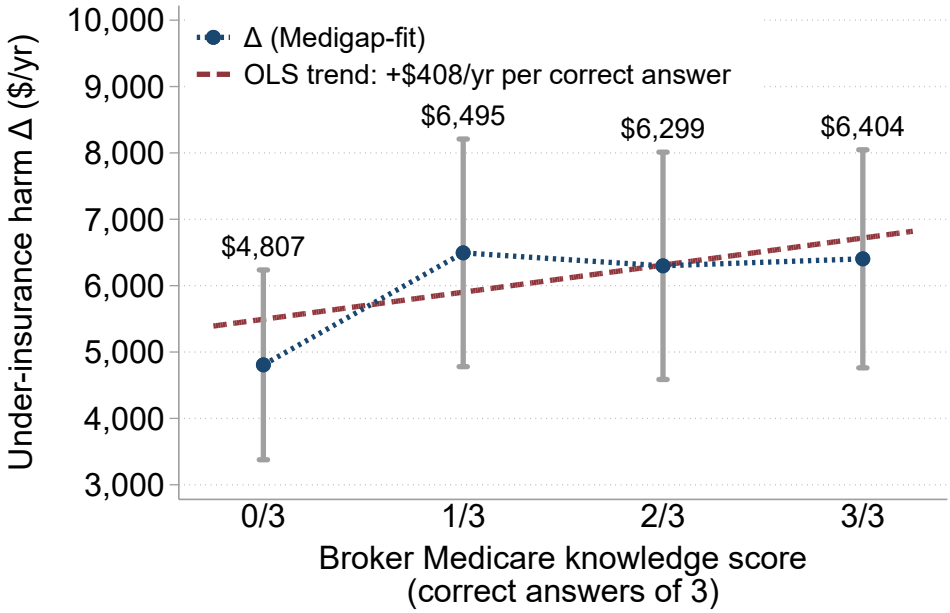
Figure A.8: Broker Survey Knowledge Check Questions

Figure A.9: Broker Knowledge Score by Question



Notes: Figure A.9 shows the share of brokers who answered each question in the multiple choice broker knowledge check correctly.

Figure A.10: The under-insurance harm Δ by broker knowledge



Notes: Figure A.10 shows Δ, the mean point-of-sale under-insurance harm (before premium savings) when a Medigap-fit senior enrolls in MA, by broker knowledge score. The grey bars show the 95% confidence intervals. The knowledge score counts the number of Medicare knowledge check questions the broker answers correctly (out of three). The dashed line is the fitted OLS trend: Δ rises $408/yr per correct answer (s.e. $364, $p = 0.26$).

to use the full-sample estimate to be conservative.

C.2 Calculating lost consumer surplus from the Medigap markup

This appendix gives the back-of-the-envelope calculation behind the lifetime Medigap markup loss reported in Section 5.1 for Outcome 2. We project the monetary loss forward over an expected enrollment horizon for our hypothetical 65-year-old senior. All values are in constant 2025 dollars. We make two simplifying assumptions in our calculation. The first concerns switching: the senior remains enrolled each year with persistence φ, and the headline calculation sets $\varphi = 1$, so the senior stays in the same Medicare plan for the remainder of their life. Given underwriting restrictions on switching outside the initial Medicare enrollment window, consumer inertia, and the low rates of switching behavior observed in other Medicare markets, we believe this is a reasonable benchmark.

The second simplifying assumption is that both the low-cost Medigap plan and the broker-recommended plan follow the same premium growth rates over time. This is reasonable given the institutional setting; whether a Medigap policy is community-rated, issue-age-rated, or attained-age-rated, the annual rate revision is always reported as a percent increase relative to the existing premium. This percentage increase is then reviewed and approved by state insurance commis-

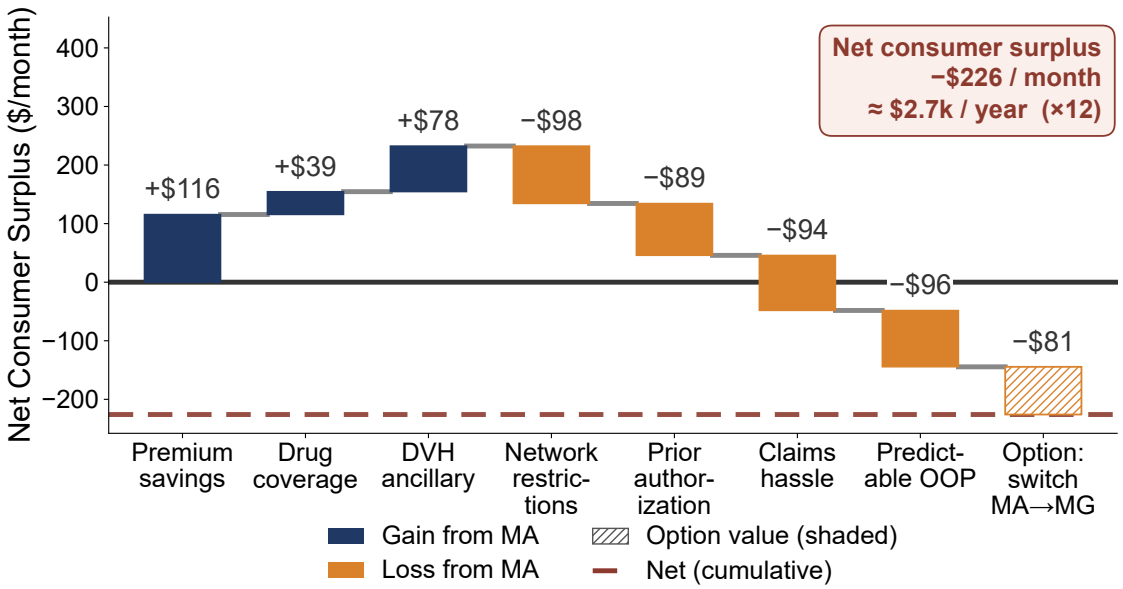


Figure A.11: MA consumer surplus waterfall: average senior (DVH wanted)

Notes: Figure A.11 shows the components of a net annualized consumer surplus loss of −$2,711 from MA restrictions by Medicare plan attribute for the average senior (versus the Medigap-fit senior in Figure A.7). Measures for each attribute are elicited as a money-metric from a survey of brokers. Premium savings calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study regions. Drug coverage valued at the 2025 national average monthly stand-alone Part D (PDP) premium from Cubanski and Damico (2026).

sioners. In interviews with former actuaries, individuals who work at insurance companies, and employees at state department of insurance offices, individuals reported that most rate increases were approved as long as they were reasonably in line with the rest of the market.

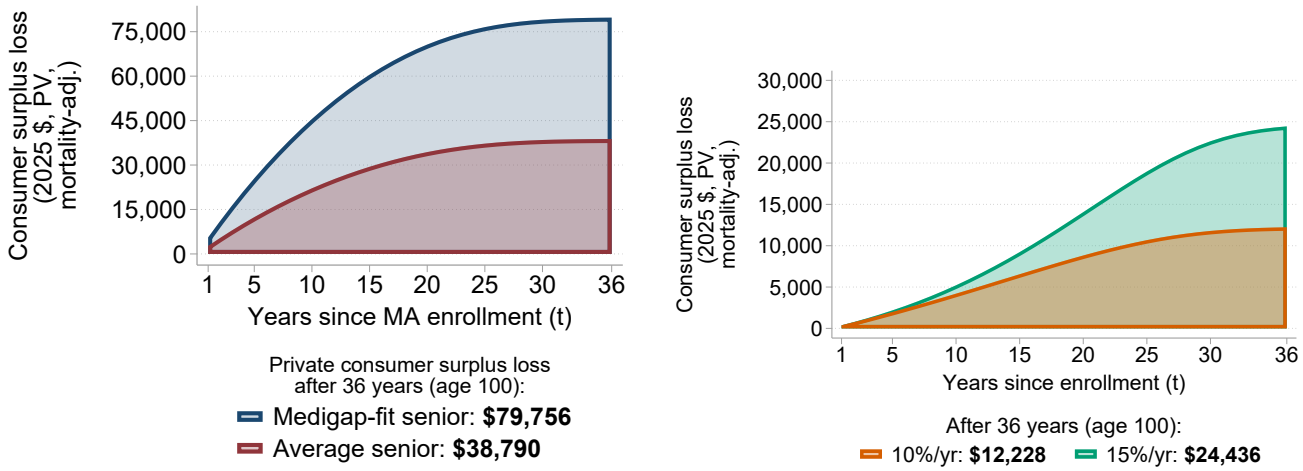
With these assumptions, we calculate the survival-weighted present value of the loss from the Medigap markup as:

$$L_{\mathrm{PV},S} \;=\; \sum_{t=1}^{T} S(t)\,\varphi^{\,t-1}\,\frac{\bar{m}\,(1+g_{\mathrm{real}})^{t-1}}{(1+r)^{t-1}}, \qquad 1+g_{\mathrm{real}} \;=\; \frac{1+g}{1+\pi}, \tag{21}$$

where $\bar{m}$ is the average observed year-1 markup in 2025 dollars, g is the annual nominal Medigap premium growth rate, π is expected inflation, φ is the annual plan persistence, r is the real discount rate, and $S(t)$ is the conditional probability that an enrollee alive at age 65 is still alive in year t. Because carriers file rate revisions as nominal percent increases while r is a real rate, we deflate each growth scenario by expected inflation before compounding, keeping the numerator and denominator of (21) in the same units; we set $\pi = 2.5\%$, the intermediate inflation assumption of the Medicare Trustees.[80] The horizon T runs from age 65 to age 100, the last age in the life table. Since few enrollees survive to these ages, the later years contribute little to the total. We use a real discount rate of $r = 3\%$ and a survival rate constructed from the 2022 NCHS United States Life

[80]Equivalently, discounting the nominal wedge at the nominal counterpart rate $(1+r)(1+\pi)-1 \approx 5.6\%$ yields identical values; we present the calculation in real terms following the OMB and cost-effectiveness conventions.

Tables.[81] We calculate the consumer surplus loss over a range of nominal premium growth rates $g \in \{5\%, 10\%, 15\%, 20\%\}$ and focus on the 10% and 15% scenarios, which better represent recent rate increases.[82] The methodology parallels the welfare-loss calculations for dominated financial products in Handel (2013) and Hortaçsu and Syverson (2004).



(a) Outcome 1: Medicare Advantage

(b) Outcome 2: Medigap Markup

Figure A.12: Lifetime Consumer Surplus Loss

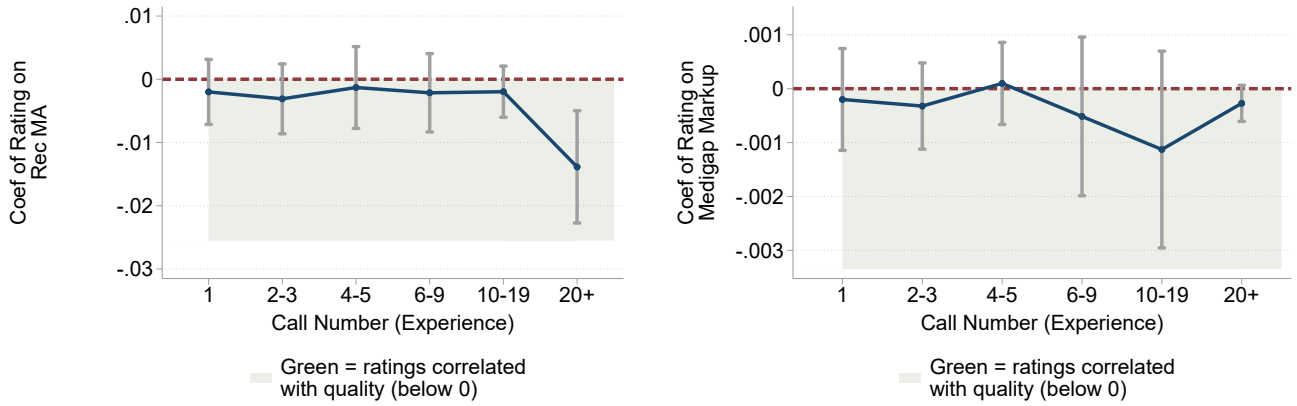
Notes: Figure A.12(a) and A.12(b) show the cumulative mortality-adjusted present value of the annual consumer surplus loss from an incorrect MA recommendation and from the Medigap markup, respectively, in constant 2025 dollars. Figure A.12(a) shows the consumer surplus loss for both a Medigap-fit senior and for the average senior. Figure A.12(b) shows the consumer surplus loss from a high-markup Medigap plan with a growth rate of 10%/year and 15%/year. The annual loss is held flat in real terms and discounted at $r = 3\%$ real; survival weights use the 2022 NCHS life tables.

Annual premium growth (g)	Cumulative welfare loss ($, PV, mortality-adj.)	
	$r = 3\%$ (primary) $L_{\mathrm{PV},S}(36;\, g,\, 0.03)$	$r = 5\%$ (robustness) $L_{\mathrm{PV},S}(36;\, g,\, 0.05)$
5% (lower end)	$6,884	$5,652
10%	$12,228	$9,473
15%	$24,436	$17,791
20% (higher end)	$54,235	$37,235

Table A8: Cumulative lifetime monetary loss by premium-growth rate and discount rate.

Notes: Table A8 reports the survival-weighted present value of the Medigap markup monetary loss, Equation (21), from age 65 through the life-table terminal age of 100, in constant 2025 dollars. Each row is a nominal premium-growth scenario from carrier rate filings, deflated by expected inflation ($\pi = 2.5\%$) before compounding; columns show the primary ($r = 3\%$) and robustness ($r = 5\%$) real discount rates.

Figure A.13: Caller Learning



(a) Outcome 1: Recommend MA

(b) Outcome 2: Medigap Markup

Notes: Figure A.13 plots the coefficients $\hat{\beta}_n$ from separate regressions of the senior's 1–5 broker rating on broker quality for each caller-experience bucket (the caller's 1st call, 2nd–3rd, ..., 20th-or-later). Panel (a) measures broker quality with an MA recommendation and Panel (b) with the Medigap markup (dollars per year). In both panels a higher value is worse, so a coefficient below zero means ratings fall as quality worsens. The green shading marks this region. Both specifications include the baseline controls. Vertical bars are 95% confidence intervals from standard errors clustered at the caller level.

C.3 Caller Learning

Information frictions could also ease over time if callers learn to identify recommendation quality as they gain experience with the market, even if they are initially unable to identify high-quality advice. As an indirect test for caller learning, we examine how the correlation between a caller's rating and our objective measures of broker quality evolves as the caller gains greater exposure to the market. We split calls into six caller experience buckets indexed by n, from the caller's first call to the twentieth or later call. The lower-experience call buckets are kept narrow to show the regions where caller learning is most likely. Higher-experience call buckets are pooled to maintain adequate sample size. Within each bucket we estimate:

$$\text{Rating}_i \;=\; \alpha_n \;+\; \beta_n\, Q_i \;+\; \mathbf{X}_i'\boldsymbol{\gamma}_n \;+\; \varepsilon_i, \tag{22}$$

where i indexes the audit call, Rating_i is the rating (1 to 5) the caller assigns the broker at the end of call i, Q_i is either Outcome 1 (whether the broker recommended MA) or Outcome 2 (the Medigap markup), and $\mathbf{X}_i$ is the vector of baseline controls. Estimating a separate β_n for each bucket allows the relationship between ratings and quality to change flexibly with caller experience.

If caller ratings better reflect recommendation quality over time, we would expect β_n to become increasingly negative as n increases (that is, an increase in recommending MA and recommending

[81] We also report the estimates with $r = 5\%$ in Table A8.

[82] The 5% scenario reflects the lower end of the distribution and corresponds to the long-run historical average until roughly 2022. The 10% and 15% scenarios reflect the bulk of the rate-increase mass observed in 2023–2026 carrier filings. The 20% scenario reflects the higher end of carrier rate filings observed.

a higher markup Medigap plan should be correlated with lower ratings). Panel (a) of Figure A.13 shows $\hat{\beta}_n$ for the broker's MA recommendation rate; Panel (b) shows the results for the Medigap Markup.

Through the nineteenth call $\hat{\beta}_n$ is statistically indistinguishable from zero in both panels and exhibits no systematic trend. Only at twenty or more calls does the Medicare Advantage estimate become meaningfully negative in the direction predicted by learning, but with a wide confidence interval. The corresponding Medigap estimate is slightly more negative, but is not significant at the 5% level. Since this bucket contains only a handful of callers and the confidence intervals are wide, we interpret this as suggestive evidence of modest learning. However, this learning only appears after far more calls than seniors typically make.

Taken together, the panels suggest that our senior callers' stated willingness to recommend a broker is not correlated with either one of our two main measures of recommendation quality, even after repeated exposure to the market. If consumer ratings are thought of as a proxy for consumer feedback and the primary mechanism through which consumers might discipline low-quality brokers, the results raise concerns about how the mechanism may work in practice. The results highlight that demand-side learning is an unlikely source of market discipline in this setting.

C.4 Broker Tenure: Learning or Selection

Section 5.1 shows that high-tenure brokers are less likely to recommend MA and recommend Medigap plans with lower markups. This result is consistent with two explanations. Brokers may learn which plan fits a senior as they gain experience, or brokers who provide low-quality advice may be more likely to leave the market. This appendix reports three tests that bear on the two explanations but cannot separate them, and describes the licensure data we will use to separate them.

First, we test whether high- and low-tenure brokers respond differently to the perceived competition treatment. We regress whether the broker recommends MA on the treatment, the high-tenure indicator, their interaction, and the baseline controls, excluding the no-commission call-center calls and the post-carrier-exit calls of Experiment A. The interaction is -8.1 percentage points (SE 8.8). With 80% power at the 5% level, the smallest interaction we could detect is 24.6 percentage points, about twice the 12.4 percentage point effect of the treatment on these calls. We therefore cannot rule out large differences by tenure in the response to the treatment.

Second, we test whether the difference between high- and low-tenure brokers shrinks when the MA commission is removed (Section 5.3). Among call-center calls, high-tenure brokers are 24.1 percentage points more likely than low-tenure brokers to recommend at least one Medigap plan before the commission removal (SE 7.1) and 31.9 percentage points more likely after it (SE 8.9). If high-tenure brokers recommended Medigap more often only because they respond less to the MA

commission, the difference would shrink once the commission is removed. It does not shrink. The share of high-tenure brokers among call-center calls also rises after the commission removal, so we do not interpret the increase in the difference.

Third, in the broker survey, each additional year a broker has sold Medicare is associated with a 0.027 point higher score on the three knowledge-check questions ($p = 0.038$). The association is smaller and not statistically significant once we control for broker characteristics ($p = 0.130$). The survey brokers are not the audited brokers, and the survey measures tenure as the number of years the broker reports selling Medicare rather than the number of years licensed.

Learning and selection predict the same result in all three tests. Under either explanation, high-tenure brokers know more about the two products at the time of the call, and that knowledge does not depend on the MA commission or on the perceived competition treatment. The survey also includes only brokers who still sell Medicare, so the survey result is subject to the same selection as the audit result.

To separate the two explanations, we follow Farber (2015) and compare the advice of brokers who later leave the market with the advice of brokers who stay, holding tenure fixed. If the tenure result reflects selection, brokers who later leave should have given lower-quality advice in the audit. If it reflects learning, high-tenure brokers should still give better advice than low-tenure brokers among the brokers who stay.

We match 540 of the 551 audited brokers with a licensure record to the Texas Department of Insurance licensure database, which we downloaded on 19 dates between April 2024 and October 2026.[83] Of these, 535 held a Texas license at the time of their audit call. Texas health insurance licenses must be renewed every 2 years. We take the license expiration date in the first download after a broker's audit call as that broker's renewal date, so the renewal decision follows the advice we observe. 268 brokers have passed their renewal date. Of these, 249 renewed, 6 still appear in the database without a new expiration date, and 13 no longer appear. Absence from the database does not mean a broker has left the market. Texas allows a broker to reinstate a license for up to one year after it expires, and 7 audited brokers returned to the database after a gap, each with a new expiration date. We therefore count a broker as leaving the market only if the broker no longer appears in the database and the license expired more than one year before the latest download. 2 brokers meet this definition so far, therefore we choose not to interpret the comparison. Nearly all of the remaining licenses expire by the end of 2027.

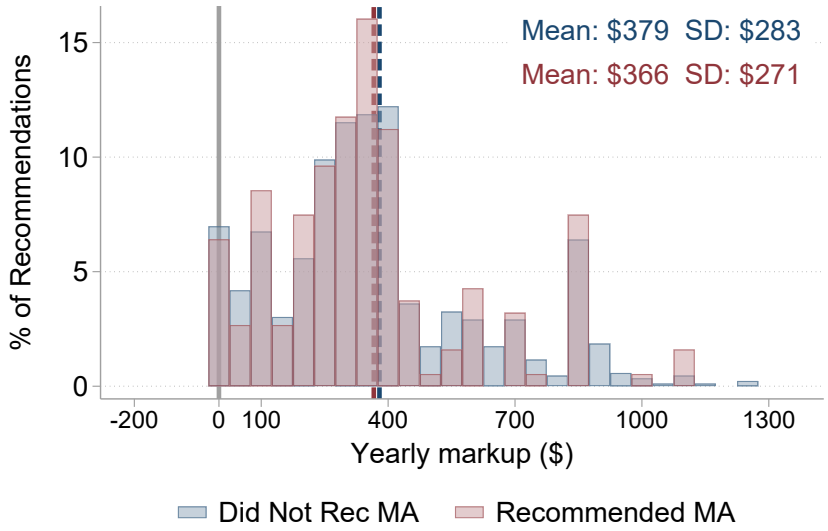
[83]California licensure data must be purchased, so we observe renewals in Texas only. Most audited brokers are licensed in several states, so a broker who does not renew a Texas license may still sell in other states. 117 of the matched brokers live in Texas, and we will report the comparison separately for these brokers.

C.5 Medigap Markup by MA Recommendation

To test whether the Medigap markup is intentionally exaggerated when brokers aim to steer the senior into Medicare Advantage, we plot the Medigap markup outcome in Decision 2 by broker recommendation in Decision 1.

Figure A.14 plots the distribution of the Medigap markup separately for calls in which the broker did and did not recommend MA. The two distributions are nearly identical, and the average markup is slightly lower among brokers who recommend MA. This suggests brokers are not inflating Medigap prices to justify an MA recommendation.

Figure A.14: Medigap Markup by Whether the Broker Recommended Medicare Advantage



Notes: Figure A.14 plots the distribution of the annual Medigap markup separately for baseline (normal-incentive) audit calls in which the broker recommended Medicare Advantage (MA) and calls in which the broker did not. The markup is the difference between the premium of the broker-recommended Medigap plan and the low-cost identical plan available to the senior. The overlaid histograms report the percent of recommendations in each markup bin, and the dashed vertical lines mark the group means. The two distributions are nearly identical, and the mean markup is marginally lower among brokers who recommended MA than among those who did not. This indicates that the Medigap markup is not inflated to rationalize an MA recommendation, so Outcome 2 is not mechanically driven by Outcome 1.

C.6 Caller Experience & Time Trends

Given concern around non-compliance when data collection for the treatment and control arms was conducted at the same time, for the majority of the study we collected data in waves. Control call waves were generally implemented before the treatment call waves; however, the waves did partially overlap in calendar time since the two secret shopper firms we used had different implementation timelines.[84] In this section, we include several additional time and caller experience controls to ensure we are fully isolating the impact of the competition treatment.

[84]We choose to implement the control arm first given concerns that once shoppers have delivered the competition line, they may be tempted to repeat the competition sentence if they feel it makes them seem more sophisticated.

Table A9: Robustness Check: Controlling for Caller Experience & Time Trends

	Recommend MA (%)				Medigap Markup ($)			
	(1) Caller Experience (Linear)	(2) Caller Experience FE	(3) + Linear Experience × Perceived Competition	(4) + Linear Time Trend	(5) Caller Experience (Linear)	(6) Caller Experience FE	(7) + Linear Experience × Perceived Competition	(8) + Linear Time Trend
Perceived Competition Assignment	11.35**	15.71***	18.13***	17.52***	-84.69***	-99.30***	-130.46***	-145.06***
	(4.93)	(5.56)	(6.72)	(6.44)	(19.89)	(25.22)	(34.37)	(37.18)
Call-Center Brokers	16.80***	18.20***	17.98***	18.57***	-8.98	-17.38	-12.55	-24.46
	(4.85)	(5.74)	(5.68)	(5.78)	(18.51)	(23.37)	(23.61)	(25.08)
Caller Experience	-0.12				0.87*			
	(0.08)				(0.50)			
Linear Experience x Perceived Competition			-0.41	-0.41			4.93	6.16
			(0.66)	(0.65)			(3.82)	(4.01)
Linear Time Trend				0.02				0.38
				(0.06)				(0.25)
Control Mean	28.13	31.82	31.82	31.82	407.25	407.01	407.01	407.01
Caller Experience FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	637	532	532	532	1,055	1,043	1,043	1,043
R^2	0.16	0.17	0.17	0.17	0.15	0.25	0.25	0.26

Notes: Table A9 supplements our main specification by adding additional controls to account for caller experience and time trends. Columns (1) and (5) are the same as our baseline specification Equation (14) and control linearly for caller experience. Columns (2) and (6) replace the linear trend with indicators for each level of caller experience. Columns (3) and (7) keep the caller experience fixed effects from Columns (2) and (6) and add an interaction between Competition Assignment and Linear Experience. Finally, Columns (4) and (8) add both the interaction between competition assignment and linear experience and a linear time trend for the date of the call. Baseline controls, with the caller-experience specification varying by column as described above; the Medigap columns (5)–(8) omit the Annual Enrollment Period interaction. Standard errors are clustered at the caller level. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$.

Table A9 summarizes the following regression specifications. Columns (1) and (5) are the same as our baseline specification Equation (14), and include linear controls for caller experience (i.e., the linear trend of the number of calls the shopper has previously completed). Columns (2) and (6) replace the linear trend with indicators for each level of caller experience as in Equation (23)

$$Y_{is} = \alpha + \beta_1(\text{Perceived Competition Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \gamma' \mathbf{X}_{is} + \epsilon_{is} \quad (23)$$

where Y_{is} represents a recommendation outcome from broker i to shopper s and $\text{Caller Experience}_{is}$ is the number of calls the senior has completed (i.e. call number minus 1). Columns (3) and (7) keep the caller experience fixed effects from Columns (2) and (6) and add an interaction between Competition Assignment and Linear Experience. This interaction allows us to isolate the impact of competition assignment from shopper learning over time.

$$\begin{aligned} Y_{is} = \alpha + \beta_1(\text{Perceived Comp Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \\ \beta_3(\text{Perceived Comp Treatment}_{is} \times \text{Caller Experience}_{is}) + \gamma' \mathbf{X}_{is} + \epsilon_{is} \end{aligned} \quad (24)$$

Finally, Columns (4) and (8) add both the interaction between competition assignment and linear experience and a linear time trend for the date of the call. All regressions cluster standard errors at the caller level.

$$\begin{aligned} Y_{is} = \alpha + \beta_1(\text{Perceived Comp Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \\ \beta_3(\text{Perceived Comp Treatment}_{is} \times \text{Caller Experience}_{is}) + \beta_4 \text{Date} + \gamma' \mathbf{X}_{is} + \epsilon_{is} \end{aligned} \quad (25)$$

By controlling for experience (or number of previous calls completed), we interpret the co-

efficient on the competition treatment as the impact of competition for the *first phone call* the shopper makes. In all specifications, the competition coefficient remains significant and if anything increases in magnitude and significance as additional controls are added. The results suggest that the competition treatment has a larger impact on the first call seniors make. Because 66% of seniors who use a broker consult only one (Appendix B.4), the first-call effect is the one most seniors would experience.

C.7 Perceived Competition IV Analysis

Table A10 reports the effect of the Perceived Competition Treatment instead of the Perceived Competition Assignment. The Perceived Competition Treatment is endogenous because shoppers may not follow their randomized perceived competition assignment, that is, whether they were assigned to deliver the treatment line of the script.[85]

Columns (1)–(4) report as-treated OLS estimates. Columns (5)–(8) instrument the Perceived Competition Treatment with the randomly assigned Perceived Competition Assignment. This identifies the effect of delivering the script for compliers if shoppers accurately report whether they delivered it. The instrument is strong, with first-stage coefficients of 0.74–0.85 and F-statistics of 170–407. The OLS and IV results are also similar in magnitude. For every outcome, the two estimates have the same sign and statistical significance, and the differences between them are small relative to the standard errors. For the Medigap outcomes the IV estimates are slightly larger than the OLS estimates, which is what we would expect if misreporting of delivery biases the OLS estimates toward zero. Overall, the similarity suggests that selection into delivering the script introduces little bias. Note, however, that the OLS and IV comparison cannot separate selection from misreporting.

C.8 LLM analysis and additional results

We use a large language model (LLM) to conduct text analysis on the call transcripts from the audit study. Where a call was recorded, we transcribe it and pass the transcript to GPT-4.1-mini (`gpt-4.1-mini-2025-04-14`) through the OpenAI Batch API with the temperature set to zero and with a fixed seed.

The model receives one context line naming the caller's and the broker's gender, followed by the full transcript, and returns a single JSON record per call. It receives nothing about the treatment

[85]Noncompliance is two-sided. A shopper assigned to control may deliver the competition script, and a shopper assigned to treatment may forget to. Random assignment therefore identifies the average effect of delivering the script for compliers (a local average treatment effect) rather than the effect of treatment on the treated. The complier interpretation also requires the standard monotonicity (no-defiers) assumption, or that no shopper would deliver the script only when assigned to control. This is plausible because delivery follows a scripted protocol and shoppers have no reason to do the opposite of their instructions. The strong first stage also leaves little room for defiers.

Table A10: OLS and IV Comparison

	OLS				Instrumental Variable (IV)			
	(1) Recommend MA (%)	(2) Medigap Markup ($)	(3) Medigap Broker Markup ($)	(4) Medigap % Markup	(5) Recommend MA (%)	(6) Medigap Markup ($)	(7) Medigap Broker Markup ($)	(8) Medigap % Markup
Panel A: OLS & IV Second Stage								
Perceived Competition Treatment	16.06***	-83.88***	-85.11***	-6.79***	15.29**	-98.74***	-99.52***	-8.30***
	(4.68)	(19.14)	(16.77)	(1.57)	(6.96)	(24.71)	(21.95)	(2.07)
Control Mean	28.13	407.25	217.69	31.47				
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055	637	1,055	1,002	1,055
R^2	0.17	0.16	0.16	0.23	0.17	0.16	0.16	0.23
Panel B: First Stage								
First Stage Coefficient					0.74***	0.85***	0.84***	0.85***
					(0.06)	(0.04)	(0.04)	(0.04)
First-Stage F-stat					*169.85*	*406.53*	*357.71*	*406.53*

Notes: Table A10 reports the effect of the Perceived Competition Treatment, an indicator for whether the shopper reports actually delivering the competition script during the call. This variable is endogenous because shoppers may deviate from their randomized Perceived Competition Assignment, that is, from whether the shopper is randomly assigned to deliver or not deliver the treatment line of the script. Columns (1) - (4) report as-treated OLS estimates. Columns (5) - (8) report IV estimates that instrument Perceived Competition Treatment with Perceived Competition Assignment. Because noncompliance is two-sided, the IV identifies the average effect of delivering the competition script for compliers, assuming shoppers report delivery accurately and no shopper would deliver the script only when assigned to control (the monotonicity assumption). Panel B reports the first stage; first-stage F-statistics are shown in italics. The identifying assumption is that random assignment affects broker recommendations only through delivery of the competition script. Baseline controls. Standard errors are clustered at the caller level. $^{*}p < 0.10,^{**} p < 0.05,^{***} p < 0.01$.

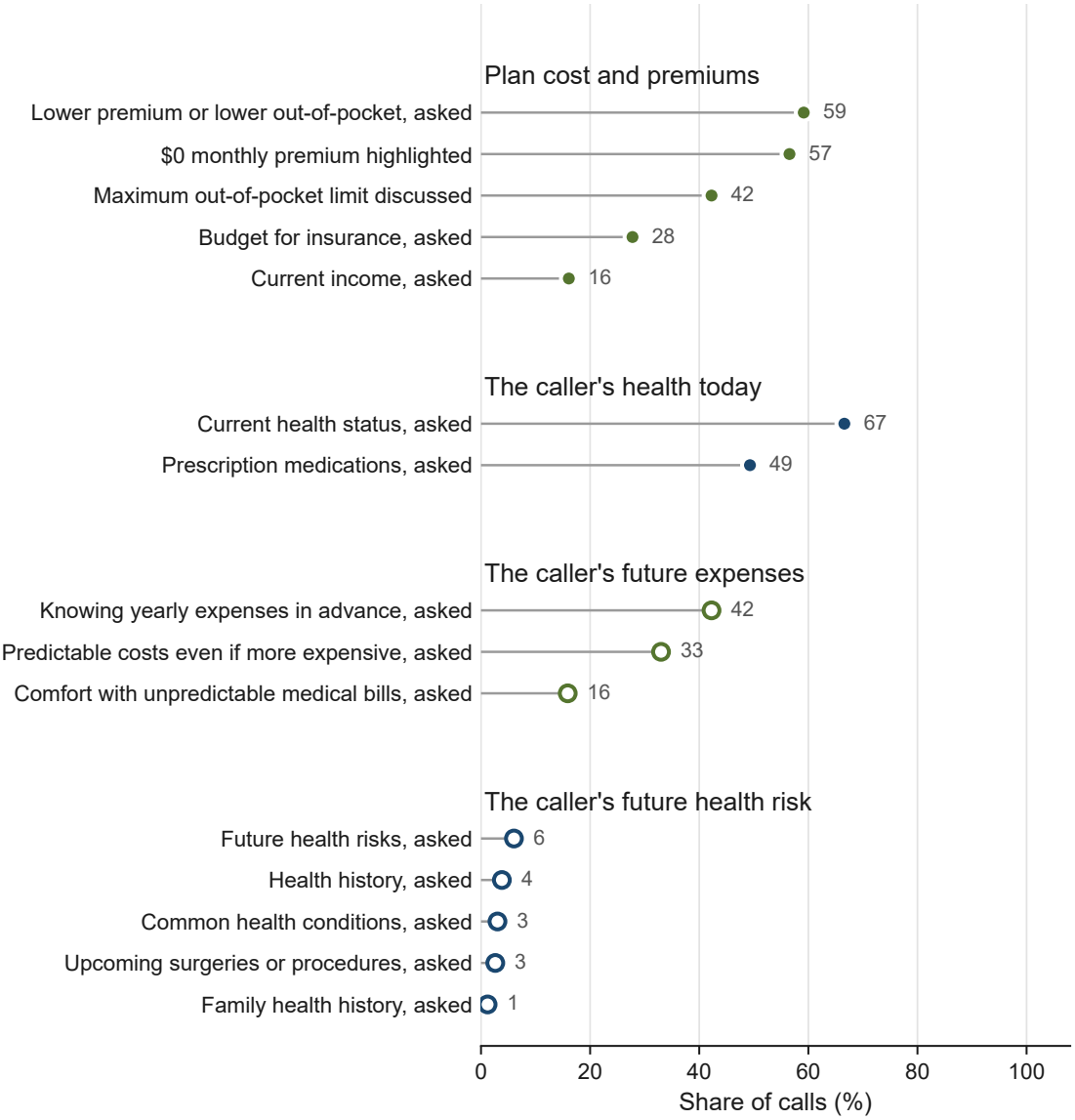
arm, the recommendation the shopper recorded after the call, or any other outcome, so the coding is blind to treatment except through the conversation itself. Transcripts are available for 497 of the 637 calls.

The measure of caller price sensitivity used in Section 5.4 comes from the following prompt, quoted in full:

> "Rate the caller's overall price sensitivity on a scale of 1 to 5 based on the entire conversation. 1=Not at all price sensitive (cost never mentioned, budget not a concern); 2=Slightly price sensitive; 3=Moderately price sensitive (mentioned cost a few times); 4=Quite price sensitive; 5=Very price sensitive (cost was a dominant concern, repeatedly mentioned budget constraints or inability to afford higher premiums)."

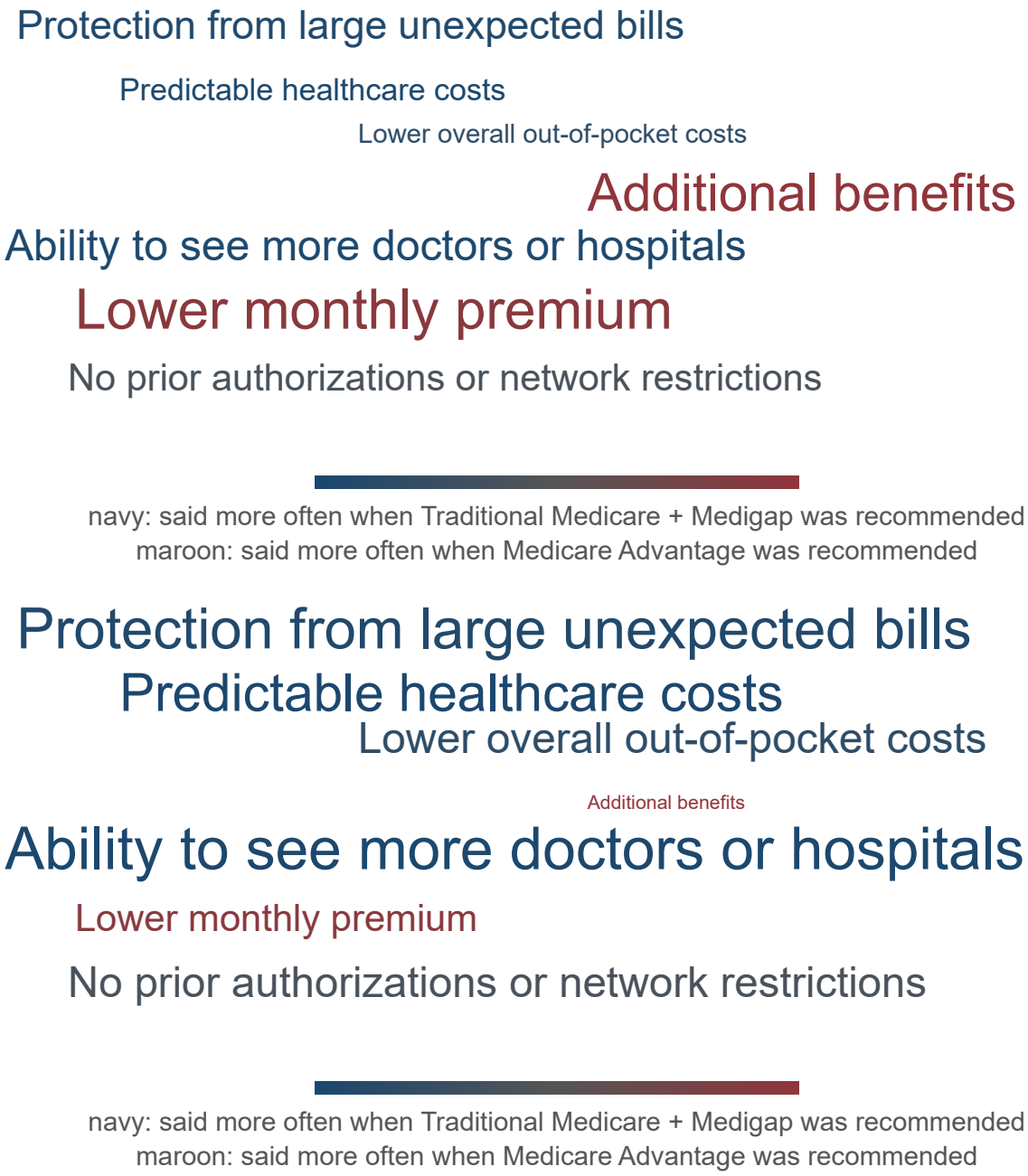
Table 8 tests whether the perceived competition treatment shifts perceived price sensitivity and finds no significant difference between the treatment and control calls. Instrumenting for whether the shopper delivered the perceived competition line with treatment assignment gives similarly small and insignificant estimates of -0.05 (SE 0.06) without controls and 0.12 (SE 0.08) with baseline controls.

Figure A.15: Topics Covered on the Call, by Theme



Notes: Figure A.15 reports the share of calls where each topic was discussed, grouped into four themes. The analysis is done on the 497 calls for which we have transcripts. Topics are ordered by frequency within each theme. Green markers are used to denote financial topics and blue markers are used for health topics. Solid markers are used for "present-focused" topics and hollow markers are used for "future-focused" topics.

Figure A.16: Reasons Given for the Decision 1 Recommendation, by Recommended Plan



Notes: Figure A.16 reports the reasons brokers gave for the Decision 1 recommendation, split by the plan recommended: 154 calls recommending Medicare Advantage (upper panel) and 313 recommending Traditional Medicare plus Medigap (lower panel), of the 497 LLM-coded transcripts; 30 recommended neither and are dropped. The split is on the broker's first recommendation, so the Medicare Advantage group includes 66 calls that also recommended Medigap later in the call. Type size is proportional to the square root of the share of calls in that panel's group citing the reason, on one scale and one layout common to both panels, so a reason sits in the same place in each and only its size changes. A reason is drawn only if cited on at least 5% of calls in one group.

Table A11: Coverage of Key Recommendation Topics by Broker Tenure

	All Mean (SE)	High Tenure Mean (SE)	Low Tenure Mean (SE)	Difference Δ	OLS p-value (robust)
9 Core Topics Covered (Z-Score)	0.00	6.44	-14.07	20.50	0.03
	(4.49)	(5.57)	(7.39)		
Network Restrict	90.14	89.15	92.31	-3.16	0.25
	(1.34)	(1.69)	(2.14)		
Prior Auth	24.95	27.57	19.23	8.34	0.04
	(1.94)	(2.42)	(3.17)		
Switching Plans	69.42	73.02	61.54	11.48	0.01
	(2.07)	(2.41)	(3.91)		
Cost Predictability	92.96	92.67	93.59	-0.92	0.70
	(1.15)	(1.41)	(1.97)		
Drug Plan	73.84	73.31	75.00	-1.69	0.69
	(1.97)	(2.40)	(3.48)		
Travel	93.56	92.96	94.87	-1.91	0.40
	(1.10)	(1.39)	(1.77)		
Medigap Standardized	57.34	62.17	46.79	15.38	0.00
	(2.22)	(2.63)	(4.01)		
OOP Costs	42.25	44.57	37.18	7.40	0.12
	(2.22)	(2.70)	(3.88)		
Plan Tradeoffs	98.19	97.95	98.72	-0.77	0.52
	(0.60)	(0.77)	(0.90)		
N (Calls)	497	341	156		

Notes: Table A11 reports means (robust standard errors in parentheses) of the standardized nine-topic coverage index and of each individual coverage topic, split by the high-tenure indicator, together with the raw difference and the p-value from a bivariate OLS test. Entries are percentages except the first row, which is the standardized index. A positive difference indicates greater coverage by high-tenure brokers.

D Enrollment Effects of the MA Commission Removal: Evidence from CMS Administrative Data

We supplement the natural experiment within the audit study with an analysis using the monthly MA enrollment and contract data collected by CMS to test whether the MA commission removals affect actual MA enrollments at the county level. Recall from Section 4 that we intentionally choose rural areas with few MA plans for the audit study. As a result, there are few MA insurance companies in the zip codes we choose, potentially leaving the actions of the few national carriers visible in enrollments at the county level.

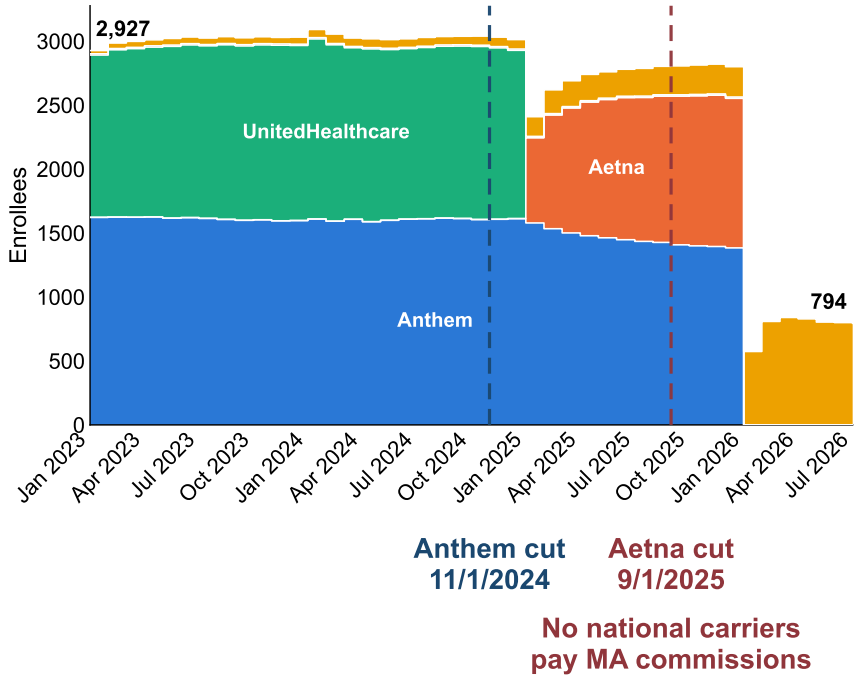
Data and sample We use the publicly available CMS Monthly Enrollment by Contract, Plan, State, and County (CPSC) data files for January 2021 through September 2026. The data report enrollment for every MA plan in every county in the US. We combine this dataset with the State and County Penetration (SCP) files, which give us the number of Medicare-eligible individuals in the county and total MA enrollment.[86]

[86]CPSC suppresses enrollment count cells with ten or fewer observations, but continues to report the plan-county row. For these suppressed cells we use the midpoint of $[0, 10]$. We assume missing plan-month observations have zero enrollees. Since CPSC reports enrollment by the enrollee's county of residence instead of the plan's service

Our sample is the county's individual MA market or standard, general-enrollment MA plans only, excluding special needs plans and employer or union group plans. These are the MA plans sold by the brokers in our audit study to our audit study senior, and they are the object of focus in Outcome 1. The excluded MA plans, which include the employer and special needs plans described below, create a comparison group that we use below.

4 carriers offer individual MA plans in the county. Anthem's MediBlue Access PPO (H8552-029) is the incumbent, offered every year from 2021 and holding just over half the market (53% in October 2024). UnitedHealthcare offers 3 HMOs over the window, 2 of which are still in the county in 2024. Aetna enters for plan year 2025 with 2 PPOs. Imperial Health Plan, a California-only carrier that no national call center is appointed to sell, offers 4 HMOs throughout. Anthem stopped paying commissions on H8552-029 effective November 1, 2024 and Aetna on its plans effective September 1, 2025. Imperial offered MA plans that paid commissions during the entire period. UnitedHealthcare's plans also stopped paying commissions, but we were unable to confirm the date of this change, so it is not marked in the figures.

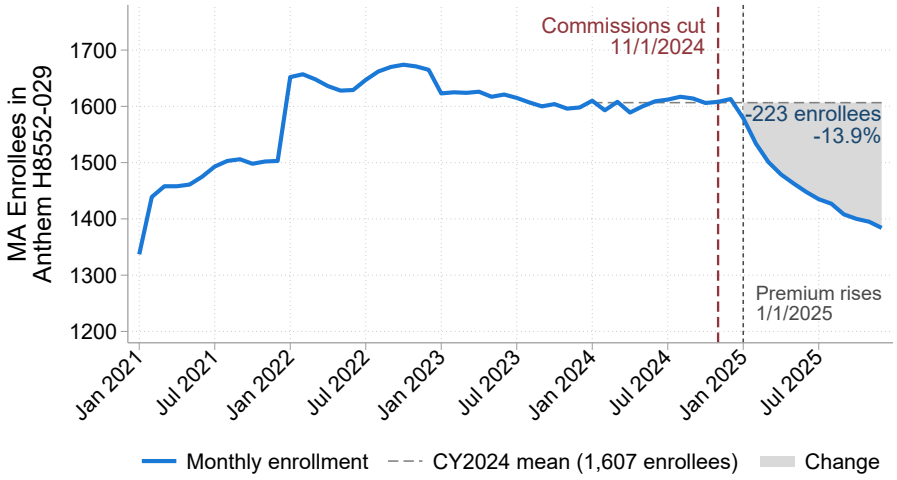
Figure A.17: Individual MA enrollment in the audit county, by carrier



Notes: Figure A.17 shows monthly CPSC enrollment in the county's individual MA plans by carrier, January 2023 through September 2026. The height of the stack is the size of the county's individual MA market, which falls from 2,927 in January 2023 to 786 in September 2026. Special needs plans and employer or union group plans are excluded. The thin gold band is Imperial Health Plan. The dashed vertical lines mark the Anthem and Aetna commission removals. Plan non-renewals take effect January 1, 2025 (UnitedHealthcare) and January 1, 2026 (Anthem and Aetna). Anthem's monthly premium was $30.00 in plan years 2021 through 2024 and $44.00 in 2025.

area, a missing plan-county row does not necessarily mean the plan left the county. We therefore verify every plan entry and exit using the CMS landscape files.

Figure A.18: Anthem MA Enrollment



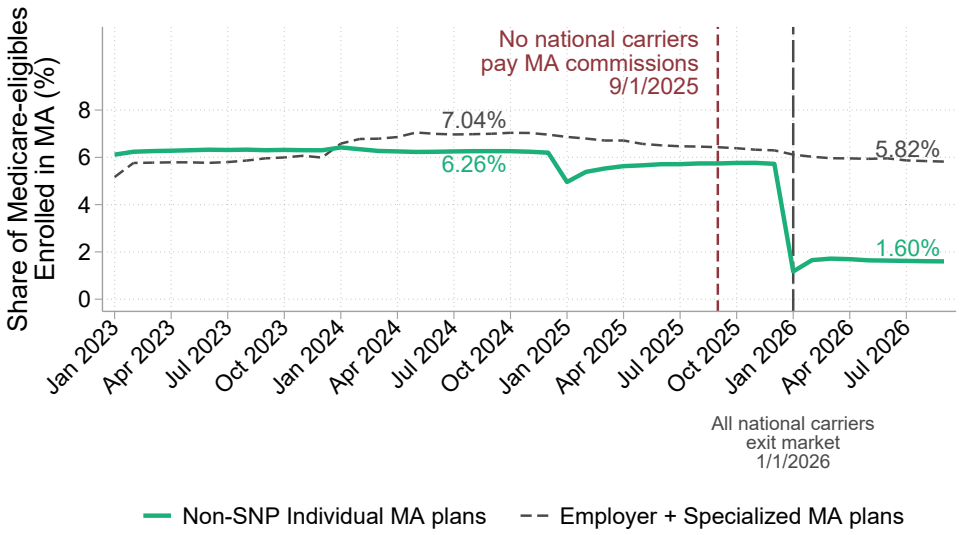
MA enrollment response after the AEP enrollment window Figure A.17 shows the county's individual MA market month by month. Anthem's commission ended on November 1, 2024, but Medicare beneficiaries can move only in the Annual Enrollment Period, which runs from October 15 to December 7 for coverage starting January 1, or through more limited windows. November and December 2024 are therefore the only two months in which Anthem had stopped paying commissions but the plan and its premium were unchanged. Anthem's enrollment does not move in those two months: net changes of +2 and +5, against a mean net change of −0.8 per month over January 2023 to October 2024. The impact of the commission change becomes visible in January 2025: net enrollment flows into the plan are −34, −45, and −33 in the first three months of 2025, or −19.1 per month on average across 2025.

This is consistent with the institutional setting: since consumers cannot switch plans outside of enrollment windows (October through December), a commission change cannot move *existing* enrollees out of the plan during periods outside these enrollment windows. Thus the commission changes outside an enrollment window can only influence the flow of *new* enrollees who are choosing an MA plan.

Two features of this setting make the administrative data results comparable to our audit study. First, the size of the MA market is small enough that the entire individual MA market is easily identified. Because we targeted rural counties with few plans, the entire MA market is 10 plans from 4 carriers over the two-year window. This allows us to manually verify the date and scope of each market change against carrier documents and the CMS landscape files. Second, the enrollment data cover the Anthem cut of November 2024 as well as the September 2025 cut that identifies Experiment A. This allows us to observe two commission removal changes in the same market and isolate the potential confounds specific to each change.

Individual versus Employer and Special Needs Plans This analysis compares the share of Medicare-eligible individuals in non-Special Needs Plan (SNP) individual MA plans to a placebo group of employer and specialized MA plans.[87] The individual non-SNP plan population is the population our audit study seniors fall into; the plans these beneficiaries are enrolled in are the exact Anthem, Aetna, and Imperial Health MA plans described above. To create a placebo group of Medicare enrollees that was not affected by MA commission changes in the same county, we use the share of Medicare-eligible individuals in employer and specialized plans. This group includes all other MA plans in the county not available for general enrollment. These plans may also be sold through brokers, but they face different commission schedules and were not subject to the same MA commission change natural experiment.[88]

Figure A.19: Individual versus employer and specialized MA enrollment in the audit county



Notes: Figure A.19 plots MA enrollment as a percent of Medicare-eligible individuals in Shasta County over time. The solid green line shows the individual MA market, which excludes special needs plans (SNPs), employer and union group plans, and a very small group of cost and PACE contracts. The grey dashed line shows the excluded plans. The vertical dashed lines mark September 1, 2025, when no national carrier pays MA commissions, and January 1, 2026, when the national carriers exit the market.

Figure A.19 plots the share of Medicare-eligible individuals enrolled in MA in the individual MA market against the share of individuals enrolled in employer and specialized MA plans. Both are for Shasta County, our California audit study county of interest.

The two shares are at similar levels before the national carrier change: 6% and 7% of Medicare-eligibles in October 2024 (a ratio of 0.89); however, the two lines begin to diverge after the 2024 AEP enrollments take effect in January 2025 and continue to diverge through 2025, the months

[87]We exclude Special Needs Plans (SNP) MA plans since these are different plans that are only available to Medicare beneficiaries with special needs and not available to the general population. These plans often face different broker commission schedules.

[88]One SNP MA plan did become non-commissionable, but there were several other SNP MA plans, including ones offered by national carriers, available for enrollment in the region. In contrast, all of the national insurance companies that offer MA plans in Shasta County stopped paying MA commissions.

in which the commissions were removed. Both national carriers (Anthem and Aetna) exit the MA market in January 2026; thus, the gap after that date reflects both the removal of national MA carrier plans that pay commissions and the plan exits themselves.

Imperial, a local carrier, remained in the market after January 2026 and throughout 2026. Imperial continued to pay MA commissions and had very similar (in some cases identical) provider networks and higher star ratings. Imperial's enrollment rose only slightly when the Anthem and Aetna plans closed (Figure A.17). The plans closed and their enrollees received guaranteed-issue rights to Medigap, but we cannot observe Medigap enrollment. We note, however, that we do not see full substitution of enrollees into the Imperial plan despite its similar networks and higher star ratings. We therefore treat this as suggestive evidence that a portion of MA enrollments may have been driven by commissions.

E Structural Model: Estimation Details

Why the baseline sale-probability ratio is calibrated Since the audit study variation is insufficient to identify R_0, we choose to calibrate R_0 using the broker survey. We calibrate $R_0 = 0.61$ using the ratio of the mean sale close probabilities reported by brokers in our survey for a Medigap sale and for a Medicare Advantage sale. We then use the experimental variation in the perceived competition treatment to estimate R_1 against R_0. Because R_0 comes from the survey and not from the audit, the estimation does not impose $\hat{R}_1 < R_0$, and the comparison of $\hat{R}_1$ with R_0 tests the direction of the perceived competition response (Online Appendix A.10).

Perceived harms The two broker types have different beliefs about which plan is the misfit plan and therefore which sale harms the senior. The informed type believes MA does not match the senior's preferences and harms the senior. The misinformed type believes Medigap does not match the senior's preferences and harms the senior. Only the product of each harm and the Decision 1 weight κ_1 enters the likelihood. As a result, a freely estimated harm for the misinformed type cannot be separately identified from the misinformed share using the share of MA recommendations when MA pays no commission. For this reason, we calibrate both harms.

In the main specification, we set the misinformed type's perceived harm equal to the informed type's net harm N^{MA}. This calibration does not add a parameter and preserves the identification of $\bar{\lambda}$. The misinformed type's perceived cost of a Medigap sale may be better described as an effort or information cost than as an over-insurance harm (Online Appendix A.8). Table A14 reports an asymmetric-harm specification that instead calibrates the misinformed type's harm using the survey's over-insurance harm. The main specification is parsimonious and does not add an additional parameter while preserving the identification of $\bar{\lambda}$. Table A14 reports the comparison: $\Delta \log L = -3.5$ for the asymmetric specification and a Vuong statistic of $z = 1.93$, where a positive

z favors the main specification.

Appointment portfolios Each broker's portfolio is measured from state insurance carrier appointment records. Both Texas and California, the two states our audit study targets, require appointments before brokers can sell a plan from an insurance company. However, both allow just-in-time appointments. We match the broker in our audit study to the appointment data using national producer number (NPN), carrier (NAIC number), and call date. We match 95% of observations. For the remaining observations, we treat plans priced below the cheapest Medigap plan the broker is licensed to sell as outside the broker's portfolio.

Commission multipliers and the Part D stream Each enrollment pays the broker a first-year commission and, in each later year the enrollee stays in the plan, a renewal commission of about half that amount (Section 2). We value the stream over its first 6 years and assume that the enrollee stays in the plan for the full horizon. Historically commissions are not required to be paid after year 6. CMS dropped the six-year limit for MA in a rule codified in 2021 (CMS, 2021), and renewals may continue after year 6, so the horizon is conservative. We discount the stream with an annual discount factor of $\delta = 0.95$, a conventional value in the literature, and hold each renewal commission constant in real terms. This is conservative: CMS raises the MA and Part D caps every year (the national MA cap rose from \$611 in 2024 to \$694 in 2026), and Medigap commissions are a fixed share of premiums that carriers raise every year, so both streams have grown faster than inflation. For Medicare Advantage and Part D, we use the first-year and renewal commissions published by CMS. Medigap pays a percentage z of the premium, so the Medigap stream is $W_{\mathrm{MG},i} = z\,\tilde{\rho}_i \mathcal{M}_{\mathrm{MG}}$, where $\tilde{\rho}_i$ is the median Plan G or Plan N premium available to the senior on the date of call i: the median of each plan letter's distinct premiums on the senior's menu, weighted by the letter's share of those premiums. We use the median because the broker has not yet chosen the exact Medigap plan when making Decision 1, so we assume the expected commission is roughly equal to the commission for the median Medigap policy. Comparing the lowest, average and median premiums, the median is also the closest to the premiums of the plans actually recommended by brokers in Decision 2. $\mathcal{M}_{\mathrm{MG}}$ is the present value of a stream that pays 1 in year 1 and 1/2 in each renewal year. A Medigap commission stream is therefore worth 3.15 times its year-1 payment.

Since MA and Part D renewals are also approximately half of the year-1 payment, every product's stream is approximately the same multiple of its year-1 payment. The horizon and the discount factor therefore change the units of the dollar parameters but not the relative pay between products or the fit of the model. A Medigap sale is also accompanied by a stand-alone Part D plan. Both Medigap commission terms in the model, $W_{\mathrm{MG},i}$ in Decision 1 and the commission in Equation (18), include the present value of the Part D commission stream. In Decision 2, we use the average of the 2024 and 2025 Part D streams. The Part D stream is the same for every plan in

the choice set, so it does not affect the plan choice on control calls. On treated calls, the deflator multiplies the Part D stream together with the Medigap commission.

The Decision 2 choice utility The Decision 2 weight κ_2 is a separate parameter from the Decision 1 weight κ_1. In Decision 2, the suitability cost $K_{2,ij}$ is a common cost per dollar of markup that the broker incurs when making the recommendation. In Decision 1, the suitability cost varies across calls and is incurred only when the sale closes. The appointment cost a_{appt} is the cost to the broker of selling a plan from an insurance company the broker is not yet appointed to sell. Since brokers can become appointed after a sale using "just-in-time appointments," this is a cost the broker can avoid and not a binding constraint. In Equation (18), the cheapest plan ($m_j = 0$) plays the role of plan L and every marked-up plan plays the role of plan H. The equation therefore extends the two-plan comparison of Section 3.4 to the consumer's full choice set.

Units of the commission sensitivity Section 6.1 writes Decision 1 in dollars (Equation (17)). We maximize the likelihood in the equivalent index form, which multiplies out the bracket in Equation (17) and replaces the dollar location with the location in index units, $\beta\bar{K}_1$. The dollar spread of the suitability cost is therefore $\mu_K = \$1{,}000/\beta$. By assumption, Equations (17) and (18) value a PV \$1,000 of commission at the same rate β. The plan-choice data identify only the net markup slope α and the appointment cost relative to the logit error, so we assume the two scales are equal instead of estimating them separately (Section 6.2).

The calibrated harm enters Decision 1 only through the dollar suitability cost $K_1 = \kappa_1 \mathcal{A}N$, and Equation (17) is written in terms of this dollar cost. Under the main specification, where both types perceive the same harm, the calibrated harm therefore does not enter the Decision 1 likelihood. A different calibrated harm changes only the implied weight $\kappa_1 = K_1/(\mathcal{A}N)$ and leaves β, $\bar{K}_1$, and every predicted recommendation unchanged. It does change the dollar value of the counterfactuals in Section 7, which use the net harm to value each averted MA recommendation.

Plan rows in the likelihood A call can include more than one Medigap quote, since the audit records a quote for each plan letter (G or N) in each of the two audit states the broker priced. Each alternative in the Decision 2 choice set is a premium level offered in the senior's state, plan letter, date, and gender cell, and we match each quote to the premium level of the recommended plan. The recommended plan's premium matches a premium level in the choice set for 99.7% of quotes. We weight the quotes from each call so that each call counts once in the likelihood.

Tenure assumptions Tenure enters as $\log(1 + \text{tenure in years})$, standardized over the brokers the licensure database matches. Three brokers, all in call centers, have no licensure record and are assigned the median tenure of call-center brokers.

Estimation We maximize the likelihood over a vector of unrestricted parameters. Three parameters are positive by definition and enter in logs: the suitability cost location, the treated sale-probability ratio R_1, and the appointment cost. The suitability cost location enters through its log in index units, $\log(\beta\bar{K}_{1,i}) = \log(\beta\bar{K}_1) + \delta'(x_i - \bar{x})$, where $\log(\beta\bar{K}_1)$ is that log when each caller characteristic is at its mean over calls ($x_i = \bar{x}$). Dividing $\beta\bar{K}_{1,i}$ by β gives the dollar location $\bar{K}_{1,i}$. The appointment cost enters as the log of βa_{appt}, its value in index units. The commission sensitivity β and the treatment deflator η are unrestricted, so either can be negative. Re-estimating the main specification from 60 different starting values does not find a log likelihood more than 0.001 above the main specification.

We do not estimate the Decision 2 suitability cost weight κ_2 directly. On control calls, the plan recommendation data identify only the net coefficient on the markup m_j in Equation (18), or the net markup slope $\alpha = \beta(z\mathcal{M}_{\text{MG}} - \kappa_2\mathcal{A})$ (Section 6.1). We maximize the likelihood over α, which combines the PV commission per dollar of annual markup, $z\mathcal{M}_{\text{MG}}$, and the PV suitability cost per dollar of annual markup, $\kappa_2\mathcal{A}$. We then calculate $\kappa_2 = (z\mathcal{M}_{\text{MG}} - \alpha/\beta)/\mathcal{A}$ from the estimates of α and β.

A common suitability-cost location The main specification lets call-center status enter through the broker type probability but not through the suitability-cost location. The difference between call-center and non-call-center brokers that cannot be explained by commissions therefore runs through the broker type probability, not through the suitability cost. The data support this choice (Table A14). While call-center brokers may work under conditions that could lower the cost of recommending a plan that does not fit the senior, such as sales targets, adding a call-center shift to the location does not improve the fit (likelihood-ratio statistic 0.92, $p = 0.34$), and the estimated log ratio of the call-center to the non-call-center location is small, -0.14. Call-center status itself does matter: removing it from the model worsens the fit (likelihood-ratio statistic 22.9, $p < 0.001$). In the model with both channels, dropping call-center status from the type probability while keeping it in the location is also rejected (likelihood-ratio statistic 10.6, $p = 0.001$).

The treated sale-probability ratio at the boundary The likelihood is highest as the sale-probability ratio on perceived competition-treated calls, R_1, approaches zero, and it falls as R_1 rises. Because R_1 enters in logs, its estimate can approach zero but cannot reach it. We therefore report the estimate of R_1 as zero, the lower limit of its range. Standard errors are not informative at this boundary because they rely on the curvature of the likelihood around an interior maximum. Neither the normal approximation nor the bootstrap is reliable when a parameter lies on the boundary of the parameter space (Andrews, 2000). For this reason, Table 10 reports the 95% confidence interval for R_1 from inverting the likelihood-ratio test rather than a standard error.

To construct the interval, we fix R_1 at each value on a grid, re-estimate all other parameters,

and record the maximized log likelihood. The interval, $R_1 \leq 0.30$, contains the values of R_1 whose log likelihood is close enough to the maximum that a likelihood-ratio test does not reject them at the 5% level. Because R_1 cannot fall below zero, the interval is conservative (Self and Liang, 1987). We can therefore reject values of R_1 above this bound, but not values below it.

The interval excludes the calibrated control-call ratio $R_0 = 0.61$. Equivalently, we test the hypothesis that the perceived competition treatment has no effect on Decision 1, $R_1 = R_0$, with a likelihood-ratio test. The test rejects this hypothesis with a statistic of 9.7 ($p = 0.002$). Since we are interested in whether the treatment lowers the ratio ($R_1 < R_0$), the relevant test is one-sided, and the one-sided p-value is half this value.

The hypothesis $R_1 = R_0$ sets R_1 to 0.61, an interior value of the parameter space. The usual χ^2_1 critical values therefore apply in large samples, even though the estimate $\hat{R}_1$ lies on the boundary. The test statistic measures how much better the model fits at the best value of R_1 than at $R_1 = R_0$. Because R_1 cannot fall below zero, the best value is zero. Allowing negative values could only improve the fit at the best value and raise the statistic, so the bound at zero makes the test conservative.

Inference We estimate standard errors and intervals for the structural model using a caller-cluster bootstrap. This matches the caller-level clustering in the reduced-form analysis, which allows the recommendations that different brokers give the same caller to be correlated. The caller characteristics in $\bar{K}_1$ (Section 6.1) also absorb part of this correlation.

Each broker must stay in a single cluster, since the broker's type is fixed across the broker's calls and the likelihood multiplies over those calls (Equation (19)). 51 brokers received calls from more than one caller, so we assign each broker to the caller who placed most of the broker's audit calls. A caller's cluster contains every call to the brokers assigned to that caller, including calls placed by other callers, so every call is in exactly one cluster and no call is dropped. Since 8 callers did not place most of the calls to any broker, their 10 calls enter other callers' clusters, and the 109 callers form 101 clusters. Each bootstrap sample draws 101 clusters with replacement. A cluster drawn twice enters the sample twice, and we treat its brokers in each copy as distinct brokers. We re-estimate the model on each of 1000 bootstrap samples, and we call each re-estimate a refit. Each refit starts the optimizer from nine values and keeps the one with the highest likelihood (Handel and Kolstad, 2015). The 95% interval for a parameter runs from the 2.5th to the 97.5th percentile of that parameter across the refits.

We keep every refit. In a small number of bootstrap samples, the data do not pin down every parameter, and the refit's estimates of those parameters take very large values. These refits make the standard deviation across refits very large but change the 2.5th and 97.5th percentiles little. We therefore report 95% intervals instead of standard errors (Horowitz, 2019). Appendix

Table A13 reports intervals from a bootstrap that resamples the 554 brokers instead of the caller clusters.

We use the bootstrap refits instead of the normal approximation around the maximum because the misinformed share among non-call-center brokers is identified only indirectly (Section 6.2). It comes from the non-call-center calls placed after October 1, 2025. Since only one of the three carriers in the county still paid MA commissions, we assume these brokers earn one third of the MA commission stream. This share reaches zero in 10% of resamples. A percentile interval can capture this boundary, but a normal approximation cannot.

The sale-probability ratio on perceived competition-treated calls, R_1, is the exception because its estimate lies on the boundary of its parameter space, and we report its likelihood-ratio interval instead (described above). The refits nonetheless re-estimate R_1 subject to $R_1 \geq 0$, so the counterfactual intervals in Appendix E.3 include its uncertainty.

Table A12: Structural Model Parameter Estimates: All Parameters

Parameter	Estimate	95% interval
β (commission sensitivity, per PV-$1,000)	0.90	[0.39, 1.55]
$\log(\beta \bar{K}_1)$ (log suitability-cost location)	1.21	[0.80, 1.81]
γ_{ncc} (informed-type logit, intercept for non-call-center brokers)	2.67	[2.01, 40.02]
γ_{cc} (informed-type logit, call-center shift)	-1.79	[-39.04, -1.06]
γ_{ten} (informed-type logit, broker tenure, z-scored log years)	0.59	[0.21, 1.15]
R_1 (treated sale-probability ratio)[a]	0.00	[0.00, 0.30]
η (perceived competition deflator, per $1,000 of annual markup)	0.67	[0.19, 2.30]
α (net markup slope, per $1,000 of annual markup)	-1.75	[-2.11, -1.48]
$\log(\beta a_{\text{appt}})$ (log appointment cost)	0.57	[0.43, 0.70]
Caller characteristics in $\log(\beta \bar{K}_{1,i})$ *(centered)*		
δ_{college} (college education)	-0.24	[-0.74, 0.10]
δ_{white} (white)	0.23	[-0.30, 0.68]
δ_{female} (female)	-0.08	[-0.48, 0.52]
δ_{age} (age, per decade)	0.08	[-0.23, 1.08]
δ_{exper} (caller experience, per 10 prior calls)	0.11	[-0.01, 0.40]
Log likelihood	-1,563.45	
Calls / brokers / callers	637 / 554 / 109	

Notes: Table A12 reports all 14 estimated parameters of the main specification, each on the scale at which it is estimated. β is the change in the log-odds of an MA recommendation per present-value (PV) $1,000 of the expected commission gap. The suitability-cost location and the appointment cost are estimated in logs and in logit-index units, which is why each is multiplied by β. The γ are coefficients of the logit for the probability that a broker is of the informed type, and η and α are per $1,000 of annual Medigap markup. The last five rows are the coefficients δ on the callers' college education, race (white), gender (female), age (in decades) and experience (in tens of prior calls) in the log location of the Decision 1 suitability cost, each centered at its mean over calls (Section 6.1). β cancels from these coefficients, so each is the change in $\log \bar{K}_1$ for a one-unit change in the characteristic. Each 95% interval runs from the 2.5th to the 97.5th percentile of the parameter across 1000 caller-cluster bootstrap refits, and the same refits give the intervals of Table 10. [a]The estimate of R_1 is zero to numerical precision, the lower limit of its range. Its interval is the 95% confidence interval from inverting the likelihood-ratio test (Appendix E).

Table A13: Dollar-Equivalent Parameters: 95% Intervals under Alternative Clustering

Parameter	Estimate	95% bootstrap interval, clustered by Caller (headline)	95% bootstrap interval, clustered by Broker
$\mu_K = \$1{,}000/\beta$ (scale of the suitability cost, PV \$)	1,107	[633, 2,545]	[527, 2,930]
$\bar{K}_1$ (suitability-cost location, PV \$)	3,696	[2,834, 8,638]	[2,715, 8,949]
$\bar{\lambda}$ (misinformed share)	0.17	[0.11, 0.22]	[0.11, 0.22]
$\bar{\lambda}_{cc}$ (misinformed share, call-center brokers)	0.39	[0.29, 0.49]	[0.29, 0.47]
$\bar{\lambda}_{ncc}$ (misinformed share, non-call-center brokers)	0.06	[0.00, 0.11]	[0.00, 0.12]
R_1 (treated sale-probability ratio)[a]	0.00	[0.00, 0.34]	[0.00, 0.34]
$e^{-0.5\eta}$ (relative probability that a senior who signals talking to multiple brokers accepts a plan \$500/yr above the cheapest)	0.72	[0.32, 0.91]	[0.35, 0.94]
K_2 (Decision 2 suitability cost per \$1,000 of annual markup, PV \$)	2,571	[1,723, 5,364]	[1,522, 6,011]
a_{appt} (appointment cost, PV \$)	1,966	[1,106, 4,650]	[965, 5,295]
Caller characteristics: percent change in $\bar{K}_1$, $100(e^{\delta}-1)$			
δ_{college} (college education)	-22	[-52, +11]	[-45, +1]
δ_{white} (white)	+26	[-26, +97]	[-13, +84]
δ_{female} (female)	-8	[-38, +67]	[-39, +33]
δ_{age} (age, per decade)	+9	[-21, +191]	[-18, +87]
δ_{exper} (caller experience, per 10 prior calls)	+12	[-1, +49]	[+2, +45]
Refits		1,000	1,000

Notes: Table A13 reports the dollar equivalents of Table 10 with two 95% bootstrap intervals, and the percent change in $\bar{K}_1$, $100(e^{\delta}-1)$, for a one-unit change in each caller characteristic (Table A12). Each interval runs from the 2.5th to the 97.5th percentile of the parameter across refits of the model on bootstrap samples. The first resamples the 101 caller clusters and is the interval of Table 10 (Appendix E). The second resamples the 554 brokers. Each refit starts the optimizer from nine values and keeps the one with the highest likelihood (Handel and Kolstad, 2015). [a]The estimate of R_1 is zero to numerical precision, the lower limit of its range. Each refit re-estimates R_1, and we report its bootstrap percentile interval in both columns as a check. The bootstrap is not reliable when a parameter lies on the boundary of the parameter space (Andrews, 2000), so Table 10 reports the 95% confidence interval from inverting the likelihood-ratio test instead (Appendix E).

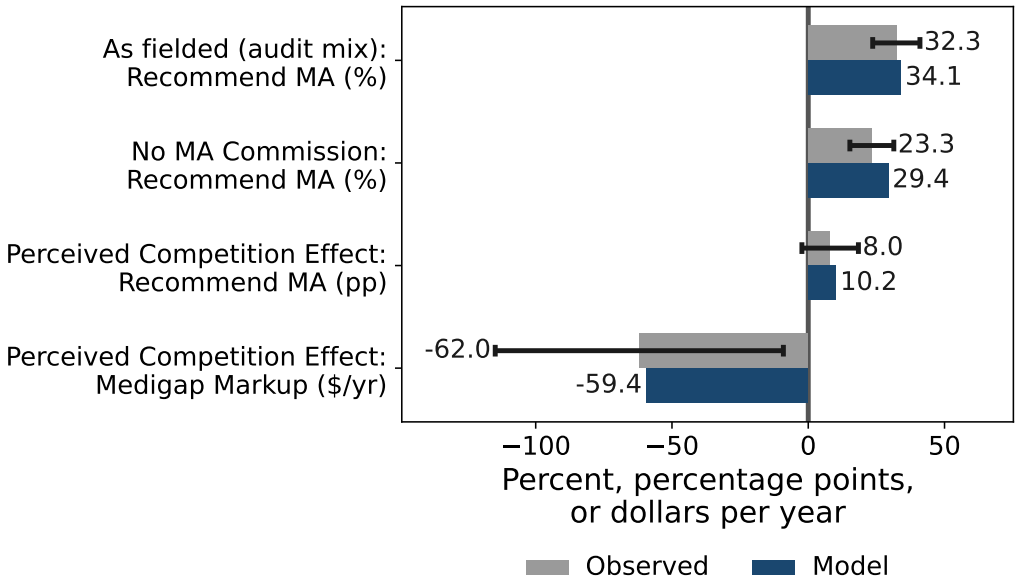


Figure A.20: Observed versus predicted treatment effects

Notes: Each pair compares a moment in the audit data with the model's prediction at the estimated parameters. The moments are cell means, so they differ from the reduced-form regression coefficients, which include controls. The share of brokers recommending MA on commission-paying calls, the share on no-MA-commission calls, and Experiment B's effects on Decisions 1 and 2 are the main sources of variation used to estimate $\bar{K}_1$, $\bar{\lambda}$, R_1 and η.

E.1 Model Fit

Figure A.20 plots each moment observed in the audit data against the model's prediction at the estimated parameters. Four of these moments are the main sources of variation used to estimate $\bar{K}_1$, $\bar{\lambda}$, η and R_1. They are the baseline share of brokers recommending MA (34% fitted), the no-MA-commission floor (29% fitted against 23% observed), Experiment B's effect on Decision 2 (a \$59 annual reduction in the mean recommended markup against \$62 observed, 96% of the effect), and Experiment B's effect on Decision 1 (10.2 percentage points fitted against 8.0 observed, 127% of the effect). The rest of this subsection compares the model with moments and reduced-form estimates that the estimation does not target.

Since the model mixes over two latent broker types, maximum likelihood does not force the fitted shares to equal the observed shares. As a result, the fitted floor differs from the observed floor. The observed floor still lies inside the range of floors the model produces across 1000 audits simulated at the estimated parameters (simulated $p = 0.346$, or twice the smaller of the shares of simulated audits above or below the observed floor).

Fit of the perceived competition treatment's effect on Decision 1 In the model, the perceived competition treatment affects Decision 1 only through the sale-probability ratio R, or how likely a Medigap recommendation is to end in a sale relative to a Medicare Advantage recommendation. The treatment lowers R from $R_0 = 0.61$ on control calls to R_1 on treated calls. Since R is a ratio of two probabilities, it cannot fall below zero. This limits how far the treatment can move a broker of the informed type toward Medicare Advantage. At $R_1 = 0$, the broker expects a Medigap recommendation never to result in a sale, so the Medigap commission no longer enters the informed type's cutoff in Equation (16).[89] The estimate is at this limit. At $\hat{R}_1 = 0$, the model fits a 10.2 percentage-point effect against the observed 8.0 percentage points.

We perform two checks on whether this limit distorts the model fit. First, removing the bound at $R_1 = 0$ (the free competition coefficient row of Table A14) improves the log likelihood by 1.17 points with the same number of parameters, but implies $R_1 = -0.36$, which is not a valid ratio of probabilities. Second, since R_1 is fitted to this effect, we simulate 1000 audits from the estimated model and re-estimate the paper's reduced-form regression on each simulated audit. The perceived competition coefficient on the share of brokers recommending Medicare Advantage in the actual audit lies well inside the simulated distribution (simulated $p = 0.460$, and $p = 0.671$ for the within-cell contrast). It also lies inside the simulated distribution under the asymmetric-harm calibration of Table A14 ($p = 0.080$).

The estimate $\hat{R}_1 = 0$ does not imply that brokers believe a senior who is shopping around never

[89] This limit applies only to the informed type. A broker of the misinformed type already believes Medicare Advantage matches the senior's preferences, so as R_1 approaches zero the model predicts that this broker recommends Medicare Advantage on every treated call that pays an MA commission.

buys Medigap. The data bound R_1 only from above ($R_1 \leq 0.30$, Table 10), and Appendix E.3 reports the multiple-broker-signal counterfactual across that range.

The survey bound on the misinformed share The broker survey provides an additional check on the model that is not used in the likelihood. In the model, we assume a broker recommends MA when MA pays no commission primarily due to information frictions that lead the broker to believe MA matches the senior's preferences. At the model estimates, the misinformed type recommends MA on 83% of no-commission calls and the informed type on only 2%. Since the misinformed type is defined by this belief, the share of surveyed brokers whose rating of how well Medigap fits the Medigap-fit vignette seniors, read as a probability, falls below the broker's indifference threshold is an upper bound on the misinformed share. The threshold is 0.38 under the main specification, where both types perceive the same harm, and 0.20 under the asymmetric-harm calibration of Table A14. Averaging each of the 157 respondents' two vignette answers, 11% of brokers fall below the first threshold and 2% fall below the second.

Under the main specification, the estimated misinformed share, 0.17 (SE 0.03), exceeds the bound by 1.5 standard errors of the difference, which accounts for the sampling uncertainty in both the estimate and the survey bound. The entire gap comes from call centers. Among call-center respondents, the bound is 15% against an estimated 39%, while among non-call-center respondents it is 9% against 6%. Under the asymmetric calibration, the estimated share, 17%, is far above its bound.

There are two caveats to this comparison. First, the bound and the estimate are not independent. Under the main specification, the threshold equals $R_0/(1 + R_0)$, so it uses the same survey close probabilities that calibrate R_0. Under the asymmetric-harm calibration, the threshold also depends on the calibrated harms. For example, lowering the over-insurance harm in that calibration both tightens the bound and raises the estimated share. Second, the broker survey and the audit study cover different brokers. A broker who sells only Medicare Advantage is a misinformed type by construction and has no Medigap belief to elicit. In addition, the audited call-center brokers work at two specific firms, while the survey's call-center respondents may not. If taken at face value, the bound suggests that the model may overstate the misinformed share among call-center brokers.

E.2 Alternative Specifications

Table A14 reports the structural model under alternative specifications, grouped by the assumption each one changes. Each row re-estimates the model with one change to the main specification and re-solves the baseline loss and the policy counterfactuals in the table's last five columns at that row's estimates.

Table A14: Alternative Specifications of the Structural Model

Specification	k	$\Delta \log L$	Test	Parameter estimates					$ per Medigap-fit senior per year				
				β	$\bar{K}_1$ (PV $)	$\bar{\lambda}$ ($\bar{\lambda}_{cc}$ / $\bar{\lambda}_{ncc}$)	$e^{-0.5\eta}$	K_2 (PV $)	Baseline loss	Signal talking to multiple brokers	High ratings	Pay parity	Markup cap (25%)
Main specification	14	–	–	0.90	3,696	0.17 (0.39 / 0.06)	0.72	2,571	2,154	−357	43	377	128
A. Calibrated inputs													
Asymmetric perceived harms	14	−3.5	Vuong z=1.93	0.47	5,281	0.17 (0.40 / 0.06)	0.50	4,466	2,107	−171	39	222	130
R_0 at the broker-level survey mean	14	+0.5	non-nested	0.86	3,800	0.17 (0.39 / 0.06)	0.70	2,679	2,145	−400	43	361	128
Non-call-center misinformed share fixed at survey value	12	−8.1	LR 7.01 (p=.008)[a]	1.14	3,735	0.22 (0.40 / 0.13[†])	0.78	2,167	2,246	−383	52	352	124
Medigap commission at the cheapest premium instead of the median	14	−0.6	Vuong z=0.67	0.95	3,633	0.17 (0.38 / 0.06)	0.73	2,469	2,166	−277	41	545	127
Medigap commission at the average premium instead of the median	14	+0.1	Vuong z=−0.81	0.89	3,710	0.17 (0.39 / 0.06)	0.71	2,592	2,151	−370	43	346	128
B. Broker heterogeneity													
Call-center shift in the suitability-cost location, $\bar{K}_{1,cc} \neq \bar{K}_{1,ncc}$	15	+0.5	LR 0.92 (p=.34)	0.91	3,872 / 3,381	0.17 (0.37 / 0.07)	0.72	2,565	2,148	−359	45	376	128
No call-center term in the type probability or the suitability cost	13	−11.5	LR 22.9 (p <.001)	0.69	4,448	0.16 (0.24 / 0.12)	0.62	3,207	2,108	−282	42	327	129
Call-center status in $\bar{K}_1$ instead of the type probability	14	−4.9	Vuong z=1.11	0.54	6,068 / 3,049	0.12 (0.17 / 0.09)	0.55	3,921	2,080	−245	37	304	131
One broker type	11	−28.1	LR 56.2[b]	0.33	3,944	–	0.42	6,297	2,036	−174	0	264	132
Broker tenure in $\bar{K}_1$ instead of the type probability	14	−1.6	Vuong z=0.51	0.93	3,797	0.17 (0.36 / 0.07)	0.72	2,510	2,148	−371	38	383	128
Call-center status as a common shift in both Decision 1 indices	15	+0.8	LR 1.6 (p=.20)	0.89	3,857	0.16 (0.34 / 0.07)	0.71	2,595	2,149	−371	42	390	128
C. Competition response and Decision 2													
No competition response, $R_1 = R_0$	13	−4.8	LR 9.7 (p=.002)	0.58	4,183	0.18 (0.40 / 0.06)	0.57	3,684	2,129	20	37	284	129
Free competition coefficient (no bound at $R_1 = 0$)	14	+1.2	non-nested	0.79[c]	–	0.18 (0.41 / 0.06)	0.67	2,853	2,118	−498	45	335	129
Quadratic markup term in Decision 2	15	+0.0	LR 0.02 (p=.88)	0.90	3,701	0.17 (0.39 / 0.06)	0.71	2,508	2,153	−356	43	376	127
D. Caller characteristics													
No caller characteristics in $\bar{K}_1$	9	−16.6	LR 33.2 (p <.001)	0.73	3,558	0.16 (0.41 / 0.04)	0.65	3,053	2,146	−265	24	347	128
Caller characteristics in $\bar{K}_1$ without caller experience	13	−2.1	LR 4.1 (p=.04)	0.84	3,499	0.17 (0.39 / 0.05)	0.69	2,713	2,151	−319	40	358	128
Adding caller age squared (the reduced form's full caller controls)	15	+0.1	LR 0.18 (p=.67)	0.98	3,777	0.18 (0.39 / 0.07)	0.74	2,421	2,159	−372	47	384	127
Caller characteristics as a common shift in both Decision 1 indices	14	+0.2	Vuong z=−0.11	0.86	3,615	0.16 (0.37 / 0.05)	0.70	2,672	2,165	−369	36	390	127

Notes: Table A14 reports the structural model under alternative specifications. The main specification is the one in Table 10, with the caller characteristics in $\bar{K}_1$, and the parameters are defined in Section 6.1. Panel D removes or relocates the caller characteristics. k is the number of estimated parameters. $\Delta \log L$ is the row's maximized log likelihood minus that of the main specification. The Test column reports a likelihood-ratio (LR) test for rows nested in the main specification and a Vuong test for rows that are not: the asymmetric-harm calibration, the two Medigap commission rows, and the rows that move a term from one part of Decision 1 to another. We report no test for the two rows marked non-nested. The Vuong z divides the difference in log likelihoods by its standard error across the 101 caller clusters of the bootstrap (Appendix E), computed from each cluster's log likelihood. A positive z favors the main specification. β is per present-value $1,000 of commission, and $e^{-0.5\eta}$ and K_2 are defined as in Table 10. $\bar{K}_1$ is the location when each caller characteristic is at its mean over calls. The two rows in Panel B with a separate call-center location report $\bar{K}_1$ for non-call-center (ncc) and call-center (cc) brokers, ncc / cc. The misinformed share is averaged over all brokers, with the call-center and non-call-center shares in parentheses. Dollar columns are per Medigap-fit senior per year. [a]LR test against the main specification without broker tenure in the type probability ($\log L = -1568.02$, 13 parameters), since a fixed share for one firm type is not nested in the specification with tenure. [†]Fixed at the share of non-call-center survey respondents licensed to sell only Medicare Advantage. [b]The one-type null lies on the boundary of the parameter space, so the statistic has no χ^2 reference distribution. A parametric bootstrap with 200 draws gives $p < 0.005$. The ratings screen acts only through the share of misinformed brokers among highly rated brokers, so with one type it is worth zero by construction. [c]Not normalized by the perceived harm, so $\bar{K}_1$ is not reported.

E.3 Additional Counterfactuals and Welfare Details

Medigap markup benchmark Table A15 compares the $386 annual recommended markup with four benchmarks and splits it into the markup of the cheapest plan the broker is appointed to sell and the markup the broker adds above that plan. This split is an accounting decomposition of the observed markups and does not identify the causes of the markup. Table A17 reports how much each force in the model contributes to the markup.

Table A15: Benchmarks for the Medigap markup

	$/yr	95% CI
Benchmarks		
Average plan on the menu	584	[575, 592]
Average plan the broker is appointed to sell	631	[622, 642]
Recommended plan	386	[355, 407]
Cheapest plan the broker is appointed to sell	207	[186, 222]
Cheapest plan on the menu	0	
Recommended plan above the cheapest plan the broker is appointed to sell		
All recommendations	179	[154, 200]
broker is appointed to sell the recommended plan	154	[137, 174]
broker is not appointed to sell the recommended plan	25	[10, 36]

Notes: Table A15 reports the annual markup over the cheapest coverage-identical plan in the consumer's choice set, per Medigap purchaser. Every benchmark uses the same choice sets and weights. The second panel reports the part of the recommended markup that lies above the cheapest plan the broker is appointed to sell at the time of the call, split by whether the broker is appointed to sell the recommended plan. Brackets are 95% caller-cluster bootstrap intervals, which resample callers and recompute each benchmark on the resampled calls.

We compare the recommended markup to four benchmarks. The average plan on the senior's menu has a markup of $584 a year, and the average plan the broker is appointed to sell has a markup of $631, because the plans a broker is appointed to sell are more expensive on average than the full choice set available to the senior. Brokers recommend plans with an average markup of $386, and the cheapest plan the broker is appointed to sell would have cost $207.

We further split the markup above the cheapest appointed plan by where the recommended plan lies. Of this amount, $154 comes from calls on which the broker recommended a plan the broker was appointed to sell, priced above the cheapest such plan, and $25 comes from calls on which the broker recommended a plan from an insurance company the broker was not appointed to sell. Only 5% of recommendations are the cheapest plan in the senior's choice set, and 18% of recommendations match the cheapest plan the broker is appointed to sell.

Accounting Commissions are transfers from insurance companies to brokers and net out when calculating total surplus. Markups are transfers from consumers to brokers and insurance companies, so a markup reduction is a dollar-for-dollar consumer gain and exactly zero in total surplus. The steering component does affect total surplus. A Medigap-fit senior steered into Medicare Advantage loses both the value of the coverage and the option to switch plans later, net of the pre-

mium savings from enrolling in MA. We value the unsuitable MA recommendation at the broker's perceived consumer harm elicited from the broker survey and not at the consumers' revealed or stated willingness to pay for the plan attributes. The choice is in line with existing literature that relies on expert valuations when well-documented consumer information frictions distort choice (Abaluck and Gruber, 2011; Handel and Kolstad, 2015; Handel and Schwartzstein, 2018). We use the broker-elicited harm range in every estimate.

Partial equilibrium All counterfactuals hold premiums, commission schedules, plan choice sets, and the population of practicing brokers fixed. With the exception of the two policies that change how seniors search (the multiple-broker signal and the ratings screen) and consumer education, which changes how likely brokers believe a senior is to accept each recommendation, the counterfactuals also hold fixed how seniors search for and evaluate advice. The evidence suggests that seniors would respond very little to a policy change, so their responses would have little effect on broker behavior. Among seniors who enrolled through a broker, 66% consulted only one broker (Appendix B.4), and our audit study results suggest that senior ratings of brokers are not significantly correlated with recommendation quality and that this correlation does not strengthen with experience (Section 5.2 and Appendix C.3). Enrollees also rarely change plans, in part because changes are limited to annual enrollment windows and a move from Medicare Advantage to Medigap requires medical underwriting (Section 2).

Given the evidence of a muted consumer response, we assume that a policy that changes broker pay or the plans brokers can recommend changes enrollment primarily through the recommendations brokers make, which is the channel our counterfactuals model. On the supply side, CMS caps Medicare Advantage commissions and state regulators review Medigap premium increases (Appendix C.2), which limits how far insurance companies can respond in the short run. The one supply response we observe in the audit is in the provision of advice. When insurance companies stopped paying Medicare Advantage commissions, the call centers removed the non-commissionable plans from their internal search tools (Section 4.2). This response weighs against removing the Medicare Advantage commission entirely and in favor of pay parity, which keeps every plan commissionable.

Over longer horizons, insurance companies could respond to pay parity or a markup cap by changing their premiums or the plans they offer. Changes in plan offerings and commissions could also affect which brokers enter or stay in the market, and a policy that changes who enrolls in which plan can change premiums through adverse selection (Handel, 2013). We do not model these responses.

Inference and Calibration We calculate the confidence intervals on the policy counterfactuals using the same caller-cluster bootstrap refits as the intervals on the structural parameters (Ap-

pendix E). We recompute each counterfactual at the estimated parameters of each of the 1,000 refits and report the 2.5th and 97.5th percentiles across refits. In every refit, we hold the calibrated parameters at their calibrated values. We report sensitivity to these calibrations separately (for example, in Table A14). Each refit also re-estimates the treated sale-probability ratio R_1 subject to $R_1 \geq 0$, and we recompute the counterfactuals at each refit's own R_1. The bootstrap does not give a reliable interval for R_1 itself, which lies on the boundary of the parameter space (Appendix E), but letting it vary across refits carries its uncertainty into the counterfactual intervals.

The counterfactual in which every senior signals talking to multiple brokers is also outside the support of Experiment B. In the audit study, the perceived competition treatment is a statement the broker cannot verify, and it is delivered on calls to randomly assigned brokers. We estimate the broker response, R_1, by comparing treated and control calls. This counterfactual applies the statement to every caller and holds the broker response at R_1. Note that this counterfactual assumes brokers do not update their beliefs when every senior says they are talking to multiple brokers. If every senior made the statement but did not actually search more, the statement would carry no information and brokers might stop responding to it. For this reason, we interpret this counterfactual as an upper bound on the response to the treatment.

Consumer education is outside the support of the data for different reasons. The audit does not identify the sale-probability ratio on control calls. This is instead calibrated from the broker survey (Appendix E), so the counterfactual moves a calibrated input to $R = 1$ and relies on the model's structure to estimate the broker response.

Signal talking to multiple brokers: alternative baselines, the reduced-form valuation, and the R_1 range Section 7.1 values the "signal talking to multiple brokers" counterfactual as the contrast between the all-control and the all-treated state to match the estimand of Experiment B. Figure 6 instead values it against the audit's observed mix of treated and control calls, which is the baseline for every other policy in the figure. On that basis, the policy lowers consumer surplus by \$357 per senior per year. The reduced-form treatment effects give the same sign and a larger magnitude.

The model-based estimate is smaller because it values the treatment's fitted effects. These are a \$56 reduction in the mean recommended markup and a rise in the share of brokers recommending Medicare Advantage from 31% to 41% of calls. Since the treated sale-probability ratio lies on the boundary of its parameter space, we also report the change at the upper end of its likelihood-ratio interval. At $R_1 = 0.30$, the change in consumer surplus is $-$\$215 per senior per year.

The "consumer education + multiple-broker signal" bar of Figure 6 applies the same treatment to a senior who brokers believe is equally likely to accept either recommendation. In this case, $R_1 = 1$ regardless of the strength of competition, so the bar adds only the Decision 2 markup

reduction to consumer education and does not depend on the range of R_1.

The full policy roster Figure A.21 shows the main counterfactuals in Figure 6, plus the additional counterfactuals described below.

Broker composition counterfactuals The broker composition counterfactuals restrict the pool of brokers to those with a given characteristic. These counterfactuals can be thought of as redirecting consumers to brokers with certain characteristics, assuming the alternative brokers have capacity to absorb the additional demand. We re-weight the broker pool and evaluate each remaining broker over the full distribution of calls in the audit.

Non-call-center brokers only This counterfactual removes every broker who works in a call center. In the type probability, working in a call center raises a broker's probability of being misinformed (Section 6.2), so the brokers who remain are more often informed and steer less. The screen leaves commissions unchanged, so an informed broker who remains still recommends Medicare Advantage whenever the commission gap outweighs the suitability cost. Restricting the pool to non-call-center brokers recovers \$409 per senior per year, 19% of the loss.

Experience of more than 4 years This counterfactual removes every broker licensed for 4 years or less. In the type probability, a broker's probability of being misinformed falls with years licensed (Section 6.2), so the experienced brokers who remain steer less, again with commissions unchanged. Restricting the pool to brokers with experience of > 4 years recovers \$288 per senior per year, 13% of the loss.

Call-center onboarding This counterfactual changes the information of call-center brokers instead of removing them from the broker pool. It sets the call-center term in the type probability to zero, so a call-center broker is as likely to be misinformed as a non-call-center broker with the same experience. Every call-center broker stays in the market and keeps the estimated commission response and suitability cost. The policy represents an onboarding program that trains call-center hires to the non-call-center standard. It recovers \$241 per senior per year (11% of the loss). This is less than the gain from restricting the pool to non-call-center brokers. The screen also removes the other differences between call-center and non-call-center brokers in the model, such as the lower suitability cost of call-center brokers, while onboarding changes only their information.

Removing appointment frictions This counterfactual acts on Decision 2 only. It sets the cost of recommending a plan the broker is not appointed to sell to zero, so plans outside the broker's portfolio compete with plans inside it on equal terms. The commission incentive to recommend higher-premium plans is unchanged. The expected markup falls by \$46 a year per

Medigap purchaser, worth $30 per senior per year (1% of the loss). The gain is small even though the cheapest plan in the broker's portfolio carries a markup of $207 a year, since brokers who can recommend any plan still face the same commission incentive. As with the markup cap, the gain is a transfer from brokers and insurance companies to seniors, so it is zero in total surplus.

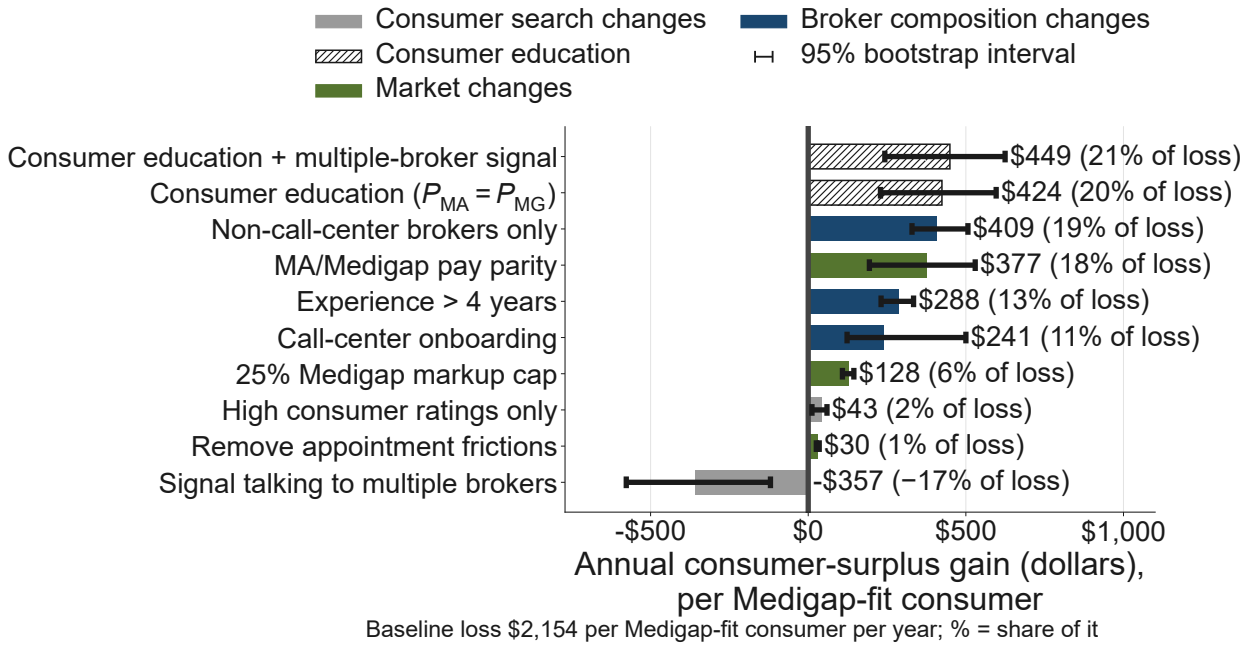


Figure A.21: The full set of counterfactual policies

Notes: Figure A.21 shows the full set of counterfactual policies that we simulate. These include the counterfactuals in Figure 6 and add counterfactuals that change broker composition (e.g. non-call-center brokers only, experience over four years, and call-center onboarding (training)). We also add a counterfactual removing Medigap appointment frictions. Confidence intervals show 95% caller-cluster bootstrap percentile intervals from the refits (Appendix E.3).

What drives broker advice Tables A16 and A17 report the counterfactuals summarized in Section 6.2. Each row switches off one force (or two in the third panel of Table A16) relative to the same baseline. The rows are therefore separate counterfactuals and do not add up. The perceived competition rows compare the all-control and all-treated states, which is the contrast in Experiment B, and are divided by the same baseline as the other rows. With an infinite suitability cost, only brokers of the misinformed type recommend MA in Decision 1, and every broker recommends the cheapest plan in Decision 2. Each refit re-estimates the treated sale-probability ratio R_1, which lies on the boundary of its parameter space (Appendix E). At the upper end of its likelihood-ratio interval, $R_1 = 0.30$, the perceived competition treatment raises the share of brokers recommending MA by 5 percentage points instead of 10.

In Table A17, only the difference in commissions across Medigap plans affects the plan choice, so the flat-commission row also shows the markup with no pay for either product. There is no information-frictions row because broker type does not enter Decision 2 (Section 6.1). The

commission and suitability cost rows rely on the assumption that the commission sensitivity is the same in both decisions, while the perceived competition and appointment cost rows do not.

Table A16: What Drives Medicare Advantage Recommendations to a Medigap-Fit Senior

Force switched off	Change (pp)	Change relative to baseline	95% interval (pp)
Commissions			
Pay parity	−7	−21%	−10 to −4
No pay for either product	−13	−38%	−17 to −8
No MA commission	−17	−49%	−22 to −11
Information			
Every broker of the informed type	−14	−41%	−19 to −9
Commissions and information together			
Pay parity, every broker informed	−23	−66%	−28 to −15
No pay, every broker informed	−29	−84%	−34 to −19
No MA commission, every broker informed	−31	−90%	−36 to −21
Perceived competition			
Treatment on every call rather than none	+10[†]	+30%	+4 to +16
Suitability cost			
No suitability cost	+49	+145%	+33 to +59
Infinite suitability cost	−16	−48%	−22 to −12

Notes: Table A16 reports the change in the model-implied share of commission-paying calls on which the broker recommends Medicare Advantage (MA) to a Medigap-fit senior when one force in the model is switched off (two in the third panel). The baseline is 34%, averaged over the audit's mix of treated and control calls. Pay parity sets the MA year-1 commission equal to the Medigap year-1 commission on a policy at the median premium. No suitability cost sets its location to zero and keeps its dispersion across calls. The perceived competition row is measured from the share with no call treated, 31%. Intervals are 95% caller-cluster bootstrap percentile intervals (Appendix E.3). [†]At the upper end of the 95% likelihood-ratio interval for R_1, $R_1 = 0.30$, this change is +5 percentage points.

Table A17: What Drives the Medigap Markup

Force switched off	Change ($/yr)	Change relative to baseline	95% interval ($/yr)
Commissions			
Flat Medigap pay (equivalent to no pay)	−31	−8%	−55 to −12
Perceived competition			
Treatment on every call rather than none	−56	−14%	−94 to −23
Appointment cost			
No appointment cost	−46	−12%	−52 to −39
Suitability cost			
No suitability cost	+294	+76%	+242 to +389
Infinite suitability cost	−386	−100%	–

Notes: Table A17 reports the change in the model-implied mean markup over the cheapest coverage-identical plan in the senior's choice set, in dollars per year per Medigap purchaser, when one force in the model is switched off. The baseline is $386, averaged over the audit's mix of treated and control calls. A flat Medigap commission pays the broker the same amount on every plan in the choice set. The perceived competition row is measured from the markup with no call treated, $408. Intervals are 95% caller-cluster bootstrap percentile intervals (Appendix E.3).

E.4 Scaling to the Medicare Population

Section 7.2 uses a back-of-the-envelope calculation to scale the consumer surplus loss per advised senior to two populations: the seniors who turn 65 each year and current Medicare beneficiaries. We multiply the loss we estimate from the audit by population counts, a broker-use share, and plan-fit shares that the audit does not identify. The goal is to show the scale of the loss rather than to produce an estimate with a sampling distribution. The annual consumer surplus loss in population u is

$$L_u \;=\; N_u \times b \times \sum_g \pi_g \, \ell_g, \tag{26}$$

where N_u is the number of seniors in the population, b is the share who enroll through a broker, π_g is the share of seniors of plan-fit type g, and ℓ_g is the annual consumer surplus loss per advised senior of that type. The gain from a policy is the change in L_u when we recalculate ℓ_g under the policy. For current beneficiaries, we also check the plan-fit shares against observed enrollment, as we describe below. For seniors turning 65, we also report the loss in present-value wealth over their lifetime. As in Appendix C.1, we hold the annual loss fixed in real terms from age 65 through age 100, weight each year by survival, and discount at a real rate of 3%. This multiplies the annual loss by 14.3.

Assumptions Both calculations rest on the following assumptions. First, the brokers seniors use nationally make the same recommendations as the brokers in our audit study on calls where the broker is paid the full Medicare Advantage commission. Second, seniors enroll in the plan the broker recommends. Third, the plan-fit shares are the average answers to a broker survey question asking what share of seniors is a clear fit for Medicare Advantage, a clear fit for Traditional Medicare plus Medigap, and a clear fit for neither. The question asks about seniors in general, not the broker's own clients. Fourth, the broker-use share is the same for every plan-fit type. We cannot observe plan-fit type in the UAS, but broker use does not differ between Medicare Advantage and Medigap enrollees ($p = 0.91$). Fifth, a senior enrolled in the plan type that fits them loses only the Medigap markup, if they are enrolled in Medigap; we do not count losses from the choice among Medicare Advantage plans. Sixth, ambiguous-fit seniors are recommended Medigap 58% of the time under every policy, so the policies change only the markup these seniors pay. We describe the assumptions for MA-fit seniors and for current beneficiaries below.

Population: seniors turning 65 The main population is Medicare beneficiaries who choose between Medicare Advantage and Medigap. We start from the 60.7 million Medicare beneficiaries aged 65 and over in 2024 (CMS, 2025a) and subtract two groups. The first group is the 6.5 million beneficiaries with full Medicaid benefits. Medicaid pays their cost sharing, and federal law prohibits selling them a Medigap policy. We calculate this count as the 13.6 million beneficiaries who were dually eligible in some month of 2022, times the share aged 65 and over (65%), times the

share of that group with full benefits (73%). Both shares come from the December 2025 MedPAC and MACPAC data book on dually eligible beneficiaries.

The second group is the 12.6 million beneficiaries with employer or union coverage, whose employer selects their plan. We calculate this count as the 14.1 million beneficiaries of all ages with employer or union coverage in 2023, times the share of Medicare beneficiaries aged 65 and over (60.7 of 67.8 million in 2024). The 2023 count includes 8.2 million beneficiaries with an employer supplement to Traditional Medicare and 5.9 million in employer-group Medicare Advantage plans, from KFF's analysis of the Medicare Current Beneficiary Survey. Partial-benefit dual eligibles, whose Medicaid coverage is limited to Medicare premiums and cost sharing, remain in the population.

This leaves $60.7 - 6.5 - 12.6 = 41.6$ million beneficiaries, or 69% of those aged 65 and over. Since there is no count of coverage at the time of enrollment, we apply the same ratio to the 4.1 million people who turn 65 each year. This gives us $4.1 \times 41.6/60.7 = 2.81$ million seniors. We also assume that 50% of seniors use a broker to enroll in Medicare. This gives us an estimate of 1.41 million broker-advised Medicare enrollees per year.

Population: current beneficiaries For current beneficiaries, we count the beneficiaries who hold each plan using KFF's analysis of the 2023 Medicare Current Beneficiary Survey.[90] Among beneficiaries aged 65 and over, 28.3 million are enrolled in Medicare Advantage and 12.0 million hold a Medigap plan. As in the first calculation, we exclude beneficiaries with employer or union plans and beneficiaries with full Medicaid benefits. We subtract the 5.28 million beneficiaries aged 65 and over in employer-group Medicare Advantage plans, which we calculate as the 5.9 million beneficiaries of all ages in these plans times the share of Medicare beneficiaries aged 65 and over. We also subtract the 3.94 million beneficiaries aged 65 and over with full Medicaid benefits who are enrolled in Medicare Advantage, which we calculate as the 8.3 million dual eligibles in Medicare Advantage times the same age and full-benefit shares as above. This assumes that the age and full-benefit shares of all dual eligibles also apply to dual eligibles in Medicare Advantage, and that no beneficiary is counted in both groups. Partial-benefit dual eligibles remain in the population. This leaves $28.3 - 5.28 - 3.94 = 19.1$ million Medicare Advantage enrollees, or 31.1 million beneficiaries who hold either plan. Since we assume 50% of seniors use a broker, we estimate that 15.6 million current beneficiaries used a broker to enroll in their plan. We leave out beneficiaries in Traditional Medicare without a Medigap plan. Brokers are paid only when a senior enrolls in a plan, so these beneficiaries are unlikely to have used a broker. Since they can still enroll in either plan, leaving them out understates the loss.

Both the population restriction and the 50% broker-use share are conservative. In our UAS

[90]The survey covers beneficiaries enrolled in both Part A and Part B, so its count of beneficiaries aged 65 and over is below the CMS count we use for seniors turning 65.

survey, 73% of respondents report neither Medicaid eligibility nor access to an employer plan (Table A6). Of these respondents, 56% used a broker (61% at ages 65 to 69), which is higher than the 50% we assume throughout. Broker use is also above 50% among UAS respondents who hold Medicare Advantage and among those who hold Medigap, the plans in our calculation for current beneficiaries. All of these shares count respondents who answer "Not sure" as not using a broker. Without the population restriction, the baseline loss is $1.82 billion a year for the 2.05 million seniors who turn 65 each year and use a broker.

Loss per advised senior A Medigap-fit senior loses the net harm when the broker recommends Medicare Advantage and the Medigap markup when the broker recommends Medigap. We value the markup at $386 a year, the average markup the structural model predicts when the broker recommends Medigap. At the estimates in Section 6.1, this loss is $2,154 a year. An MA-fit senior loses the over-insurance harm plus the markup when the broker recommends Medigap. The survey elicits this harm with both plans priced at $0, so it excludes the higher Medigap premium, and this loss is a lower bound. Under our main assumption, brokers recommend Medigap to 14.4% of MA-fit seniors, and the loss is $235 a year. An ambiguous-fit senior loses only the markup when the broker recommends Medigap, or $225 a year. Weighting by the plan-fit shares, the loss is $890 per advised senior per year.

MA-fit seniors No audited call presents an MA-fit senior, so we use the structural model to predict the broker's recommendation to this senior. A broker of the informed type knows that MA fits the senior and recommends Medigap only when the commission gap outweighs the suitability cost. At the estimated parameters, this almost never happens. We assume that a broker of the misinformed type, who believes that MA fits the audited Medigap-fit senior, also believes that Medigap fits an MA-fit senior. We cannot test this assumption with the audit. The 14.4% of MA-fit seniors recommended Medigap averages the two broker types over the same audit calls as the Medigap-fit senior, weighting each broker by the structural model's estimated probability that the broker is misinformed. Without the assumption about misinformed brokers, the loss of an MA-fit senior is $9 a year, the loss per advised senior is $776, and the baseline loss for seniors who turn 65 each year is $1.09 billion rather than $1.25 billion.

The loss could also be larger. On 21.0% of transcribed calls, the broker reported selling only Medigap. If we add this share of brokers, who recommend Medigap to every senior, the loss per advised senior rises to $1,062 and the baseline loss rises to $1.49 billion. The broker survey supports this higher value. In the survey, 41% of brokers recommend Medigap to a vignette senior who is an MA fit, compared with 14.4% under our main assumption.

The order of the two policies In Figure 7, we add the 25% Medigap markup cap after pay parity. Since the cap applies only to seniors who buy Medigap and pay parity increases the number

of these seniors, the cap is worth $124 million after pay parity and $105 million on its own. Of the $124 million, about $19 million comes from seniors whom pay parity moves into Medigap, where the markup cap now applies. The gain from the two policies together does not depend on the order.

Taxpayer savings The Medicare Payment Advisory Commission estimates that Medicare spends $84 billion more on Medicare Advantage enrollees than it would spend on the same beneficiaries in Traditional Medicare (Medicare Payment Advisory Commission, 2025), or $2,407 for each of the 34.9 million enrollees. This comparison uses the spending of Traditional Medicare beneficiaries, 42% of whom have Medigap (Freed et al., 2024). We make two assumptions. First, we assume a senior who receives a Medigap recommendation enrolls in Medigap, so the change in recommendations equals the change in MA enrollment. This may overestimate the number of seniors who leave MA if some seniors do not follow the broker's recommendation. Second, we assume that a senior who leaves MA enrolls in Traditional Medicare with Medigap rather than Traditional Medicare alone, which is plausible for seniors turning 65, who can buy Medigap without medical underwriting during their open-enrollment period. Medigap coverage raises Medicare spending by 22.2% (Cabral and Mahoney, 2019). We subtract the part of this additional spending that is not already in the MedPAC comparison, $1,561 a year, which leaves $846 per enrollee per year. Because the second assumption assigns this additional spending to every senior who leaves MA, it increases the estimated additional spending and understates the federal saving per senior.

We apply this value to every senior whom a policy moves out of Medicare Advantage, of either plan-fit type. As a result, part of the taxpayer savings in bar (4) comes from MA-fit seniors whom pay parity moves into Medigap, and these seniors lose consumer surplus. Under the two policies, MA-fit seniors lose $4 million a year because pay parity increases the share of brokers who recommend Medigap to an MA-fit senior. Note that bar (0) does not include any taxpayer cost, so bar (4) is a change in federal spending rather than a remaining loss.

Checking the plan-fit shares against enrollment Since we observe the plans current beneficiaries hold, we can check the plan-fit shares from the broker survey against actual enrollment. Let r_g be the share of brokers who recommend MA to a senior of plan-fit type g. For a Medigap-fit senior, $r_M = 34\%$ is the share of brokers the structural model predicts recommend MA to the senior in our audit study, on calls where the broker is paid the full MA commission (Section 7). For an MA-fit senior, r_A is one minus the 14.4% of brokers the structural model predicts recommend Medigap under the assumptions in the MA-fit seniors paragraph above. For an ambiguous-fit senior, r_X is one minus the 58% of brokers who recommend Medigap for the ambiguous-fit senior profile in our broker survey. Since we assume seniors enroll in the plan the broker recommends,

the predicted share of broker users enrolled in MA is

$$s_{\text{MA}} \;=\; \pi_M\, r_M \;+\; \pi_A\, r_A \;+\; \pi_X\, r_X. \tag{27}$$

At the broker survey shares, Equation (27) predicts that 61% of current beneficiaries who used a broker are enrolled in Medicare Advantage. In the 2023 Medicare Current Beneficiary Survey, 61% of beneficiaries who hold either plan are enrolled in MA (19.1 of 31.1 million). Since we assume the same broker-use share for both plans, this is also the observed MA share among broker users. Note that the predicted share is not the MA-fit share: it adds the Medigap-fit and ambiguous-fit seniors to whom brokers recommend MA and leaves out the MA-fit seniors to whom brokers recommend Medigap. Most of the predicted MA share comes from MA-fit seniors, so this comparison mainly checks the broker survey's plan-fit shares. It says little about the share of brokers who recommend MA to a Medigap-fit senior.

Alternative plan-fit shares Table A18 reports two checks on the loss for current beneficiaries. The first also uses the broker survey shares, but uses the observed number of beneficiaries in each plan. At the broker survey shares, Equation (27) gives the share of broker users in each plan who are of each plan-fit type. For example, $\pi_M r_M / s_{\text{MA}} = 19\%$ of broker users enrolled in Medicare Advantage are Medigap-fit seniors. We apply these shares to the observed number of beneficiaries in each plan, which gives a loss of $13.9 billion a year. The second holds the ambiguous-fit share at either end of its 95% confidence interval in the broker survey, 13% or 18%, and keeps the ratio of Medigap-fit to MA-fit seniors at its broker survey value. This gives a loss of $14.2 billion or $13.6 billion, respectively. At the average senior's net harm of $2,711 a year, the loss is $8.7 billion.

Table A18: Consumer Surplus Loss Among Current Beneficiaries Under Alternative Plan-Fit Shares

	Medigap-fit senior's net harm ($5,574)	Average senior's net harm ($2,711)
Broker survey plan-fit shares	$13.9	$8.7
Broker survey mix within each plan	$13.9	$8.7
Ambiguous-fit share of 13% to 18%	$13.6–14.2	$8.5–8.8

Notes: Table A18 reports the annual consumer surplus loss, in billions of dollars, for the 15.6 million current beneficiaries aged 65 and over who hold Medicare Advantage or Medigap and used a broker. The first row is the main estimate in Figure 8, at the broker survey plan-fit shares. The second row applies the share of each plan-fit type within each plan, calculated from Equation (27) at the broker survey shares, to the observed number of beneficiaries in each plan. The last row holds the ambiguous-fit share at the ends of its 95% confidence interval in the broker survey and keeps the ratio of Medigap-fit to MA-fit seniors at its broker survey value. This range reflects only sampling error in the survey share and is not a confidence interval for the loss.

Other assumptions for current beneficiaries We make two further assumptions for current beneficiaries. First, we assume that their plans reflect the broker recommendations we observe

today. Note that many of these beneficiaries enrolled when MA commissions were lower and the MA market was smaller. Second, we hold the Medigap markup at its level in the year of enrollment. This ignores two changes after enrollment. Seniors can switch to a cheaper plan with the same plan letter, which a birthday rule such as California's allows without medical underwriting. Ignoring these switches overstates the loss. The markup also grows with premiums, and ignoring this growth understates the loss.

Inference The national estimates depend on inputs we do not estimate: the population counts, the plan-fit shares, the broker-use share, and the two harms brokers report in our survey: the harm to a Medigap-fit senior enrolled in Medicare Advantage and the harm to an MA-fit senior enrolled in Medigap. We hold these inputs at their reported values. A confidence interval from the bootstrap refits would reflect only the uncertainty in the estimated parameters, so we do not report one for the baseline loss. Instead, we report the baseline loss under alternative assumptions for MA-fit seniors (above) and, for current beneficiaries, under alternative plan-fit shares (Table A18). The range in Table A18 from the ambiguous-fit share reflects only the sampling error in the broker survey's ambiguous-fit share. For the policy counterfactuals, the caller-cluster bootstrap refits of Appendix E.3 give a 95% confidence interval of \$61 million to \$226 million for the \$151 million gain from pay parity among seniors who turn 65 each year. This interval also holds the inputs above at their reported values.